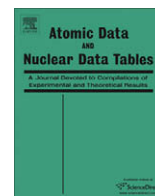


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## Atomic Data and Nuclear Data Tables

journal homepage: [www.elsevier.com/locate/adt](http://www.elsevier.com/locate/adt)Hartree–Fock–Roothaan energies and expectation values for the neutral atoms He to Uuo: The *B*-spline expansion methodShiro L. Saito<sup>\*</sup>

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## ABSTRACT

The ground state energies and expectation values of atoms are given by Hartree–Fock–Roothaan calculations with one *B*-spline set. For the neutral atoms He to Uuo, the total energies, kinetic energies, potential energies, and virial ratios are tabulated. Our total energies are in excellent agreement with the highly accurate 10-digit numerical Hartree–Fock energies given by Koga and Thakkar [T. Koga, A.J. Thakkar, J. Phys. B 29 (1996) 2973]. The virial ratios are in complete agreement to within 12-digits of the exact value  $-2$ . Orbital energies, electron densities at the nucleus, electron–nucleus cusp ratios, and radial expectation values  $\langle r^n \rangle$  ( $n = 2, 1, -1, -2, -3$ ) are also given.

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## 1. Introduction

The Hartree–Fock (HF) equations were first proposed by Fock [1] in 1930. Since then, the HF method has taken a central role in studies of atomic and molecular electronic structure. The development of effective computational methods for HF equations is also a subject of study. There are two computational methods for HF equations: numerical and algebraic methods. Although numerical methods are standard, their application is restricted to atoms and diatomic molecules. Numerical computational methods for atomic HF equations have been developed by Froese Fischer [2,3]. Algebraic methods can also be applied to polyatomic molecules. Algebraic methods approximate wavefunctions by a linear combination of appropriate basis functions, which has been developed by Roothaan [4–6]. This method is often referred to as the Hartree–Fock–Roothaan (HFR) method.

Atomic HF equations give physical properties of various atomic systems in the non-relativistic independent particle approximation. These data are also important for molecular calculation, for it is impossible to develop basis sets and pseudo-potentials for use in molecular calculations without them. Moreover, solutions of atomic HF equations are needed for atomic and molecular calculations, including electron correlation or relativistic effects. It is therefore important to tabulate accurate atomic HF data.

In 1974, a monumental compilation of atomic HFR wavefunctions was published by Clementi and Roetti [7]. Their atomic HFR wavefunctions were calculated with optimized Slater-type functions. Unfortunately, their results include some errors. Tatewaki and Sekiya [8–10], Bunge et al. [11,12], and Koga et al. [13] have improved Clementi–Roetti’s results by careful selection of principal quantum numbers and full optimization of nonlinear parameters in Slater-type functions.

Basis sets commonly used for atomic HFR calculations are Slater-type functions and Gaussian-type functions. All necessary atomic integrals over these functions can be evaluated analytically. However, their nonlinear parameters must be optimized and the principal quantum number chosen; this is laborious and tedious work. Since these functions are non-orthogonal sets, problems of computational linear dependence may arise.

Piecewise polynomial functions also be used as a basis set. The advantage of piecewise polynomial functions is that they are very flexible and relatively free from computational linear dependence. Several types of piecewise polynomial functions have been applied to atomic calculations by Shore [14–16], Gilbert and co-workers [17,18], and Silverstone’s group [19–21]. In particular, the *B*-spline set is frequently used for atomic calculations. The *B*-splines of order  $K$   $\{B_{i,K}\}$  are piecewise polynomials of degree  $K - 1$  on a knot sequence in a cavity of radius  $R$  [22]. The knot sequence  $\{t_i\}$  is a set of points defined on an interval  $[0, R]$ .  $B_{i,K}(r)$  is nonzero in the interval  $[t_i, t_{i+K})$ . The first application of *B*-splines to atomic calculations was by Shore [14–16]. They have been further applied by Bottcher and

Strayer [23], Johnson and co-workers [24,25], Froese Fischer and co-workers [26,27], Jeng and Hsue [28], Saito [29], and others.

Since the *B*-splines are very flexible, it is possible to apply a common *B*-spline set to all atoms, irrespective of orbital symmetry, without loss of accuracy. Consequently, *B*-splines can act as a universal basis set. This has been shown by the present author for atomic systems with nuclear charge  $Z \leq 54$  [29]. Such universality makes it possible to store all necessary atomic integrals over the *B*-splines so that they can be used for various atomic calculations. The *B*-spline set is therefore a very valuable basis set for atomic calculations.

Our present purpose is to give highly accurate atomic data by the HFR method with the *B*-spline set. Clementi and Roetti [7], Bunge et al. [11,12], and Koga et al. [13] gave atomic HFR data for  $Z \leq 54$ , because relativistic effects are comparatively small. We give non-relativistic HF limits for  $Z \leq 118$ . To this end, we carried out HFR calculations with the *B*-spline set for the ground state energies of neutral atoms from He to Uuo, and obtained the total, kinetic, and potential energies to 12-digit accuracy. These energies, together with virial ratios, the electron densities at the nucleus, electron–nucleus cusp ratios, orbital energies, and radial expectation values  $r^n$  ( $n = 2, 1, -1, -2, -3$ ) are reported below (see Table 1).

2. *B*-spline Hartree–Fock–Roothaan method

The atomic HFR orbital with symmetry  $\lambda$  can be written as

$$\psi_{n\lambda m}(r, \theta, \varphi) = r^{-1} P_{n\lambda}(r) Y_{\lambda}^m(\theta, \varphi), \quad (1)$$

where  $P_{n\lambda}$  and  $Y_{\lambda}^m(\theta, \varphi)$  respectively denote the radial function and the spherical harmonics. In the present method, the  $P_{n\lambda}(r)$  are expanded by the *B*-spline set defined in an interval  $[0, R]$  with the boundary conditions  $P_{n\lambda}(0) = 0$  and  $P_{n\lambda}(R) = 0$ . A single *B*-spline set is employed regardless of atomic systems and symmetries of the orbitals, in view of its high flexibility.

To construct the *B*-spline set for atomic calculations, the knot sequence must be generated in  $[0, R]$  with  $K$ -fold multiple end-points. Since the first and last terms of such *B*-splines are nonzero at  $r = 0$  and at  $r = R$ , respectively, they were removed from the basis set in order to satisfy the boundary conditions. An  $N$ -term *B*-spline set  $\{B_{i,K}\}_{i=2, \dots, N+1}$  is thereby obtained, and  $P_{n\lambda}(r)$  is expanded as

$$P_{n\lambda}(r) = \sum_{i=1}^N C_{n\lambda,i} B_{i+1,K}(r). \quad (2)$$

The one-electron integrals for HFR calculations are

$$I_{ij} = \int_0^R dr B_{i+1,K}(r) \mathcal{O} B_{j+1,K}(r), \quad (3)$$

where  $\mathcal{O}$  is a one-electron operator and  $ij = 1, \dots, N$ . Evaluation of a total of  $\frac{1}{2}N(N+1)$  one-electron integrals (OEs) is needed in the HFR calculations according to Roothaan and Bagus [6]. The number of OEs ( $N_{\text{OEI}}$ ) needed in the *B*-spline HFR calculations is smaller, how-

ever, because the products  $B_{i,K}(r)B_{j,K}(r)$  ( $i \geq j$ ) vanish outside  $[t_i, t_{j+K})$  due to the definition of the  $B$ -splines. Hence, only OEIs which satisfy the conditions  $i - j < K$  and  $i \geq j$  need to be evaluated, and  $N_{\text{OEI}}$  is expressed as

$$N_{\text{OEI}} = \frac{1}{2}K(2N - K + 1). \quad (4)$$

The supermatrices  $\mathcal{P}_{ij,\mu kl}$  and  $\mathcal{Q}_{ij,\mu kl}$  defined by Roothaan and Bagus [6] are rewritten by using the two-electron integrals (TEIs) over the  $B$ -spline set

$$R_{ij,kl}^v = \int_0^R dr_1 \int_0^R dr_2 B_{i+1,K}(r_1) B_{j+1,K}(r_1) \frac{r_1^v}{r_1^{v+1}} B_{k+1,K}(r_2) B_{l+1,K}(r_2) \quad (5)$$

as follows:

$$\mathcal{P}_{ij,\mu kl} = R_{ij,kl}^0 - \frac{1}{2} \sum_{v=|\lambda-\mu|}^{\lambda+\mu} A_{\lambda\mu\nu} R_{ik,jl}^v \quad (6)$$

and

$$\mathcal{Q}_{ij,\mu kl} = \sum_{v=0}^{2(\lambda,\mu)_<} J_{\lambda\mu\nu} R_{ij,kl}^v - \frac{1}{2} \sum_{v=|\lambda-\mu|}^{\lambda+\mu} K_{\lambda\mu\nu} R_{ik,jl}^v, \quad (7)$$

where  $A_{\lambda\mu\nu} J_{\lambda\mu\nu}$ , and  $K_{\lambda\mu\nu}$  are numerical coefficients [6,30]. Since the supermatrices contain redundant TEIs, we need only evaluate the distinct TEIs, which satisfy the conditions

$$i \geq j, \quad k \geq l, \quad (8)$$

$$i - j < K, \quad k - l < K, \quad (9)$$

and

$$\frac{1}{2}i(i-1) + j \geq \frac{1}{2}k(k-1) + l. \quad (10)$$

These conditions arise from the permutational symmetries of the indices  $i, j, k$ , and  $l$  of the TEIs. The number of TEIs ( $N_{\text{TEI}}$ ) needed in constructing the supermatrices is

$$N_{\text{TEI}} = \frac{1}{2}N_{\text{OEI}}(N_{\text{OEI}} + 1)(2\lambda_{\text{max}} + 1), \quad (11)$$

where  $\lambda_{\text{max}}$  is the maximum azimuthal quantum number of the atom under consideration. All the integrals needed in the HFR calculations were evaluated by a Gaussian integration procedure.

### 3. Computational details

Our previous work showed that a 40-term  $B$ -spline set with  $K=9$  and  $R=40$  gives results accurate to at least 10-digits [29]. From many test calculations, we found that a  $B$ -spline set with  $N \geq 80$  gives accuracy to 12-digits. Hence, in this work, we used a 100-term  $B$ -spline set with  $K=9$  and  $R=40$  on an exponential-type knot sequence [17] with the initial interval  $R_1 = 10^{-4}$ . From the HFR wavefunctions, we calculated the spherical averaged electron densities  $\rho(r)$ , their values at the nucleus  $\rho_0 = \rho(0)$ , and the electron–nucleus cusp ratio [31]

$$C_{\text{EN}} = -\frac{1}{2Z} \frac{d \log \rho(r)}{dr} \Big|_{r=0}. \quad (12)$$

The exact value of  $C_{\text{EN}}$  is unity for the exact HF wavefunction.

### 4. Accuracy of results

As far as we know, the 10-digit numerical HF energies for atoms from He to Lr given by Koga and Thakkar [32] are considered to be very accurate. The present energies are in agreement with their results. Also, the present virial ratios for all atoms are in agreement to 12-digits and the  $C_{\text{EN}}$  values to 8-digits of the exact values. Our HFR wavefunctions are therefore highly accurate, while the corresponding energies exhibit 12-digit accuracy, two more digits than the most accurate results available to date [32].

### Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.adt.2009.06.001.

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## Explanation of Tables

**Table 1. Hartree–Fock–Roothaan energies and expectation values of the ground state for He–Uuo.**

Each atomic HFR data are collected as a block. The first row lists the element symbol, the atomic number  $Z$ , the configuration of the ground state, and its term. The configurations are described with the corresponding noble gas symbol. The second and third row list the Total, Kinetic, and Potential energies in hartree units, together with the Virial ratio. The fourth row gives the electron density at the nucleus  $\rho_0$  and the cusp ratio of the electron–nucleus  $C_{EN}$ . The orbital energy,  $E_{ORB}$ , and radial expectation values of  $r^n$  ( $n = 2, 1, -1, -2, -3$ ),  $\langle r^n \rangle$ , are tabulated for each HFR orbital, 1s, 2s, 2p, etc.

**Table 1**

Hartree-Fock-Roothaan energies and expectation values of the ground state for He–Uuo. See page 839 for Explanation of Tables.

|                                                                            |              |                       |                     |                          |                          |                          |
|----------------------------------------------------------------------------|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| He, $Z = 2$ , $1s^2$ ( $^1S$ )                                             |              |                       |                     |                          |                          |                          |
| Total energy = $-2.86167999561$ ;      Kinetic energy = $2.86167999561$    |              |                       |                     |                          |                          |                          |
| Potential energy = $-5.72335999122$ ;      Virial ratio = $-2.00000000000$ |              |                       |                     |                          |                          |                          |
| $\rho_0 = 3.5959183$ ; $C_{EN} = 1.0000000$                                |              |                       |                     |                          |                          |                          |
|                                                                            | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ |                          |
| 1s                                                                         | $-0.917956$  | $1.184828$            | $0.927273$          | $1.687282$               | $5.995504$               |                          |
| Li, $Z = 3$ , [He] $2s^1$ ( $^2S$ )                                        |              |                       |                     |                          |                          |                          |
| Total energy = $-7.43272693073$ ;      Kinetic energy = $7.43272693073$    |              |                       |                     |                          |                          |                          |
| Potential energy = $-14.8654538615$ ;      Virial ratio = $-2.00000000000$ |              |                       |                     |                          |                          |                          |
| $\rho_0 = 13.814820$ ; $C_{EN} = 1.0000000$                                |              |                       |                     |                          |                          |                          |
|                                                                            | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ |                          |
| 1s                                                                         | $-2.477741$  | $0.446803$            | $0.573125$          | $2.685034$               | $14.888255$              |                          |
| 2s                                                                         | $-0.196323$  | $17.738462$           | $3.873664$          | $0.345391$               | $0.435419$               |                          |
| Be, $Z = 4$ , [He] $2s^2$ ( $^1S$ )                                        |              |                       |                     |                          |                          |                          |
| Total energy = $-14.5730231683$ ;      Kinetic energy = $14.5730231683$    |              |                       |                     |                          |                          |                          |
| Potential energy = $-29.1460463366$ ;      Virial ratio = $-2.00000000000$ |              |                       |                     |                          |                          |                          |
| $\rho_0 = 35.387717$ ; $C_{EN} = 1.0000000$                                |              |                       |                     |                          |                          |                          |
|                                                                            | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ |                          |
| 1s                                                                         | $-4.732670$  | $0.232955$            | $0.414994$          | $3.681877$               | $27.753365$              |                          |
| 2s                                                                         | $-0.309270$  | $8.426431$            | $2.649414$          | $0.522522$               | $1.055647$               |                          |
| B, $Z = 5$ , [He] $2s^2 2p^1$ ( $^2P$ )                                    |              |                       |                     |                          |                          |                          |
| Total energy = $-24.5290607285$ ;      Kinetic energy = $24.5290607285$    |              |                       |                     |                          |                          |                          |
| Potential energy = $-49.0581214571$ ;      Virial ratio = $-2.00000000000$ |              |                       |                     |                          |                          |                          |
| $\rho_0 = 71.921371$ ; $C_{EN} = 1.0000000$                                |              |                       |                     |                          |                          |                          |
|                                                                            | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                         | $-7.695335$  | $0.143361$            | $0.325866$          | $4.674339$               | $44.538021$              |                          |
| 2s                                                                         | $-0.494706$  | $4.709135$            | $1.977064$          | $0.712882$               | $2.024533$               |                          |
| 2p                                                                         | $-0.309856$  | $6.146091$            | $2.204758$          | $0.605005$               | $0.529886$               | $0.775586$               |
| C, $Z = 6$ , [He] $2s^2 2p^2$ ( $^3P$ )                                    |              |                       |                     |                          |                          |                          |
| Total energy = $-37.6886189630$ ;      Kinetic energy = $37.6886189630$    |              |                       |                     |                          |                          |                          |
| Potential energy = $-75.3772379259$ ;      Virial ratio = $-2.00000000000$ |              |                       |                     |                          |                          |                          |
| $\rho_0 = 127.45797$ ; $C_{EN} = 1.0000000$                                |              |                       |                     |                          |                          |                          |
|                                                                            | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                         | $-11.325519$ | $0.097199$            | $0.268443$          | $5.664439$               | $65.234324$              |                          |
| 2s                                                                         | $-0.705627$  | $3.052065$            | $1.589344$          | $0.896798$               | $3.255510$               |                          |
| 2p                                                                         | $-0.433341$  | $3.746799$            | $1.714494$          | $0.783503$               | $0.892063$               | $1.691811$               |
| N, $Z = 7$ , [He] $2s^2 2p^3$ ( $^4S$ )                                    |              |                       |                     |                          |                          |                          |
| Total energy = $-54.4009342085$ ;      Kinetic energy = $54.4009342085$    |              |                       |                     |                          |                          |                          |
| Potential energy = $-108.801868417$ ;      Virial ratio = $-2.00000000000$ |              |                       |                     |                          |                          |                          |
| $\rho_0 = 205.96835$ ; $C_{EN} = 1.0000000$                                |              |                       |                     |                          |                          |                          |
|                                                                            | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                         | $-15.629060$ | $0.070265$            | $0.228297$          | $6.653236$               | $89.841983$              |                          |
| 2s                                                                         | $-0.945324$  | $2.149438$            | $1.332277$          | $1.078176$               | $4.753636$               |                          |
| 2p                                                                         | $-0.567589$  | $2.547657$            | $1.409632$          | $0.957690$               | $1.336282$               | $3.099635$               |
| O, $Z = 8$ , [He] $2s^2 2p^4$ ( $^3P$ )                                    |              |                       |                     |                          |                          |                          |
| Total energy = $-74.8093984700$ ;      Kinetic energy = $74.8093984700$    |              |                       |                     |                          |                          |                          |
| Potential energy = $-149.618796940$ ;      Virial ratio = $-2.00000000000$ |              |                       |                     |                          |                          |                          |
| $\rho_0 = 311.66089$ ; $C_{EN} = 1.0000000$                                |              |                       |                     |                          |                          |                          |
|                                                                            | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                         | $-20.668657$ | $0.053146$            | $0.198589$          | $7.642171$               | $118.382564$             |                          |
| 2s                                                                         | $-1.244315$  | $1.581225$            | $1.141965$          | $1.265272$               | $6.591452$               |                          |
| 2p                                                                         | $-0.631906$  | $1.974975$            | $1.232198$          | $1.111111$               | $1.818711$               | $4.974052$               |
| F, $Z = 9$ , [He] $2s^2 2p^5$ ( $^2P$ )                                    |              |                       |                     |                          |                          |                          |
| Total energy = $-99.4093493867$ ;      Kinetic energy = $99.4093493867$    |              |                       |                     |                          |                          |                          |
| Potential energy = $-198.818698773$ ;      Virial ratio = $-2.00000000000$ |              |                       |                     |                          |                          |                          |
| $\rho_0 = 448.32228$ ; $C_{EN} = 1.0000000$                                |              |                       |                     |                          |                          |                          |
|                                                                            | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                         | $-26.382760$ | $0.041612$            | $0.175747$          | $8.630362$               | $150.834170$             |                          |
| 2s                                                                         | $-1.572535$  | $1.216566$            | $1.001094$          | $1.449751$               | $8.697084$               |                          |
| 2p                                                                         | $-0.730018$  | $1.543522$            | $1.084786$          | $1.271674$               | $2.394764$               | $7.545112$               |

Table 1 (continued)

| Ne, $Z = 10$ , [He]2s <sup>2</sup> 2p <sup>6</sup> ( <sup>1</sup> S) |            |                       |                     |                          |                          |                          |
|----------------------------------------------------------------------|------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| Total energy = −128.547098109 ; Kinetic energy = 128.547098109       |            |                       |                     |                          |                          |                          |
| Potential energy = −257.094196219 ; Virial ratio = −2.00000000000    |            |                       |                     |                          |                          |                          |
| $\rho_0 = 619.92207$ ; $C_{EN} = 1.0000000$                          |            |                       |                     |                          |                          |                          |
|                                                                      | $E_{ORB}$  | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                   | −32.772443 | 0.033470              | 0.157631            | 9.618054                 | 187.197184               |                          |
| 2s                                                                   | −1.930391  | 0.967084              | 0.892113            | 1.632553                 | 11.071316                |                          |
| 2p                                                                   | −0.850410  | 1.228456              | 0.965274            | 1.435350                 | 3.058851                 | 10.906183                |
| Na, $Z = 11$ , [Ne]3s <sup>1</sup> ( <sup>2</sup> S)                 |            |                       |                     |                          |                          |                          |
| Total energy = −161.858911617 ; Kinetic energy = 161.858911617       |            |                       |                     |                          |                          |                          |
| Potential energy = −323.717823234 ; Virial ratio = −2.00000000000    |            |                       |                     |                          |                          |                          |
| $\rho_0 = 833.75753$ ; $C_{EN} = 1.0000000$                          |            |                       |                     |                          |                          |                          |
|                                                                      | $E_{ORB}$  | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                   | −40.478501 | 0.027481              | 0.142858            | 10.607384                | 227.533835               |                          |
| 2s                                                                   | −2.797026  | 0.731495              | 0.779068            | 1.867338                 | 14.453163                |                          |
| 3s                                                                   | −0.182103  | 20.704973             | 4.208771            | 0.301398                 | 0.398380                 |                          |
| 2p                                                                   | −1.518140  | 0.822134              | 0.798485            | 1.696598                 | 4.187700                 | 17.004822                |
| Mg, $Z = 12$ , [Ne]3s <sup>2</sup> ( <sup>1</sup> S)                 |            |                       |                     |                          |                          |                          |
| Total energy = −199.614636425 ; Kinetic energy = 199.614636425       |            |                       |                     |                          |                          |                          |
| Potential energy = −399.229272849 ; Virial ratio = −2.00000000000    |            |                       |                     |                          |                          |                          |
| $\rho_0 = 1093.7178$ ; $C_{EN} = 1.0000000$                          |            |                       |                     |                          |                          |                          |
|                                                                      | $E_{ORB}$  | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                   | −49.031736 | 0.022956              | 0.130594            | 11.597954                | 271.847138               |                          |
| 2s                                                                   | −3.767721  | 0.571084              | 0.690335            | 2.107821                 | 18.382362                |                          |
| 3s                                                                   | −0.253053  | 12.418651             | 3.252948            | 0.399387                 | 0.788451                 |                          |
| 2p                                                                   | −2.282226  | 0.597696              | 0.684998            | 1.951690                 | 5.469806                 | 24.920088                |
| Al, $Z = 13$ , [Ne]3s <sup>2</sup> 3p <sup>1</sup> ( <sup>2</sup> P) |            |                       |                     |                          |                          |                          |
| Total energy = −241.876707251 ; Kinetic energy = 241.876707251       |            |                       |                     |                          |                          |                          |
| Potential energy = −483.753414502 ; Virial ratio = −2.00000000000    |            |                       |                     |                          |                          |                          |
| $\rho_0 = 1402.8456$ ; $C_{EN} = 1.0000000$                          |            |                       |                     |                          |                          |                          |
|                                                                      | $E_{ORB}$  | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                   | −58.501027 | 0.019459              | 0.120258            | 12.589248                | 320.129366               |                          |
| 2s                                                                   | −4.910672  | 0.458897              | 0.620026            | 2.349007                 | 22.797468                |                          |
| 3s                                                                   | −0.393420  | 7.890676              | 2.599278            | 0.506786                 | 1.359444                 |                          |
| 2p                                                                   | −3.218303  | 0.455357              | 0.600543            | 2.205108                 | 6.915849                 | 34.938537                |
| 3p                                                                   | −0.209951  | 14.006309             | 3.433893            | 0.379283                 | 0.310545                 | 1.088344                 |
| Si, $Z = 14$ , [Ne]3s <sup>2</sup> 3p <sup>2</sup> ( <sup>3</sup> P) |            |                       |                     |                          |                          |                          |
| Total energy = −288.854362517 ; Kinetic energy = 288.854362517       |            |                       |                     |                          |                          |                          |
| Potential energy = −577.708725034 ; Virial ratio = −2.00000000000    |            |                       |                     |                          |                          |                          |
| $\rho_0 = 1765.6069$ ; $C_{EN} = 1.0000000$                          |            |                       |                     |                          |                          |                          |
|                                                                      | $E_{ORB}$  | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                   | −68.812456 | 0.016701              | 0.111431            | 13.581150                | 372.382224               |                          |
| 2s                                                                   | −6.156538  | 0.377257              | 0.562941            | 2.590401                 | 27.691122                |                          |
| 3s                                                                   | −0.539842  | 5.676258              | 2.207086            | 0.603231                 | 1.997903                 |                          |
| 2p                                                                   | −4.256054  | 0.359681              | 0.535408            | 2.456382                 | 8.520325                 | 47.267269                |
| 3p                                                                   | −0.297115  | 8.980959              | 2.752220            | 0.478031                 | 0.501193                 | 2.054174                 |
| P, $Z = 15$ , [Ne]3s <sup>2</sup> 3p <sup>3</sup> ( <sup>4</sup> S)  |            |                       |                     |                          |                          |                          |
| Total energy = −340.718780975 ; Kinetic energy = 340.718780975       |            |                       |                     |                          |                          |                          |
| Potential energy = −681.437561951 ; Virial ratio = −2.00000000000    |            |                       |                     |                          |                          |                          |
| $\rho_0 = 2186.3141$ ; $C_{EN} = 1.0000000$                          |            |                       |                     |                          |                          |                          |
|                                                                      | $E_{ORB}$  | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                   | −79.969715 | 0.014488              | 0.103805            | 14.573553                | 428.606531               |                          |
| 2s                                                                   | −7.511096  | 0.315921              | 0.515660            | 2.831708                 | 33.056869                |                          |
| 3s                                                                   | −0.696416  | 4.347110              | 1.932687            | 0.694730                 | 2.712754                 |                          |
| 2p                                                                   | −5.400959  | 0.291798              | 0.483401            | 2.706271                 | 10.283993                | 62.144536                |
| 3p                                                                   | −0.391708  | 6.389654              | 2.322713            | 0.570152                 | 0.715755                 | 3.311114                 |
| S, $Z = 16$ , [Ne]3s <sup>2</sup> 3p <sup>4</sup> ( <sup>3</sup> P)  |            |                       |                     |                          |                          |                          |
| Total energy = −397.504895917 ; Kinetic energy = 397.504895917       |            |                       |                     |                          |                          |                          |
| Potential energy = −795.009791835 ; Virial ratio = −2.00000000000    |            |                       |                     |                          |                          |                          |
| $\rho_0 = 2669.4697$ ; $C_{EN} = 1.0000000$                          |            |                       |                     |                          |                          |                          |
|                                                                      | $E_{ORB}$  | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                   | −92.004450 | 0.012686              | 0.097151            | 15.566449                | 488.806575               |                          |
| 2s                                                                   | −9.004290  | 0.268530              | 0.475772            | 3.073197                 | 38.899577                |                          |
| 3s                                                                   | −0.879527  | 3.443217              | 1.720723            | 0.786103                 | 3.534555                 |                          |
| 2p                                                                   | −6.682509  | 0.242088              | 0.441042            | 2.954448                 | 12.202760                | 79.775192                |
| 3p                                                                   | −0.437369  | 5.065257              | 2.060722            | 0.650667                 | 0.941922                 | 4.840029                 |

(continued on next page)

Table 1 (continued)

| Cl, Z = 17, [Ne]3s <sup>2</sup> 3p <sup>5</sup> ( <sup>2</sup> P)      |             |                       |                     |                          |                          |                          |
|------------------------------------------------------------------------|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| Total energy = −459.482072393 ;      Kinetic energy = 459.482072393    |             |                       |                     |                          |                          |                          |
| Potential energy = −918.964144786 ;      Virial ratio = −2.00000000000 |             |                       |                     |                          |                          |                          |
| $\rho_0 = 3219.1904$ ; $C_{EN} = 1.0000000$                            |             |                       |                     |                          |                          |                          |
|                                                                        | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                     | −104.884421 | 0.011200              | 0.091295            | 16.559692                | 552.979472               |                          |
| 2s                                                                     | −10.607481  | 0.231193              | 0.441714            | 3.314397                 | 45.207778                |                          |
| 3s                                                                     | −1.072912   | 2.812967              | 1.555630            | 0.874884                 | 4.434402                 |                          |
| 2p                                                                     | −8.072228   | 0.204326              | 0.405715            | 3.201537                 | 14.277858                | 100.395288               |
| 3p                                                                     | −0.506400   | 4.059228              | 1.842027            | 0.733287                 | 1.201130                 | 6.768715                 |
| Ar, Z = 18, [Ne]3s <sup>2</sup> 3p <sup>6</sup> ( <sup>1</sup> S)      |             |                       |                     |                          |                          |                          |
| Total energy = −526.817512803 ;      Kinetic energy = 526.817512803    |             |                       |                     |                          |                          |                          |
| Potential energy = −1053.63502561 ;      Virial ratio = −2.00000000000 |             |                       |                     |                          |                          |                          |
| $\rho_0 = 3839.7818$ ; $C_{EN} = 1.0000000$                            |             |                       |                     |                          |                          |                          |
|                                                                        | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                     | −118.610351 | 0.009960              | 0.086104            | 17.553229                | 621.125519               |                          |
| 2s                                                                     | −12.322153  | 0.201226              | 0.412280            | 3.555317                 | 51.979683                |                          |
| 3s                                                                     | −1.277353   | 2.350432              | 1.422172            | 0.961985                 | 5.414412                 |                          |
| 2p                                                                     | −9.571466   | 0.174342              | 0.375330            | 3.449989                 | 16.525582                | 124.378607               |
| 3p                                                                     | −0.591017   | 3.310850              | 1.662959            | 0.814074                 | 1.473630                 | 8.974745                 |
| K, Z = 19, [Ar]4s <sup>1</sup> ( <sup>2</sup> S)                       |             |                       |                     |                          |                          |                          |
| Total energy = −599.164786767 ;      Kinetic energy = 599.164786767    |             |                       |                     |                          |                          |                          |
| Potential energy = −1198.32957353 ;      Virial ratio = −2.00000000000 |             |                       |                     |                          |                          |                          |
| $\rho_0 = 4538.6540$ ; $C_{EN} = 1.0000000$                            |             |                       |                     |                          |                          |                          |
|                                                                        | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                     | −133.533050 | 0.008913              | 0.081466            | 18.547363                | 693.264691               |                          |
| 2s                                                                     | −14.489958  | 0.176593              | 0.386389            | 3.797706                 | 59.263023                |                          |
| 3s                                                                     | −1.748781   | 1.883466              | 1.277058            | 1.074858                 | 6.817757                 |                          |
| 4s                                                                     | −0.147475   | 31.544762             | 5.243732            | 0.236589                 | 0.295275                 |                          |
| 2p                                                                     | −11.519280  | 0.150801              | 0.349427            | 3.697027                 | 18.926783                | 151.807393               |
| 3p                                                                     | −0.954423   | 2.440657              | 1.436846            | 0.939014                 | 1.946484                 | 12.955908                |
| Ca, Z = 20, [Ar]4s <sup>2</sup> ( <sup>1</sup> S)                      |             |                       |                     |                          |                          |                          |
| Total energy = −676.758185925 ;      Kinetic energy = 676.758185925    |             |                       |                     |                          |                          |                          |
| Potential energy = −1353.51637185 ;      Virial ratio = −2.00000000000 |             |                       |                     |                          |                          |                          |
| $\rho_0 = 5319.6071$ ; $C_{EN} = 1.0000000$                            |             |                       |                     |                          |                          |                          |
|                                                                        | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                     | −149.363726 | 0.008023              | 0.077301            | 19.541934                | 769.393189               |                          |
| 2s                                                                     | −16.822744  | 0.156172              | 0.363504            | 4.040858                 | 67.043922                |                          |
| 3s                                                                     | −2.245376   | 1.546004              | 1.159424            | 1.188772                 | 8.398846                 |                          |
| 4s                                                                     | −0.195530   | 20.453347             | 4.218478            | 0.299737                 | 0.535008                 |                          |
| 2p                                                                     | −13.629269  | 0.131725              | 0.326869            | 3.944087                 | 21.491263                | 183.005615               |
| 3p                                                                     | −1.340707   | 1.908065              | 1.274550            | 1.058812                 | 2.462375                 | 17.740481                |
| Sc, Z = 21, [Ar]4s <sup>2</sup> 3d <sup>1</sup> ( <sup>2</sup> D)      |             |                       |                     |                          |                          |                          |
| Total energy = −759.735718041 ;      Kinetic energy = 759.735718041    |             |                       |                     |                          |                          |                          |
| Potential energy = −1519.47143608 ;      Virial ratio = −2.00000000000 |             |                       |                     |                          |                          |                          |
| $\rho_0 = 6182.3043$ ; $C_{EN} = 1.0000000$                            |             |                       |                     |                          |                          |                          |
|                                                                        | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                     | −165.899899 | 0.007259              | 0.073535            | 20.537538                | 849.554833               |                          |
| 2s                                                                     | −19.080621  | 0.139111              | 0.343178            | 4.284364                 | 75.313932                |                          |
| 3s                                                                     | −2.567324   | 1.339906              | 1.079391            | 1.281937                 | 9.815223                 |                          |
| 4s                                                                     | −0.210109   | 18.070982             | 3.959738            | 0.320316                 | 0.616974                 |                          |
| 2p                                                                     | −15.668247  | 0.116092              | 0.307067            | 4.191507                 | 24.224801                | 218.306681               |
| 3p                                                                     | −1.574549   | 1.619878              | 1.174391            | 1.152126                 | 2.911816                 | 22.411307                |
| 3d                                                                     | −0.343712   | 3.634409              | 1.675448            | 0.798978                 | 0.886336                 | 1.429417                 |
| Ti, Z = 22, [Ar]4s <sup>2</sup> 3d <sup>2</sup> ( <sup>3</sup> F)      |             |                       |                     |                          |                          |                          |
| Total energy = −848.405996991 ;      Kinetic energy = 848.405996991    |             |                       |                     |                          |                          |                          |
| Potential energy = −1696.81199398 ;      Virial ratio = −2.00000000000 |             |                       |                     |                          |                          |                          |
| $\rho_0 = 7133.2303$ ; $C_{EN} = 1.0000000$                            |             |                       |                     |                          |                          |                          |
|                                                                        | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                     | −183.272759 | 0.006598              | 0.070116            | 21.533644                | 933.722829               |                          |
| 2s                                                                     | −21.422913  | 0.124734              | 0.325038            | 4.527617                 | 84.053712                |                          |
| 3s                                                                     | −2.873398   | 1.180624              | 1.013029            | 1.370325                 | 11.257740                |                          |
| 4s                                                                     | −0.220788   | 16.501038             | 3.778877            | 0.336130                 | 0.679574                 |                          |
| 2p                                                                     | −17.791188  | 0.103139              | 0.289587            | 4.438400                 | 27.117383                | 257.833330               |
| 3p                                                                     | −1.795088   | 1.405778              | 1.093772            | 1.239883                 | 3.369373                 | 27.544577                |
| 3d                                                                     | −0.440656   | 2.737548              | 1.459755            | 0.907418                 | 1.129236                 | 2.016352                 |

Table 1 (continued)

| V, $Z = 23$ , [Ar]4s <sup>2</sup> 3d <sup>3</sup> ( <sup>4</sup> F)  |             |                       |                     |                          |                          |                          |
|----------------------------------------------------------------------|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| Total energy = −942.884337738 ; Kinetic energy = 942.884337738       |             |                       |                     |                          |                          |                          |
| Potential energy = −1885.76867548 ; Virial ratio = −2.00000000000    |             |                       |                     |                          |                          |                          |
| $\rho_0 = 8176.9554$ ; $C_{EN} = 1.0000000$                          |             |                       |                     |                          |                          |                          |
|                                                                      | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                   | −201.502832 | 0.006023              | 0.067000            | 22.530119                | 1021.892402              |                          |
| 2s                                                                   | −23.874653  | 0.112497              | 0.308744            | 4.770636                 | 93.262185                |                          |
| 3s                                                                   | −3.183186   | 1.050952              | 0.955586            | 1.456687                 | 12.760334                |                          |
| 4s                                                                   | −0.230579   | 15.232738             | 3.626312            | 0.350651                 | 0.738347                 |                          |
| 2p                                                                   | −20.022491  | 0.092273              | 0.274031            | 4.684841                 | 30.168581                | 301.813371               |
| 3p                                                                   | −2.019226   | 1.235623              | 1.025157            | 1.325483                 | 3.847830                 | 33.273946                |
| 3d                                                                   | −0.509620   | 2.247781              | 1.323233            | 0.998468                 | 1.361094                 | 2.645535                 |
| Cr, $Z = 24$ , [Ar]4s <sup>1</sup> 3d <sup>5</sup> ( <sup>7</sup> S) |             |                       |                     |                          |                          |                          |
| Total energy = −1043.35637629 ; Kinetic energy = 1043.35637629       |             |                       |                     |                          |                          |                          |
| Potential energy = −2086.71275257 ; Virial ratio = −2.00000000000    |             |                       |                     |                          |                          |                          |
| $\rho_0 = 9313.7337$ ; $C_{EN} = 1.0000000$                          |             |                       |                     |                          |                          |                          |
|                                                                      | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                   | −220.386406 | 0.005520              | 0.064146            | 23.527255                | 1114.089518              |                          |
| 2s                                                                   | −26.209636  | 0.101988              | 0.294014            | 5.013676                 | 102.947511               |                          |
| 3s                                                                   | −3.285161   | 0.960479              | 0.912446            | 1.529583                 | 14.107002                |                          |
| 4s                                                                   | −0.222050   | 15.722039             | 3.675128            | 0.346490                 | 0.712112                 |                          |
| 2p                                                                   | −22.139856  | 0.083058              | 0.260079            | 4.931330                 | 33.384628                | 350.579763               |
| 3p                                                                   | −2.050932   | 1.128512              | 0.977756            | 1.394535                 | 4.266281                 | 38.787661                |
| 3d                                                                   | −0.373605   | 2.505944              | 1.367998            | 1.003724                 | 1.415314                 | 2.893481                 |
| Mn, $Z = 25$ , [Ar]4s <sup>2</sup> 3d <sup>5</sup> ( <sup>6</sup> S) |             |                       |                     |                          |                          |                          |
| Total energy = −1149.86625171 ; Kinetic energy = 1149.86625171       |             |                       |                     |                          |                          |                          |
| Potential energy = −2299.73250342 ; Virial ratio = −2.00000000000    |             |                       |                     |                          |                          |                          |
| $\rho_0 = 10560.077$ ; $C_{EN} = 1.0000000$                          |             |                       |                     |                          |                          |                          |
|                                                                      | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                   | −240.533991 | 0.005077              | 0.061527            | 24.523974                | 1210.234875              |                          |
| 2s                                                                   | −29.109475  | 0.092907              | 0.280660            | 5.256006                 | 113.080758               |                          |
| 3s                                                                   | −3.816646   | 0.852853              | 0.860446            | 1.624910                 | 15.951861                |                          |
| 4s                                                                   | −0.247871   | 13.309321             | 3.381968            | 0.376550                 | 0.845286                 |                          |
| 2p                                                                   | −24.812589  | 0.075182              | 0.247530            | 5.176571                 | 36.746010                | 404.061126               |
| 3p                                                                   | −2.479531   | 0.982778              | 0.913716            | 1.491987                 | 4.869770                 | 46.653756                |
| 3d                                                                   | −0.638847   | 1.640663              | 1.130058            | 1.167041                 | 1.850670                 | 4.152233                 |
| Fe, $Z = 26$ , [Ar]4s <sup>2</sup> 3d <sup>6</sup> ( <sup>5</sup> D) |             |                       |                     |                          |                          |                          |
| Total energy = −1262.44366540 ; Kinetic energy = 1262.44366540       |             |                       |                     |                          |                          |                          |
| Potential energy = −2524.88733080 ; Virial ratio = −2.00000000000    |             |                       |                     |                          |                          |                          |
| $\rho_0 = 11908.704$ ; $C_{EN} = 1.0000000$                          |             |                       |                     |                          |                          |                          |
|                                                                      | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                   | −261.373422 | 0.004685              | 0.059112            | 25.521202                | 1310.401031              |                          |
| 2s                                                                   | −31.935522  | 0.084984              | 0.268461            | 5.498549                 | 123.695826               |                          |
| 3s                                                                   | −4.169440   | 0.773333              | 0.819237            | 1.709911                 | 17.698008                |                          |
| 4s                                                                   | −0.258181   | 12.382061             | 3.258537            | 0.391395                 | 0.914057                 |                          |
| 2p                                                                   | −27.413712  | 0.068377              | 0.236132            | 5.422092                 | 40.273418                | 462.809803               |
| 3p                                                                   | −2.742199   | 0.882860              | 0.865859            | 1.576545                 | 5.433672                 | 54.598422                |
| 3d                                                                   | −0.646888   | 1.490006              | 1.072737            | 1.236916                 | 2.085961                 | 4.978858                 |
| Co, $Z = 27$ , [Ar]4s <sup>2</sup> 3d <sup>7</sup> ( <sup>4</sup> F) |             |                       |                     |                          |                          |                          |
| Total energy = −1381.41455298 ; Kinetic energy = 1381.41455298       |             |                       |                     |                          |                          |                          |
| Potential energy = −2762.82910595 ; Virial ratio = −2.00000000000    |             |                       |                     |                          |                          |                          |
| $\rho_0 = 13367.382$ ; $C_{EN} = 1.0000000$                          |             |                       |                     |                          |                          |                          |
|                                                                      | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                   | −283.065505 | 0.004337              | 0.056878            | 26.518657                | 1414.568762              |                          |
| 2s                                                                   | −34.868335  | 0.078042              | 0.257290            | 5.740891                 | 134.776375               |                          |
| 3s                                                                   | −4.524290   | 0.705537              | 0.782377            | 1.793468                 | 19.501569                |                          |
| 4s                                                                   | −0.267421   | 11.621454             | 3.153613            | 0.404875                 | 0.976918                 |                          |
| 2p                                                                   | −30.120174  | 0.062466              | 0.225755            | 5.667323                 | 43.958908                | 526.943686               |
| 3p                                                                   | −3.006245   | 0.799036              | 0.823547            | 1.659533                 | 6.017532                 | 63.259245                |
| 3d                                                                   | −0.675419   | 1.342955              | 1.016050            | 1.310208                 | 2.343519                 | 5.928060                 |

(continued on next page)



Table 1 (continued)

| Ni, Z = 28, [Ar]4s <sup>2</sup> 3d <sup>8</sup> ( <sup>3</sup> F)                  |             |                                |                     |                          |                          |                          |
|------------------------------------------------------------------------------------|-------------|--------------------------------|---------------------|--------------------------|--------------------------|--------------------------|
| Total energy = −1506.87090819 ;                                                    |             | Kinetic energy = 1506.87090819 |                     |                          |                          |                          |
| Potential energy = −3013.74181639 ;                                                |             | Virial ratio = −2.00000000000  |                     |                          |                          |                          |
| $\rho_0 = 14940.569$ ;                                                             |             | $C_{EN} = 1.0000000$           |                     |                          |                          |                          |
|                                                                                    | $E_{ORB}$   | $\langle r^2 \rangle$          | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                                 | −305.619035 | 0.004027                       | 0.054807            | 27.516296                | 1522.736427              |                          |
| 2s                                                                                 | −37.917832  | 0.071923                       | 0.247019            | 5.983091                 | 146.323733               |                          |
| 3s                                                                                 | −4.887837   | 0.646731                       | 0.748942            | 1.876340                 | 21.375735                |                          |
| 4s                                                                                 | −0.276251   | 10.957695                      | 3.059173            | 0.417809                 | 1.038502                 |                          |
| 2p                                                                                 | −32.941737  | 0.057298                       | 0.216263            | 5.912331                 | 47.802810                | 596.700692               |
| 3p                                                                                 | −3.277679   | 0.727203                       | 0.785486            | 1.741835                 | 6.626063                 | 72.721624                |
| 3d                                                                                 | −0.706932   | 1.215356                       | 0.964709            | 1.383487                 | 2.615216                 | 6.985146                 |
| Cu, Z = 29, [Ar]4s <sup>1</sup> 3d <sup>10</sup> ( <sup>2</sup> S)                 |             |                                |                     |                          |                          |                          |
| Total energy = −1638.96374218 ;                                                    |             | Kinetic energy = 1638.96374218 |                     |                          |                          |                          |
| Potential energy = −3277.92748436 ;                                                |             | Virial ratio = −2.00000000000  |                     |                          |                          |                          |
| $\rho_0 = 16625.258$ ;                                                             |             | $C_{EN} = 1.0000000$           |                     |                          |                          |                          |
|                                                                                    | $E_{ORB}$   | $\langle r^2 \rangle$          | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                                 | −328.792979 | 0.003748                       | 0.052880            | 28.514384                | 1634.930457              |                          |
| 2s                                                                                 | −40.818957  | 0.066499                       | 0.237540            | 6.225289                 | 158.343756               |                          |
| 3s                                                                                 | −5.011981   | 0.603463                       | 0.722765            | 1.947499                 | 23.063692                |                          |
| 4s                                                                                 | −0.238494   | 13.084422                      | 3.331145            | 0.382210                 | 0.807775                 |                          |
| 2p                                                                                 | −35.617945  | 0.052747                       | 0.207536            | 6.157564                 | 51.812434                | 672.455018               |
| 3p                                                                                 | −3.324822   | 0.678319                       | 0.757481            | 1.810481                 | 7.168326                 | 81.842500                |
| 3d                                                                                 | −0.491233   | 1.335116                       | 0.991154            | 1.391228                 | 2.702496                 | 7.504245                 |
| Zn, Z = 30, [Ar]4s <sup>2</sup> 3d <sup>10</sup> ( <sup>1</sup> S)                 |             |                                |                     |                          |                          |                          |
| Total energy = −1777.84811619 ;                                                    |             | Kinetic energy = 1777.84811619 |                     |                          |                          |                          |
| Potential energy = −3555.69623238 ;                                                |             | Virial ratio = −2.00000000000  |                     |                          |                          |                          |
| $\rho_0 = 18447.676$ ;                                                             |             | $C_{EN} = 1.0000000$           |                     |                          |                          |                          |
|                                                                                    | $E_{ORB}$   | $\langle r^2 \rangle$          | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                                 | −353.304540 | 0.003497                       | 0.051085            | 29.512054                | 1751.071823              |                          |
| 2s                                                                                 | −44.361720  | 0.061672                       | 0.228773            | 6.467081                 | 170.816767               |                          |
| 3s                                                                                 | −5.637816   | 0.550224                       | 0.690587            | 2.040101                 | 25.329381                |                          |
| 4s                                                                                 | −0.292507   | 9.869411                       | 2.897739            | 0.441877                 | 1.155410                 |                          |
| 2p                                                                                 | −38.924839  | 0.048729                       | 0.199514            | 6.401756                 | 55.965439                | 754.015496               |
| 3p                                                                                 | −3.839373   | 0.611174                       | 0.719787            | 1.904423                 | 7.914882                 | 94.151778                |
| 3d                                                                                 | −0.782537   | 1.005008                       | 0.874872            | 1.530539                 | 3.202584                 | 9.450045                 |
| Ga, Z = 31, [Ar]4s <sup>2</sup> 3d <sup>10</sup> 4p <sup>1</sup> ( <sup>2</sup> P) |             |                                |                     |                          |                          |                          |
| Total energy = −1923.26100961 ;                                                    |             | Kinetic energy = 1923.26100961 |                     |                          |                          |                          |
| Potential energy = −3846.52201923 ;                                                |             | Virial ratio = −2.00000000000  |                     |                          |                          |                          |
| $\rho_0 = 20397.202$ ;                                                             |             | $C_{EN} = 1.0000000$           |                     |                          |                          |                          |
|                                                                                    | $E_{ORB}$   | $\langle r^2 \rangle$          | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                                 | −378.818423 | 0.003271                       | 0.049408            | 30.509723                | 1871.198320              |                          |
| 2s                                                                                 | −48.168423  | 0.057352                       | 0.220632            | 6.708766                 | 183.755022               |                          |
| 3s                                                                                 | −6.394657   | 0.502652                       | 0.660472            | 2.135134                 | 27.763810                |                          |
| 4s                                                                                 | −0.424592   | 7.207005                       | 2.488893            | 0.515590                 | 1.671321                 |                          |
| 2p                                                                                 | −42.494029  | 0.045150                       | 0.192085            | 6.645972                 | 60.278535                | 841.913147               |
| 3p                                                                                 | −4.482368   | 0.550701                       | 0.684090            | 2.002594                 | 8.731276                 | 108.096079               |
| 4p                                                                                 | −0.208500   | 13.900812                      | 3.424250            | 0.372130                 | 0.342325                 | 2.890816                 |
| 3d                                                                                 | −1.193370   | 0.796159                       | 0.787887            | 1.664870                 | 3.730028                 | 11.655092                |
| Ge, Z = 32, [Ar]4s <sup>2</sup> 3d <sup>10</sup> 4p <sup>2</sup> ( <sup>3</sup> P) |             |                                |                     |                          |                          |                          |
| Total energy = −2075.35973391 ;                                                    |             | Kinetic energy = 2075.35973391 |                     |                          |                          |                          |
| Potential energy = −4150.71946783 ;                                                |             | Virial ratio = −2.00000000000  |                     |                          |                          |                          |
| $\rho_0 = 22480.118$ ;                                                             |             | $C_{EN} = 1.0000000$           |                     |                          |                          |                          |
|                                                                                    | $E_{ORB}$   | $\langle r^2 \rangle$          | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                                 | −405.244452 | 0.003066                       | 0.047837            | 31.507381                | 1995.308684              |                          |
| 2s                                                                                 | −52.150337  | 0.053468                       | 0.213048            | 6.950515                 | 197.166757               |                          |
| 3s                                                                                 | −7.190999   | 0.460062                       | 0.632246            | 2.232786                 | 30.381815                |                          |
| 4s                                                                                 | −0.553364   | 5.731085                       | 2.225815            | 0.578893                 | 2.191622                 |                          |
| 2p                                                                                 | −46.236162  | 0.041948                       | 0.185185            | 6.890253                 | 64.752278                | 936.392251               |
| 3p                                                                                 | −5.161600   | 0.497423                       | 0.650884            | 2.103809                 | 9.613239                 | 123.723785               |
| 4p                                                                                 | −0.287354   | 9.656819                       | 2.866920            | 0.445515                 | 0.511516                 | 4.801200                 |
| 3d                                                                                 | −1.634898   | 0.657085                       | 0.721234            | 1.793328                 | 4.277798                 | 14.106104                |

Table 1 (continued)

As,  $Z = 33$ , [Ar]4s<sup>2</sup>3d<sup>10</sup>4p<sup>3</sup> (<sup>4</sup>S)

Total energy = −2234.23865428 ; Kinetic energy = 2234.23865428

Potential energy = −4468.47730856 ; Virial ratio = −2.00000000000

 $\rho_0 = 24700.940$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −432.586204 | 0.002879              | 0.046363            | 32.505042                | 2123.404285              |                          |
| 2s | −56.309828  | 0.049964              | 0.205963            | 7.192370                 | 211.054600               |                          |
| 3s | −8.029623   | 0.422160              | 0.605963            | 2.332276                 | 33.170919                |                          |
| 4s | −0.685896   | 4.747516              | 2.029707            | 0.637555                 | 2.735245                 |                          |
| 2p | −50.153745  | 0.039071              | 0.178758            | 7.134625                 | 69.387266                | 1037.702602              |
| 3p | −5.880694   | 0.450758              | 0.620229            | 2.207112                 | 10.555903                | 141.061679               |
| 4p | −0.369482   | 7.371550              | 2.512258            | 0.509990                 | 0.686378                 | 6.958228                 |
| 3d | −2.112657   | 0.556645              | 0.667451            | 1.917758                 | 4.848001                 | 16.818712                |

Se,  $Z = 34$ , [Ar]4s<sup>2</sup>3d<sup>10</sup>4p<sup>4</sup> (<sup>3</sup>P)

Total energy = −2399.86761170 ; Kinetic energy = 2399.86761170

Potential energy = −4799.73522340 ; Virial ratio = −2.00000000000

 $\rho_0 = 27064.394$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −460.867406 | 0.002710              | 0.044978            | 33.502730                | 2255.487708              |                          |
| 2s | −60.668873  | 0.046790              | 0.199330            | 7.434387                 | 225.422143               |                          |
| 3s | −8.932103   | 0.388448              | 0.581534            | 2.433150                 | 36.124079                |                          |
| 4s | −0.837382   | 4.015468              | 1.869231            | 0.695413                 | 3.331128                 |                          |
| 2p | −54.268901  | 0.036478              | 0.172757            | 7.379111                 | 74.184073                | 1146.095583              |
| 3p | −6.661523   | 0.410405              | 0.592268            | 2.311199                 | 11.550346                | 160.063566               |
| 4p | −0.402855   | 6.196235              | 2.299643            | 0.562825                 | 0.857377                 | 9.280235                 |
| 3d | −2.649627   | 0.480003              | 0.622432            | 2.039610                 | 5.443415                 | 19.814174                |

Br,  $Z = 35$ , [Ar]4s<sup>2</sup>3d<sup>10</sup>4p<sup>5</sup> (<sup>2</sup>P)

Total energy = −2572.44133316 ; Kinetic energy = 2572.44133316

Potential energy = −5144.88266632 ; Virial ratio = −2.00000000000

 $\rho_0 = 29574.568$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −490.060340 | 0.002554              | 0.043672            | 34.500433                | 2391.557808              |                          |
| 2s | −65.199960  | 0.043907              | 0.193107            | 7.676539                 | 240.268730               |                          |
| 3s | −9.871894   | 0.358460              | 0.558860            | 2.534994                 | 39.233655                |                          |
| 4s | −0.992680   | 3.467922              | 1.738860            | 0.750675                 | 3.952854                 |                          |
| 2p | −58.554224  | 0.034132              | 0.167142            | 7.623705                 | 79.142829                | 1261.813789              |
| 3p | −7.478209   | 0.375144              | 0.566611            | 2.416064                 | 12.597023                | 180.813561               |
| 4p | −0.457086   | 5.223857              | 2.111625            | 0.616845                 | 1.046921                 | 11.999853                |
| 3d | −3.220175   | 0.419896              | 0.584099            | 2.159167                 | 6.063301                 | 23.100465                |

Kr,  $Z = 36$ , [Ar]4s<sup>2</sup>3d<sup>10</sup>4p<sup>6</sup> (<sup>1</sup>S)

Total energy = −2752.05497735 ; Kinetic energy = 2752.05497735

Potential energy = −5504.10995469 ; Virial ratio = −2.00000000000

 $\rho_0 = 32235.891$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −520.165468 | 0.002412              | 0.042441            | 35.498152                | 2531.615021              |                          |
| 2s | −69.903082  | 0.041280              | 0.187256            | 7.918830                 | 255.595276               |                          |
| 3s | −10.849466  | 0.331729              | 0.537802            | 2.637556                 | 42.494843                |                          |
| 4s | −1.152935   | 3.040362              | 1.629392            | 0.804188                 | 4.603706                 |                          |
| 2p | −63.009785  | 0.032004              | 0.161876            | 7.868429                 | 84.264118                | 1385.109895              |
| 3p | −8.331501   | 0.343537              | 0.542627            | 2.522775                 | 13.706201                | 203.560114               |
| 4p | −0.524187   | 4.455077              | 1.951611            | 0.669219                 | 1.238775                 | 14.886773                |
| 3d | −3.825234   | 0.371478              | 0.550880            | 2.276940                 | 6.708289                 | 26.691533                |

Rb,  $Z = 37$ , [Kr]5s<sup>1</sup> (<sup>2</sup>S)

Total energy = −2938.35745426 ; Kinetic energy = 2938.35745426

Potential energy = −5876.71490851 ; Virial ratio = −2.00000000000

 $\rho_0 = 35057.627$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −551.457338 | 0.002281              | 0.041276            | 36.495966                | 2675.668561              |                          |
| 2s | −75.049347  | 0.038879              | 0.181742            | 8.161489                 | 271.416587               |                          |
| 3s | −12.133200  | 0.307673              | 0.518105            | 2.741239                 | 45.922601                |                          |
| 4s | −1.523549   | 2.557222              | 1.499207            | 0.876307                 | 5.560348                 |                          |
| 5s | −0.137867   | 36.179136             | 5.631868            | 0.217757                 | 0.280585                 |                          |
| 2p | −67.906232  | 0.030066              | 0.156927            | 8.113293                 | 89.548437                | 1516.237721              |
| 3p | −9.487696   | 0.316060              | 0.520757            | 2.629144                 | 14.858976                | 228.072868               |
| 4p | −0.810071   | 3.477822              | 1.734948            | 0.752057                 | 1.577857                 | 20.097151                |
| 3d | −4.732292   | 0.331037              | 0.521335            | 2.394259                 | 7.383568                 | 30.625195                |

(continued on next page)

Table 1 (continued)

Sr,  $Z = 38$ , [Kr]5s<sup>2</sup> (<sup>1</sup>S)

Total energy = −3131.54568644 ; Kinetic energy = 3131.54568644

Potential energy = −6263.09137288 ; Virial ratio = −2.00000000000

 $\rho_0 = 38042.813$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −583.687885 | 0.002161              | 0.040174            | 37.493849                | 2823.716087              |                          |
| 2s | −80.390796  | 0.036678              | 0.176536            | 8.404430                 | 287.729151               |                          |
| 3s | −13.475029  | 0.286029              | 0.499698            | 2.845714                 | 49.509712                |                          |
| 4s | −1.896812   | 2.191494              | 1.390924            | 0.948080                 | 6.606745                 |                          |
| 5s | −0.178456   | 24.497140             | 4.632973            | 0.268681                 | 0.484015                 |                          |
| 2p | −72.996038  | 0.028297              | 0.152268            | 8.358311                 | 94.996284                | 1655.450904              |
| 3p | −10.699976  | 0.291693              | 0.500521            | 2.735976                 | 16.062822                | 254.549015               |
| 4p | −1.098163   | 2.857308              | 1.577620            | 0.829071                 | 1.931623                 | 25.895029                |
| 3d | −5.694396   | 0.297162              | 0.495014            | 2.510861                 | 8.087602                 | 34.908907                |

Y,  $Z = 39$ , [Kr]5s<sup>2</sup>4d<sup>1</sup> (<sup>2</sup>D)

Total energy = −3331.68416985 ; Kinetic energy = 3331.68416985

Potential energy = −6663.36833970 ; Virial ratio = −2.00000000000

 $\rho_0 = 41189.283$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −616.749351 | 0.002050              | 0.039129            | 38.491890                | 2975.769738              |                          |
| 2s | −85.810945  | 0.034657              | 0.171618            | 8.647529                 | 304.526135               |                          |
| 3s | −14.758916  | 0.266604              | 0.482564            | 2.950163                 | 53.227544                |                          |
| 4s | −2.168878   | 1.946800              | 1.311714            | 1.009637                 | 7.576251                 |                          |
| 5s | −0.196143   | 21.138711             | 4.300004            | 0.290896                 | 0.584761                 |                          |
| 2p | −78.164472  | 0.026679              | 0.147874            | 8.603493                 | 100.608639               | 1803.026564              |
| 3p | −11.854193  | 0.270039              | 0.481793            | 2.842924                 | 17.314290                | 283.001330               |
| 4p | −1.301188   | 2.480428              | 1.471169            | 0.892448                 | 2.250014                 | 31.490513                |
| 3d | −6.599480   | 0.268432              | 0.471362            | 2.626971                 | 8.821095                 | 39.560777                |
| 4d | −0.249849   | 7.265462              | 2.435212            | 0.545588                 | 0.566306                 | 1.711753                 |

Zr,  $Z = 40$ , [Kr]5s<sup>2</sup>4d<sup>2</sup> (<sup>3</sup>F)

Total energy = −3538.99506487 ; Kinetic energy = 3538.99506487

Potential energy = −7077.99012975 ; Virial ratio = −2.00000000000

 $\rho_0 = 44504.258$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −650.704986 | 0.001947              | 0.038137            | 39.490032                | 3131.823681              |                          |
| 2s | −91.377690  | 0.032798              | 0.166962            | 8.890789                 | 321.808586               |                          |
| 3s | −16.055029  | 0.249103              | 0.466569            | 3.054715                 | 57.082109                |                          |
| 4s | −2.419195   | 1.754743              | 1.245628            | 1.067325                 | 8.541528                 |                          |
| 5s | −0.207293   | 19.181758             | 4.091847            | 0.306524                 | 0.657402                 |                          |
| 2p | −83.478543  | 0.025195              | 0.143725            | 8.848787                 | 106.384469               | 1959.181351              |
| 3p | −13.019768  | 0.250756              | 0.464443            | 2.949831                 | 18.611844                | 313.470804               |
| 4p | −1.487595   | 2.198404              | 1.385570            | 0.950890                 | 2.563932                 | 37.312157                |
| 3d | −7.515841   | 0.243963              | 0.450089            | 2.742175                 | 9.581296                 | 44.579448                |
| 4d | −0.336759   | 5.359665              | 2.104493            | 0.627586                 | 0.744775                 | 2.439563                 |

Nb,  $Z = 41$ , [Kr]5s<sup>1</sup>4d<sup>4</sup> (<sup>6</sup>D)

Total energy = −3753.59772775 ; Kinetic energy = 3753.59772775

Potential energy = −7507.19545549 ; Virial ratio = −2.00000000000

 $\rho_0 = 47987.098$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −685.444022 | 0.001852              | 0.037194            | 40.488327                | 3291.885753              |                          |
| 2s | −96.974832  | 0.031083              | 0.162551            | 9.134144                 | 339.572705               |                          |
| 3s | −17.247068  | 0.233323              | 0.451645            | 3.158977                 | 61.057357                |                          |
| 4s | −2.537478   | 1.611988              | 1.193620            | 1.117520                 | 9.423660                 |                          |
| 5s | −0.215594   | 18.216695             | 3.985445            | 0.314465                 | 0.686857                 |                          |
| 2p | −88.823126  | 0.023830              | 0.139800            | 9.094202                 | 112.324436               | 2124.182296              |
| 3p | −14.081429  | 0.233523              | 0.448341            | 3.056550                 | 19.953611                | 345.989206               |
| 4p | −1.556994   | 2.011678              | 1.324290            | 0.998662                 | 2.836608                 | 42.722541                |
| 3d | −8.329295   | 0.222841              | 0.430775            | 2.856822                 | 10.369624                | 49.986228                |
| 4d | −0.300647   | 5.289768              | 2.071231            | 0.647833                 | 0.800655                 | 2.724121                 |

Table 1 (continued)

Mo,  $Z = 42$ , [Kr]5s<sup>1</sup>4d<sup>5</sup> (<sup>7</sup>S)

Total energy = −3975.54949953 ; Kinetic energy = 3975.54949953

Potential energy = −7951.09899905 ; Virial ratio = −2.00000000000

 $\rho_0 = 51652.684$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −721.202198 | 0.001763              | 0.036296            | 41.486632                | 3455.938055              |                          |
| 2s | −102.850600 | 0.029498              | 0.158364            | 9.377695                 | 357.826712               |                          |
| 3s | −18.584536  | 0.218970              | 0.437613            | 3.263694                 | 65.184562                |                          |
| 4s | −2.762914   | 1.476366              | 1.142378            | 1.171270                 | 10.415831                |                          |
| 5s | −0.222729   | 16.972045             | 3.841932            | 0.327041                 | 0.748279                 |                          |
| 2p | −94.444090  | 0.022573              | 0.136081            | 9.339690                 | 118.427019               | 2298.220566              |
| 3p | −15.286564  | 0.218033              | 0.433332            | 3.163304                 | 21.342225                | 380.673617               |
| 4p | −1.723633   | 1.821871              | 1.260415            | 1.052205                 | 3.155939                 | 49.189922                |
| 3d | −9.284225   | 0.204535              | 0.413218            | 2.970616                 | 11.183800                | 55.780312                |
| 4d | −0.357918   | 4.308618              | 1.875162            | 0.713442                 | 0.965549                 | 3.481844                 |

Tc,  $Z = 43$ , [Kr]5s<sup>2</sup>4d<sup>5</sup> (<sup>6</sup>S)

Total energy = −4204.78873702 ; Kinetic energy = 4204.78873702

Potential energy = −8409.57747404 ; Virial ratio = −2.00000000000

 $\rho_0 = 55507.101$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −758.043047 | 0.001681              | 0.035440            | 42.484927                | 3623.977929              |                          |
| 2s | −109.069787 | 0.028030              | 0.154384            | 9.621442                 | 376.571591               |                          |
| 3s | −20.131835  | 0.205872              | 0.424391            | 3.368888                 | 69.466428                |                          |
| 4s | −3.152190   | 1.347033              | 1.091530            | 1.229587                 | 11.548912                |                          |
| 5s | −0.231274   | 15.644438             | 3.684811            | 0.341739                 | 0.824181                 |                          |
| 2p | −100.406091 | 0.021411              | 0.132553            | 9.585252                 | 124.692325               | 2481.538467              |
| 3p | −16.699577  | 0.204040              | 0.419294            | 3.270182                 | 22.778756                | 417.615469               |
| 4p | −2.041219   | 1.628459              | 1.193032            | 1.113717                 | 3.540568                 | 57.127030                |
| 3d | −10.444638  | 0.188595              | 0.397210            | 3.083493                 | 12.022884                | 61.968104                |
| 4d | −0.543953   | 3.142845              | 1.620236            | 0.814480                 | 1.241293                 | 4.798233                 |

Ru,  $Z = 44$ , [Kr]5s<sup>1</sup>4d<sup>7</sup> (<sup>5</sup>F)

Total energy = −4441.53948783 ; Kinetic energy = 4441.53948783

Potential energy = −8883.07897565 ; Virial ratio = −2.00000000000

 $\rho_0 = 59535.009$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −795.513471 | 0.001604              | 0.034624            | 43.483426                | 3796.035698              |                          |
| 2s | −115.158792 | 0.026669              | 0.150598            | 9.865207                 | 395.794267               |                          |
| 3s | −21.413885  | 0.193987              | 0.412017            | 3.473377                 | 73.850330                |                          |
| 4s | −3.257128   | 1.258969              | 1.054738            | 1.275814                 | 12.484174                |                          |
| 5s | −0.222428   | 16.146641             | 3.735602            | 0.337112                 | 0.785454                 |                          |
| 2p | −106.239392 | 0.020337              | 0.129203            | 9.830944                 | 131.122791               | 2674.469975              |
| 3p | −17.848751  | 0.191402              | 0.406183            | 3.376726                 | 24.257260                | 456.761637               |
| 4p | −2.101248   | 1.517312              | 1.150566            | 1.158209                 | 3.835997                 | 63.759220                |
| 3d | −11.343342  | 0.174579              | 0.382504            | 3.195826                 | 12.888852                | 68.575579                |
| 4d | −0.412778   | 3.259056              | 1.630110            | 0.824848                 | 1.289630                 | 5.143262                 |

Rh,  $Z = 45$ , [Kr]5s<sup>1</sup>4d<sup>8</sup> (<sup>4</sup>F)

Total energy = −4685.88170428 ; Kinetic energy = 4685.88170428

Potential energy = −9371.76340855 ; Virial ratio = −2.00000000000

 $\rho_0 = 63760.709$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −834.039473 | 0.001533              | 0.033844            | 44.481916                | 3972.082489              |                          |
| 2s | −121.563647 | 0.025403              | 0.146991            | 10.109146                | 415.507685               |                          |
| 3s | −22.879612  | 0.183085              | 0.400322            | 3.578228                 | 78.384259                |                          |
| 4s | −3.503891   | 1.170000              | 1.016688            | 1.326835                 | 13.558692                |                          |
| 5s | −0.221618   | 15.905416             | 3.702456            | 0.340207                 | 0.790080                 |                          |
| 2p | −112.386179 | 0.019341              | 0.126016            | 10.076687                | 137.715792               | 2877.164709              |
| 3p | −19.179593  | 0.179916              | 0.393875            | 3.483279                 | 25.781986                | 498.262989               |
| 4p | −2.291149   | 1.396926              | 1.104033            | 1.209582                 | 4.188637                 | 71.744150                |
| 3d | −12.421289  | 0.162142              | 0.368927            | 3.307609                 | 13.781000                | 75.609024                |
| 4d | −0.450195   | 2.873235              | 1.531456            | 0.878700                 | 1.460636                 | 6.086505                 |

(continued on next page)

Table 1 (continued)

Pd,  $Z = 46$ , [Kr]4d<sup>10</sup> (<sup>1</sup>S)Total energy =  $-4937.92102407$ ; Kinetic energy =  $4937.92102407$ Potential energy =  $-9875.84204814$ ; Virial ratio =  $-2.000000000000$  $\rho_0 = 68175.459$ ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -873.315934 | 0.001466              | 0.033099            | 45.480522                | 4152.136641              |                          |
| 2s | -127.966574 | 0.024225              | 0.143552            | 10.353164                | 435.704733               |                          |
| 3s | -24.209104  | 0.173103              | 0.389298            | 3.682831                 | 83.039532                |                          |
| 4s | -3.587310   | 1.101830              | 0.985961            | 1.371554                 | 14.537329                |                          |
| 2p | -118.531104 | 0.018416              | 0.122981            | 10.322529                | 144.473178               | 3089.932664              |
| 3p | -20.374285  | 0.169459              | 0.382314            | 3.589664                 | 27.350476                | 542.135103               |
| 4p | -2.330090   | 1.311078              | 1.068569            | 1.253042                 | 4.501769                 | 79.269414                |
| 3d | -13.363440  | 0.150820              | 0.356139            | 3.420148                 | 14.708074                | 83.142266                |
| 4d | -0.336002   | 2.951711              | 1.533055            | 0.892658                 | 1.519608                 | 6.506516                 |

Ag,  $Z = 47$ , [Kr]5s<sup>1</sup>4d<sup>10</sup> (<sup>2</sup>S)Total energy =  $-5197.69847310$ ; Kinetic energy =  $5197.69847310$ Potential energy =  $-10395.3969462$ ; Virial ratio =  $-2.000000000000$  $\rho_0 = 72803.442$ ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -913.835600 | 0.001403              | 0.032386            | 46.479064                | 4336.171965              |                          |
| 2s | -134.878416 | 0.023127              | 0.140268            | 10.597352                | 456.394100               |                          |
| 3s | -25.917827  | 0.163895              | 0.378839            | 3.787881                 | 87.850743                |                          |
| 4s | -4.001498   | 1.020639              | 0.949403            | 1.427065                 | 15.794162                |                          |
| 5s | -0.219980   | 15.586670             | 3.656488            | 0.344253                 | 0.782578                 |                          |
| 2p | -125.181588 | 0.017555              | 0.120088            | 10.568378                | 151.391992               | 3312.911416              |
| 3p | -21.945436  | 0.159896              | 0.371418            | 3.696050                 | 28.964882                | 588.487009               |
| 4p | -2.676820   | 1.199857              | 1.023205            | 1.310062                 | 4.921951                 | 89.225437                |
| 3d | -14.678202  | 0.140869              | 0.344428            | 3.531194                 | 15.654407                | 91.080262                |
| 4d | -0.537400   | 2.291446              | 1.369803            | 0.981689                 | 1.809912                 | 8.137199                 |

Cd,  $Z = 48$ , [Kr]5s<sup>2</sup>4d<sup>10</sup> (<sup>1</sup>S)Total energy =  $-5465.13314253$ ; Kinetic energy =  $5465.13314253$ Potential energy =  $-10930.2662851$ ; Virial ratio =  $-2.000000000000$  $\rho_0 = 77641.559$ ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -955.315356 | 0.001345              | 0.031703            | 47.477627                | 4524.201005              |                          |
| 2s | -142.006827 | 0.022100              | 0.137128            | 10.841708                | 477.576027               |                          |
| 3s | -27.708623  | 0.155390              | 0.368910            | 3.893205                 | 92.809403                |                          |
| 4s | -4.450535   | 0.945138              | 0.914133            | 1.484942                 | 17.160502                |                          |
| 5s | -0.264856   | 12.170827             | 3.237485            | 0.390397                 | 1.070563                 |                          |
| 2p | -132.047020 | 0.016753              | 0.117327            | 10.814311                | 158.474552               | 3546.417885              |
| 3p | -23.597234  | 0.151114              | 0.361119            | 3.802547                 | 30.626730                | 637.425516               |
| 4p | -3.053504   | 1.099662              | 0.980386            | 1.368985                 | 5.375454                 | 100.263569               |
| 3d | -16.071971  | 0.131895              | 0.333489            | 3.642018                 | 16.628524                | 99.495039                |
| 4d | -0.763658   | 1.884738              | 1.252087            | 1.064243                 | 2.105667                 | 9.904103                 |

In,  $Z = 49$ , [Kr]5s<sup>2</sup>4d<sup>10</sup>5p<sup>1</sup> (<sup>2</sup>P)Total energy =  $-5740.16915577$ ; Kinetic energy =  $5740.16915577$ Potential energy =  $-11480.3383115$ ; Virial ratio =  $-2.000000000000$  $\rho_0 = 82686.165$ ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$   | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|-------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -997.800457 | 0.001290              | 0.031048            | 48.476204                | 4716.222992              |                          |
| 2s | -149.395446 | 0.021140              | 0.134125            | 11.086132                | 499.242927               |                          |
| 3s | -29.624667  | 0.147533              | 0.359491            | 3.998535                 | 97.903047                |                          |
| 4s | -4.976702   | 0.877587              | 0.881300            | 1.543155                 | 18.591106                |                          |
| 5s | -0.372663   | 9.304919              | 2.844491            | 0.444788                 | 1.468232                 |                          |
| 2p | -139.171929 | 0.016005              | 0.114689            | 11.060332                | 165.721239               | 3790.710923              |
| 3p | -25.374284  | 0.143028              | 0.351365            | 3.909187                 | 32.336703                | 689.039700               |
| 4p | -3.507216   | 1.008733              | 0.939853            | 1.429674                 | 5.862279                 | 112.434013               |
| 5p | -0.197283   | 16.647185             | 3.777948            | 0.331844                 | 0.313774                 | 4.457265                 |
| 3d | -17.589565  | 0.123768              | 0.323243            | 3.752678                 | 17.630732                | 108.402496               |
| 4d | -1.063138   | 1.590785              | 1.156917            | 1.144702                 | 2.416163                 | 11.856687                |

Table 1 (continued)

Sn,  $Z = 50$ , [Kr]5s<sup>2</sup>4d<sup>10</sup>5p<sup>2</sup> (<sup>3</sup>P)Total energy = −6022.93169531 ; Kinetic energy = 6022.93169531  
Potential energy = −12045.8633906 ; Virial ratio = −2.00000000000 $\rho_0 = 87945.280$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −1041.223349 | 0.001238              | 0.030419            | 49.474791                | 4912.237385              |                          |
| 2s | −156.977583  | 0.020241              | 0.131250            | 11.330667                | 521.398939               |                          |
| 3s | −31.598981   | 0.140251              | 0.350534            | 4.104011                 | 103.138769               |                          |
| 4s | −5.512501    | 0.816477              | 0.850435            | 1.602229                 | 20.101973                |                          |
| 5s | −0.476434    | 7.646423              | 2.585908            | 0.490494                 | 1.851115                 |                          |
| 2p | −146.489269  | 0.015304              | 0.112165            | 11.306439                | 173.132108               | 4046.034840              |
| 3p | −27.209044   | 0.135565              | 0.342112            | 4.015986                 | 34.095156                | 743.409433               |
| 4p | −3.969055    | 0.927899              | 0.902129            | 1.491372                 | 6.378510                 | 125.706458               |
| 5p | −0.265040    | 12.194092             | 3.248258            | 0.386569                 | 0.445565                 | 6.832467                 |
| 3d | −19.163358   | 0.116380              | 0.313622            | 3.863205                 | 18.661181                | 117.817597               |
| 4d | −1.369045    | 1.377102              | 1.080403            | 1.221482                 | 2.733852                 | 13.957830                |

Sb,  $Z = 51$ , [Kr]5s<sup>2</sup>4d<sup>10</sup>5p<sup>3</sup> (<sup>4</sup>S)Total energy = −6313.48532075 ; Kinetic energy = 6313.48532075  
Potential energy = −12626.9706415 ; Virial ratio = −2.00000000000 $\rho_0 = 93423.569$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −1085.589045 | 0.001189              | 0.029816            | 50.473390                | 5112.244666              |                          |
| 2s | −164.757961  | 0.019398              | 0.128494            | 11.575314                | 544.044684               |                          |
| 3s | −33.636220   | 0.133489              | 0.342004            | 4.209628                 | 108.516726               |                          |
| 4s | −6.063185    | 0.761324              | 0.821515            | 1.661857                 | 21.687264                |                          |
| 5s | −0.581773    | 6.507034              | 2.390168            | 0.532141                 | 2.237687                 |                          |
| 2p | −154.003792  | 0.014649              | 0.109749            | 11.552631                | 180.707250               | 4312.637625              |
| 3p | −29.106148   | 0.128664              | 0.333323            | 4.122931                 | 35.902038                | 800.605658               |
| 4p | −4.444718    | 0.855986              | 0.867077            | 1.553756                 | 6.922145                 | 140.080978               |
| 5p | −0.334711    | 9.668052              | 2.901138            | 0.433783                 | 0.576437                 | 9.345702                 |
| 3d | −20.798076   | 0.109644              | 0.304568            | 3.973620                 | 19.719947                | 127.754822               |
| 4d | −1.687866    | 1.212440              | 1.016401            | 1.295655                 | 3.059999                 | 16.216112                |

Te,  $Z = 52$ , [Kr]5s<sup>2</sup>4d<sup>10</sup>5p<sup>4</sup> (<sup>3</sup>P)Total energy = −6611.78405928 ; Kinetic energy = 6611.78405928  
Potential energy = −13223.5681186 ; Virial ratio = −2.00000000000 $\rho_0 = 99125.885$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −1130.917005 | 0.001143              | 0.029236            | 51.472008                | 5316.246013              |                          |
| 2s | −172.755410  | 0.018606              | 0.125850            | 11.820079                | 567.181354               |                          |
| 3s | −35.754892   | 0.127199              | 0.333871            | 4.315393                 | 114.037612               |                          |
| 4s | −6.647019    | 0.711488              | 0.794422            | 1.721886                 | 23.344058                |                          |
| 5s | −0.700561    | 5.638035              | 2.227967            | 0.572718                 | 2.651393                 |                          |
| 2p | −161.734379  | 0.014034              | 0.107434            | 11.798907                | 188.446842               | 4590.770435              |
| 3p | −31.084050   | 0.122270              | 0.324966            | 4.229998                 | 37.757132                | 860.694862               |
| 4p | −4.952569    | 0.793022              | 0.834979            | 1.615914                 | 7.486016                 | 155.435563               |
| 5p | −0.359832    | 8.338904              | 2.691369            | 0.471708                 | 0.701612                 | 11.937181                |
| 3d | −22.512352   | 0.103481              | 0.296029            | 4.083956                 | 20.807243                | 138.229973               |
| 4d | −2.038289    | 1.079919              | 0.961178            | 1.368188                 | 3.396724                 | 18.647548                |

I,  $Z = 53$ , [Kr]5s<sup>2</sup>4d<sup>10</sup>5p<sup>5</sup> (<sup>2</sup>P)Total energy = −6917.98089626 ; Kinetic energy = 6917.98089626  
Potential energy = −13835.9617925 ; Virial ratio = −2.00000000000 $\rho_0 = 105056.34$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −1177.186297 | 0.001100              | 0.028678            | 52.470640                | 5524.240699              |                          |
| 2s | −180.949223  | 0.017861              | 0.123312            | 12.064949                | 590.808426               |                          |
| 3s | −37.934467   | 0.121338              | 0.326110            | 4.421281                 | 119.700730               |                          |
| 4s | −7.244360    | 0.666420              | 0.769050            | 1.782144                 | 25.068378                |                          |
| 5s | −0.821117    | 4.971798              | 2.094428            | 0.611070                 | 3.073359                 |                          |
| 2p | −169.660349  | 0.013457              | 0.105213            | 12.045262                | 196.350863               | 4880.677884              |
| 3p | −33.122310   | 0.116335              | 0.317010            | 4.337185                 | 39.660528                | 923.751423               |
| 4p | −5.473358    | 0.737040              | 0.805271            | 1.678107                 | 8.071918                 | 171.841431               |
| 5p | −0.403181    | 7.200934              | 2.501749            | 0.510299                 | 0.837939                 | 14.872184                |
| 3d | −24.285692   | 0.097830              | 0.287964            | 4.194206                 | 21.922960                | 149.256054               |
| 4d | −2.401205    | 0.971475              | 0.913033            | 1.439107                 | 3.742973                 | 21.249533                |

(continued on next page)

Table 1 (continued)

Xe,  $Z = 54$ , [Kr]5s<sup>2</sup>4d<sup>10</sup>5p<sup>6</sup> (<sup>1</sup>S)Total energy =  $-7232.13836387$ ; Kinetic energy =  $7232.13836387$ Potential energy =  $-14464.2767277$ ; Virial ratio =  $-2.000000000000$  $\rho_0 = 111219.44$ ;  $C_{EN} = 1.00000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -1224.397777 | 0.001059              | 0.028141            | 53.469285                | 5736.228872              |                          |
| 2s | -189.340123  | 0.017160              | 0.120873            | 12.309921                | 614.926128               |                          |
| 3s | -40.175663   | 0.115870              | 0.318696            | 4.527285                 | 125.506009               |                          |
| 4s | -7.856302    | 0.625571              | 0.745267            | 1.842538                 | 26.857907                |                          |
| 5s | -0.944414    | 4.440144              | 1.980959            | 0.647891                 | 3.506783                 |                          |
| 2p | -177.782449  | 0.012915              | 0.103082            | 12.291694                | 204.419399               | 5182.608324              |
| 3p | -35.221662   | 0.110816              | 0.309425            | 4.444510                 | 41.612663                | 989.859293               |
| 4p | -6.008338    | 0.685527              | 0.777023            | 1.741492                 | 8.690113                 | 189.577625               |
| 5p | -0.457290    | 6.276874              | 2.337984            | 0.547154                 | 0.970732                 | 17.826491                |
| 3d | -26.118869   | 0.092633              | 0.280333            | 4.304379                 | 23.067107                | 160.847127               |
| 4d | -2.777881    | 0.880865              | 0.870451            | 1.508735                 | 4.099133                 | 24.028467                |

Cs,  $Z = 55$ , [Xe]6s<sup>1</sup> (<sup>2</sup>S)Total energy =  $-7553.93365766$ ; Kinetic energy =  $7553.93365766$ Potential energy =  $-15107.8673153$ ; Virial ratio =  $-2.000000000000$  $\rho_0 = 117625.45$ ;  $C_{EN} = 1.00000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -1272.768831 | 0.001021              | 0.027624            | 54.467972                | 5952.215616              |                          |
| 2s | -198.143782  | 0.016498              | 0.118527            | 12.555062                | 639.541389               |                          |
| 3s | -42.693034   | 0.110753              | 0.311598            | 4.633528                 | 131.460474               |                          |
| 4s | -8.695490    | 0.588175              | 0.722793            | 1.903413                 | 28.723519                |                          |
| 5s | -1.231607    | 3.805494              | 1.839382            | 0.699412                 | 4.170704                 |                          |
| 6s | -0.123668    | 44.989505             | 6.305902            | 0.192168                 | 0.239804                 |                          |
| 2p | -186.316176  | 0.012405              | 0.101034            | 12.538212                | 212.652856               | 5496.821483              |
| 3p | -37.595936   | 0.105677              | 0.302190            | 4.551909                 | 43.612502                | 1059.061417              |
| 4p | -6.768534    | 0.640198              | 0.751145            | 1.804090                 | 9.323029                 | 208.261256               |
| 5p | -0.683475    | 5.066603              | 2.111203            | 0.606279                 | 1.211413                 | 23.254489                |
| 3d | -28.226223   | 0.087841              | 0.273097            | 4.414549                 | 24.240362                | 173.023770               |
| 4d | -3.379542    | 0.801938              | 0.831564            | 1.578371                 | 4.470532                 | 27.025832                |

Ba,  $Z = 56$ , [Xe]6s<sup>2</sup> (<sup>1</sup>S)Total energy =  $-7883.54382733$ ; Kinetic energy =  $7883.54382733$ Potential energy =  $-15767.0876547$ ; Virial ratio =  $-2.000000000000$  $\rho_0 = 124276.86$ ;  $C_{EN} = 1.00000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -1322.093398 | 0.000984              | 0.027125            | 55.466689                | 6172.199180              |                          |
| 2s | -207.154465  | 0.015874              | 0.116269            | 12.800339                | 664.652060               |                          |
| 3s | -45.280852   | 0.105961              | 0.304800            | 4.739952                 | 137.561823               |                          |
| 4s | -9.556400    | 0.553972              | 0.701586            | 1.964571                 | 30.660024                |                          |
| 5s | -1.512722    | 3.319051              | 1.721190            | 0.750076                 | 4.885203                 |                          |
| 6s | -0.157528    | 31.270766             | 5.256816            | 0.233636                 | 0.403815                 |                          |
| 2p | -195.055960  | 0.011924              | 0.099065            | 12.784811                | 221.051289               | 5823.565584              |
| 3p | -40.039790   | 0.100883              | 0.295279            | 4.659416                 | 45.660761                | 1131.449726              |
| 4p | -7.549318    | 0.599292              | 0.726963            | 1.866728                 | 9.977890                 | 228.095086               |
| 5p | -0.903863    | 4.276114              | 1.944587            | 0.660373                 | 1.456667                 | 29.066042                |
| 3d | -30.402309   | 0.083411              | 0.266226            | 4.524707                 | 25.442640                | 185.799532               |
| 4d | -4.001496    | 0.733894              | 0.796334            | 1.647495                 | 4.854710                 | 30.232116                |

La,  $Z = 57$ , [Xe]6s<sup>2</sup>5d<sup>1</sup> (<sup>2</sup>D)Total energy =  $-8221.06670260$ ; Kinetic energy =  $8221.06670260$ Potential energy =  $-16442.1334052$ ; Virial ratio =  $-2.000000000000$  $\rho_0 = 131169.69$ ;  $C_{EN} = 1.00000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -1372.280305 | 0.000949              | 0.026644            | 56.465470                | 6396.186181              |                          |
| 2s | -216.276999  | 0.015284              | 0.114095            | 13.045702                | 690.253474               |                          |
| 3s | -47.844413   | 0.101475              | 0.298294            | 4.846337                 | 143.796938               |                          |
| 4s | -10.345537   | 0.522828              | 0.681691            | 2.025420                 | 32.646435                |                          |
| 5s | -1.704429    | 2.999037              | 1.636889            | 0.791671                 | 5.512027                 |                          |
| 6s | -0.170405    | 27.561265             | 4.931219            | 0.250135                 | 0.477202                 |                          |
| 2p | -203.907402  | 0.011470              | 0.097171            | 13.031488                | 229.614902               | 6163.101893              |
| 3p | -42.459150   | 0.096409              | 0.288678            | 4.766904                 | 47.755245                | 1207.034007              |
| 4p | -8.258908    | 0.562263              | 0.704332            | 1.929296                 | 10.653356                | 249.076391               |
| 5p | -1.049440    | 3.794689              | 1.833134            | 0.703115                 | 1.665362                 | 34.296549                |
| 3d | -32.553617   | 0.079310              | 0.259694            | 4.634844                 | 26.673859                | 199.188350               |
| 4d | -4.553592    | 0.674649              | 0.764168            | 1.716369                 | 5.252865                 | 33.663042                |
| 5d | -0.268850    | 8.942112              | 2.757891            | 0.467012                 | 0.486445                 | 2.322541                 |

Table 1 (continued)

Ce,  $Z = 58$ , [Xe]6s<sup>2</sup>4f<sup>1</sup>5d<sup>1</sup> (<sup>1</sup>G)

Total energy = −8566.87268128 ; Kinetic energy = 8566.87268128

Potential energy = −17133.7453626 ; Virial ratio = −2.00000000000

 $\rho_0 = 138302.03$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −1423.102730 | 0.000917              | 0.026180            | 57.464436                | 6624.199646              |                          |
| 2s | −225.252823  | 0.014725              | 0.111992            | 13.292150                | 716.444818               |                          |
| 3s | −50.077318   | 0.097261              | 0.292045            | 4.953101                 | 150.190906               |                          |
| 4s | −10.807065   | 0.498486              | 0.665593            | 2.076732                 | 34.358720                |                          |
| 5s | −1.749455    | 2.877540              | 1.603069            | 0.808507                 | 5.751694                 |                          |
| 6s | −0.171432    | 27.106087             | 4.888520            | 0.252144                 | 0.480646                 |                          |
| 2p | −212.618113  | 0.011039              | 0.095340            | 13.279088                | 238.370319               | 6516.618645              |
| 3p | −44.550339   | 0.092223              | 0.282353            | 4.874784                 | 49.904223                | 1286.161722              |
| 4p | −8.650582    | 0.534564              | 0.686711            | 1.980632                 | 11.224806                | 267.902981               |
| 5p | −1.073869    | 3.642537              | 1.795488            | 0.717889                 | 1.734626                 | 36.416067                |
| 3d | −34.383960   | 0.075515              | 0.253475            | 4.745424                 | 27.940848                | 213.291098               |
| 4d | −4.814578    | 0.636240              | 0.742029            | 1.768404                 | 5.569332                 | 36.578539                |
| 5d | −0.297944    | 8.028571              | 2.615862            | 0.493317                 | 0.546756                 | 2.715387                 |
| 4f | −0.713190    | 1.272280              | 1.001078            | 1.283048                 | 2.140086                 | 4.715300                 |

Pr,  $Z = 59$ , [Xe]6s<sup>2</sup>4f<sup>3</sup> (<sup>4</sup>I)

Total energy = −8921.18102813 ; Kinetic energy = 8921.18102813

Potential energy = −17842.3620563 ; Virial ratio = −2.00000000000

 $\rho_0 = 145677.79$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −1474.559431 | 0.000885              | 0.025731            | 58.463577                | 6856.239193              |                          |
| 2s | −234.080284  | 0.014193              | 0.109957            | 13.539644                | 743.228513               |                          |
| 3s | −51.978923   | 0.093300              | 0.286043            | 5.060157                 | 156.739995               |                          |
| 4s | −10.945202   | 0.479619              | 0.652711            | 2.118915                 | 35.787189                |                          |
| 5s | −1.656296    | 2.911113              | 1.611100            | 0.802084                 | 5.602335                 |                          |
| 6s | −0.163924    | 28.943916             | 5.052405            | 0.242926                 | 0.430066                 |                          |
| 2p | −221.186614  | 0.010631              | 0.093568            | 13.527576                | 247.318064               | 6884.433786              |
| 3p | −46.312714   | 0.088302              | 0.276290            | 4.982967                 | 52.106425                | 1368.879712              |
| 4p | −8.727861    | 0.514226              | 0.673270            | 2.021282                 | 11.689350                | 284.325496               |
| 5p | −0.983827    | 3.750214              | 1.819607            | 0.706194                 | 1.663365                 | 35.206971                |
| 3d | −35.892497   | 0.071998              | 0.247552            | 4.856330                 | 29.242675                | 228.118395               |
| 4d | −4.785764    | 0.613369              | 0.727906            | 1.804747                 | 5.803079                 | 38.924019                |
| 4f | −0.549223    | 1.280150              | 0.998461            | 1.299223                 | 2.210511                 | 4.990282                 |

Nd,  $Z = 60$ , [Xe]6s<sup>2</sup>4f<sup>4</sup> (<sup>5</sup>I)

Total energy = −9283.88294453 ; Kinetic energy = 9283.88294453

Potential energy = −18567.7658891 ; Virial ratio = −2.00000000000

 $\rho_0 = 153321.84$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −1527.212971 | 0.000856              | 0.025298            | 59.462618                | 7092.254187              |                          |
| 2s | −243.384418  | 0.013691              | 0.108000            | 13.786417                | 770.429412               |                          |
| 3s | −54.263196   | 0.089589              | 0.280303            | 5.166740                 | 163.398748               |                          |
| 4s | −11.395776   | 0.458764              | 0.638324            | 2.168685                 | 37.520824                |                          |
| 5s | −1.698177    | 2.803114              | 1.580646            | 0.817559                 | 5.820642                 |                          |
| 6s | −0.165783    | 28.315884             | 4.995761            | 0.245658                 | 0.437886                 |                          |
| 2p | −230.225910  | 0.010246              | 0.091867            | 13.775423                | 256.409713               | 7265.063725              |
| 3p | −48.454628   | 0.084638              | 0.270504            | 5.090611                 | 54.344862                | 1454.682427              |
| 4p | −9.108511    | 0.490676              | 0.657620            | 2.070953                 | 12.267566                | 304.295282               |
| 5p | −1.005847    | 3.614310              | 1.785827            | 0.719519                 | 1.724875                 | 37.175012                |
| 3d | −37.772687   | 0.068728              | 0.241923            | 4.966516                 | 30.564632                | 243.501985               |
| 4d | −5.035966    | 0.581284              | 0.708542            | 1.854797                 | 6.123164                 | 42.028556                |
| 4f | −0.595772    | 1.167943              | 0.954587            | 1.355493                 | 2.398983                 | 5.615494                 |

Pm,  $Z = 61$ , [Xe]6s<sup>2</sup>4f<sup>5</sup> (<sup>6</sup>H)

Total energy = −9655.09896927 ; Kinetic energy = 9655.09896927

Potential energy = −19310.1979385 ; Virial ratio = −2.00000000000

 $\rho_0 = 161228.75$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −1580.789685 | 0.000828              | 0.024879            | 60.461692                | 7332.269270              |                          |
| 2s | −252.861313  | 0.013215              | 0.106111            | 14.033317                | 798.131935               |                          |
| 3s | −56.583164   | 0.086097              | 0.274794            | 5.273247                 | 170.189966               |                          |
| 4s | −11.848184   | 0.439398              | 0.624668            | 2.217996                 | 39.277473                |                          |
| 5s | −1.739373    | 2.703024              | 1.551894            | 0.832698                 | 6.037383                 |                          |
| 6s | −0.167593    | 27.724081             | 4.941794            | 0.248328                 | 0.445619                 |                          |
| 2p | −239.437914  | 0.009881              | 0.090226            | 14.023363                | 265.668628               | 7659.589852              |
| 3p | −50.631812   | 0.081200              | 0.264961            | 5.198155                 | 56.628079                | 1543.919482              |
| 4p | −9.490685    | 0.468901              | 0.642813            | 2.120137                 | 12.853633                | 324.997453               |
| 5p | −1.027256    | 3.488773              | 1.754048            | 0.732503                 | 1.785698                 | 39.170041                |
| 3d | −39.687450   | 0.065683              | 0.236554            | 5.076538                 | 31.914106                | 259.544912               |
| 4d | −5.287197    | 0.551991              | 0.690387            | 1.904269                 | 6.448098                 | 45.261200                |
| 4f | −0.628301    | 1.081203              | 0.918496            | 1.407416                 | 2.581938                 | 6.250768                 |

(continued on next page)



Table 1 (continued)

Sm,  $Z = 62$ , [Xe]6s<sup>2</sup>4f<sup>6</sup> (<sup>7</sup>F)Total energy =  $-10034.9525472$ ; Kinetic energy =  $10034.9525472$ Potential energy =  $-20069.9050944$ ; Virial ratio =  $-2.00000000000$  $\rho_0 = 169402.78$ ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -1635.284543 | 0.000801              | 0.024473            | 61.460798                | 7576.284885              |                          |
| 2s | -262.505469  | 0.012763              | 0.104286            | 14.280348                | 826.337166               |                          |
| 3s | -58.933038   | 0.082809              | 0.269501            | 5.379669                 | 177.112898               |                          |
| 4s | -12.298023   | 0.421422              | 0.611719            | 2.266720                 | 41.051159                |                          |
| 5s | -1.778680    | 2.611552              | 1.525137            | 0.847232                 | 6.247448                 |                          |
| 6s | -0.169320    | 27.177155             | 4.891383            | 0.250885                 | 0.453096                 |                          |
| 2p | -248.817219  | 0.009535              | 0.088641            | 14.271402                | 275.095110               | 8068.275042              |
| 3p | -52.838500   | 0.077972              | 0.259646            | 5.305596                 | 58.955870                | 1636.650873              |
| 4p | -9.870087    | 0.448784              | 0.628820            | 2.168690                 | 13.445404                | 346.379989               |
| 5p | -1.047151    | 3.375044              | 1.724732            | 0.744851                 | 1.844059                 | 41.145813                |
| 3d | -41.631055   | 0.062841              | 0.231428            | 5.186403                 | 33.291064                | 276.260557               |
| 4d | -5.535428    | 0.525290              | 0.673406            | 1.952968                 | 6.776373                 | 48.610909                |
| 4f | -0.666415    | 1.003132              | 0.884983            | 1.458891                 | 2.769385                 | 6.922857                 |

Eu,  $Z = 63$ , [Xe]6s<sup>2</sup>4f<sup>7</sup> (<sup>8</sup>S)Total energy =  $-10423.5430217$ ; Kinetic energy =  $10423.5430217$ Potential energy =  $-20847.0860433$ ; Virial ratio =  $-2.00000000000$  $\rho_0 = 177848.32$ ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -1690.696544 | 0.000775              | 0.024081            | 62.459936                | 7824.301092              |                          |
| 2s | -272.315849  | 0.012334              | 0.102522            | 14.527504                | 855.045060               |                          |
| 3s | -61.311967   | 0.079709              | 0.264413            | 5.486005                 | 184.167054               |                          |
| 4s | -12.744771   | 0.404702              | 0.599422            | 2.314872                 | 42.840999                |                          |
| 5s | -1.816028    | 2.527856              | 1.500228            | 0.861159                 | 6.450191                 |                          |
| 6s | -0.170964    | 26.671183             | 4.844272            | 0.253334                 | 0.460331                 |                          |
| 2p | -258.362798  | 0.009207              | 0.087111            | 14.519533                | 284.689147               | 8491.370699              |
| 3p | -55.073825   | 0.074936              | 0.254545            | 5.412931                 | 61.328101                | 1732.938019              |
| 4p | -10.246171   | 0.430153              | 0.615577            | 2.216627                 | 14.042586                | 368.441044               |
| 5p | -1.065486    | 3.271984              | 1.697707            | 0.756553                 | 1.899734                 | 43.093069                |
| 3d | -43.602595   | 0.060186              | 0.226529            | 5.296116                 | 34.695464                | 293.661914               |
| 4d | -5.780046    | 0.500872              | 0.657487            | 2.000917                 | 7.107797                 | 52.077082                |
| 4f | -0.711724    | 0.932061              | 0.853556            | 1.510205                 | 2.962059                 | 7.634679                 |

Gd,  $Z = 64$ , [Xe]6s<sup>2</sup>4f<sup>7</sup>5d<sup>1</sup> (<sup>9</sup>D)Total energy =  $-10820.6612101$ ; Kinetic energy =  $10820.6612101$ Potential energy =  $-21641.3224201$ ; Virial ratio =  $-2.00000000000$  $\rho_0 = 186582.13$ ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -1747.334818 | 0.000751              | 0.023701            | 63.458981                | 8076.292812              |                          |
| 2s | -282.631397  | 0.011927              | 0.100820            | 14.774040                | 884.174985               |                          |
| 3s | -64.098941   | 0.076788              | 0.259528            | 5.591945                 | 191.331974               |                          |
| 4s | -13.530882   | 0.386900              | 0.586145            | 2.369629                 | 44.930192                |                          |
| 5s | -2.007425    | 2.335160              | 1.442726            | 0.897608                 | 7.071656                 |                          |
| 6s | -0.183383    | 23.870062             | 4.579670            | 0.269103                 | 0.535951                 |                          |
| 2p | -268.407828  | 0.008897              | 0.085636            | 14.767117                | 294.428421               | 8928.245714              |
| 3p | -57.714117   | 0.072082              | 0.249657            | 5.519819                 | 63.737037                | 1832.558244              |
| 4p | -10.953493   | 0.409764              | 0.600905            | 2.272347                 | 14.750424                | 394.327546               |
| 5p | -1.212481    | 2.974050              | 1.619971            | 0.794636                 | 2.109351                 | 49.115231                |
| 3d | -45.971402   | 0.057701              | 0.221850            | 5.405233                 | 36.120390                | 311.667763               |
| 4d | -6.334078    | 0.471554              | 0.638298            | 2.060403                 | 7.522224                 | 56.365512                |
| 5d | -0.287278    | 7.224250              | 2.462597            | 0.527382                 | 0.619961                 | 3.386750                 |
| 4f | -1.043538    | 0.781514              | 0.788974            | 1.607418                 | 3.313008                 | 8.889167                 |

Tb,  $Z = 65$ , [Xe]6s<sup>2</sup>4f<sup>9</sup> (<sup>6</sup>H)Total energy =  $-11226.5683738$ ; Kinetic energy =  $11226.5683738$ Potential energy =  $-22453.1367475$ ; Virial ratio =  $-2.00000000000$  $\rho_0 = 195574.59$ ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -1804.342738 | 0.000728              | 0.023333            | 64.458272                | 8332.329171              |                          |
| 2s | -292.513139  | 0.011537              | 0.099165            | 15.021991                | 913.949829               |                          |
| 3s | -66.242749   | 0.074013              | 0.254798            | 5.698530                 | 198.675052               |                          |
| 4s | -13.694011   | 0.373906              | 0.576105            | 2.411881                 | 46.562344                |                          |
| 5s | -1.901252    | 2.362278              | 1.449793            | 0.891163                 | 6.906458                 |                          |
| 6s | -0.174492    | 25.636377             | 4.746599            | 0.258552                 | 0.475951                 |                          |
| 2p | -278.029306  | 0.008600              | 0.084202            | 15.015908                | 304.374611               | 9381.588056              |
| 3p | -59.715597   | 0.069378              | 0.244930            | 5.627404                 | 66.207640                | 1936.492771              |
| 4p | -11.051204   | 0.395903              | 0.590482            | 2.313388                 | 15.286820                | 415.650120               |
| 5p | -1.109516    | 3.063206              | 1.641797            | 0.782332                 | 2.027885                 | 47.552533                |
| 3d | -47.713279   | 0.055366              | 0.217344            | 5.515179                 | 37.586791                | 330.572963               |
| 4d | -6.315975    | 0.456197              | 0.627370            | 2.098172                 | 7.803389                 | 59.575284                |
| 4f | -0.695407    | 0.851632              | 0.812592            | 1.594584                 | 3.309307                 | 9.021379                 |

Table 1 (continued)

| Dy, Z = 66, [Xe]6s <sup>2</sup> 4f <sup>10</sup> ( <sup>5</sup> I) |              |                       |                                |                          |                          |                          |
|--------------------------------------------------------------------|--------------|-----------------------|--------------------------------|--------------------------|--------------------------|--------------------------|
| Total energy = −11641.4525953 ;                                    |              |                       | Kinetic energy = 11641.4525953 |                          |                          |                          |
| Potential energy = −23282.9051906 ;                                |              |                       | Virial ratio = −2.00000000000  |                          |                          |                          |
| $\rho_0 = 204862.91 ;$                                             |              |                       | $C_{EN} = 1.0000000$           |                          |                          |                          |
|                                                                    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$            | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                 | −1862.544387 | 0.000706              | 0.022976                       | 65.457481                | 8592.343893              |                          |
| 2s                                                                 | −302.864302  | 0.011168              | 0.097567                       | 15.269390                | 944.154906               |                          |
| 3s                                                                 | −68.755713   | 0.071395              | 0.250253                       | 5.804678                 | 206.125817               |                          |
| 4s                                                                 | −14.166828   | 0.359983              | 0.565247                       | 2.459706                 | 48.452127                |                          |
| 5s                                                                 | −1.941461    | 2.288928              | 1.426872                       | 0.905420                 | 7.126378                 |                          |
| 6s                                                                 | −0.176148    | 25.171218             | 4.702029                       | 0.261017                 | 0.483423                 |                          |
| 2p                                                                 | −288.115102  | 0.008319              | 0.082819                       | 15.264211                | 314.468317               | 9849.299876              |
| 3p                                                                 | −62.083368   | 0.066831              | 0.240396                       | 5.734505                 | 68.714089                | 2043.867619              |
| 4p                                                                 | −11.451407   | 0.380508              | 0.578847                       | 2.361041                 | 15.918807                | 440.378826               |
| 5p                                                                 | −1.129563    | 2.971828              | 1.616693                       | 0.794423                 | 2.088826                 | 49.765468                |
| 3d                                                                 | −49.814272   | 0.053176              | 0.213035                       | 5.624522                 | 39.073653                | 350.110809               |
| 4d                                                                 | −6.580807    | 0.436447              | 0.613578                       | 2.145914                 | 8.157054                 | 63.520037                |
| 4f                                                                 | −0.703246    | 0.810360              | 0.791752                       | 1.638548                 | 3.494921                 | 9.785372                 |
| Ho, Z = 67, [Xe]6s <sup>2</sup> 4f <sup>11</sup> ( <sup>4</sup> I) |              |                       |                                |                          |                          |                          |
| Total energy = −12065.2898028 ;                                    |              |                       | Kinetic energy = 12065.2898028 |                          |                          |                          |
| Potential energy = −24130.5796056 ;                                |              |                       | Virial ratio = −2.00000000000  |                          |                          |                          |
| $\rho_0 = 214440.88 ;$                                             |              |                       | $C_{EN} = 1.0000000$           |                          |                          |                          |
|                                                                    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$            | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                 | −1921.671632 | 0.000685              | 0.022629                       | 66.456714                | 8856.358502              |                          |
| 2s                                                                 | −313.391050  | 0.010815              | 0.096020                       | 15.516869                | 974.859888               |                          |
| 3s                                                                 | −71.308449   | 0.068915              | 0.245870                       | 5.910764                 | 213.708329               |                          |
| 4s                                                                 | −14.644518   | 0.346872              | 0.554830                       | 2.507320                 | 50.370090                |                          |
| 5s                                                                 | −1.981618    | 2.219508              | 1.404841                       | 0.919557                 | 7.347561                 |                          |
| 6s                                                                 | −0.177781    | 24.725390             | 4.658920                       | 0.263451                 | 0.490858                 |                          |
| 2p                                                                 | −298.376375  | 0.008051              | 0.081480                       | 15.512571                | 324.728815               | 10332.393774             |
| 3p                                                                 | −64.490395   | 0.064424              | 0.236029                       | 5.841531                 | 71.265168                | 2155.066876              |
| 4p                                                                 | −11.855921   | 0.366051              | 0.567705                       | 2.408480                 | 16.560541                | 465.967920               |
| 5p                                                                 | −1.149437    | 2.885409              | 1.592595                       | 0.806390                 | 2.149980                 | 52.026938                |
| 3d                                                                 | −51.953546   | 0.051116              | 0.208899                       | 5.733754                 | 40.588013                | 370.388033               |
| 4d                                                                 | −6.848877    | 0.418041              | 0.600442                       | 2.193422                 | 8.516819                 | 67.617859                |
| 4f                                                                 | −0.708583    | 0.772993              | 0.772360                       | 1.681855                 | 3.683016                 | 10.581219                |
| Er, Z = 68, [Xe]6s <sup>2</sup> 4f <sup>12</sup> ( <sup>3</sup> H) |              |                       |                                |                          |                          |                          |
| Total energy = −12498.1527833 ;                                    |              |                       | Kinetic energy = 12498.1527833 |                          |                          |                          |
| Potential energy = −24996.3055667 ;                                |              |                       | Virial ratio = −2.00000000000  |                          |                          |                          |
| $\rho_0 = 224312.97 ;$                                             |              |                       | $C_{EN} = 1.0000000$           |                          |                          |                          |
|                                                                    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$            | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                 | −1981.724675 | 0.000664              | 0.022293                       | 67.455968                | 9124.372986              |                          |
| 2s                                                                 | −324.093618  | 0.010479              | 0.094520                       | 15.764423                | 1006.064613              |                          |
| 3s                                                                 | −73.901311   | 0.066563              | 0.241641                       | 6.016790                 | 221.422536               |                          |
| 4s                                                                 | −15.127340   | 0.334508              | 0.544824                       | 2.554744                 | 52.316655                |                          |
| 5s                                                                 | −2.021781    | 2.153663              | 1.383625                       | 0.933593                 | 7.570254                 |                          |
| 6s                                                                 | −0.179392    | 24.297268             | 4.617155                       | 0.265856                 | 0.498264                 |                          |
| 2p                                                                 | −308.813356  | 0.007795              | 0.080183                       | 15.760984                | 335.156060               | 10831.118176             |
| 3p                                                                 | −66.937020   | 0.062147              | 0.231821                       | 5.948486                 | 73.860869                | 2270.156087              |
| 4p                                                                 | −12.264984   | 0.352451              | 0.557022                       | 2.455728                 | 17.212176                | 492.435553               |
| 5p                                                                 | −1.169181    | 2.803480              | 1.569413                       | 0.818251                 | 2.211427                 | 54.339313                |
| 3d                                                                 | −54.131415   | 0.049175              | 0.204925                       | 5.842884                 | 42.129875                | 391.418068               |
| 4d                                                                 | −7.120374    | 0.400849              | 0.587910                       | 2.240723                 | 8.882782                 | 71.872296                |
| 4f                                                                 | −0.711720    | 0.738959              | 0.754234                       | 1.724595                 | 3.873771                 | 11.409980                |
| Tm, Z = 69, [Xe]6s <sup>2</sup> 4f <sup>13</sup> ( <sup>2</sup> F) |              |                       |                                |                          |                          |                          |
| Total energy = −12940.1744048 ;                                    |              |                       | Kinetic energy = 12940.1744048 |                          |                          |                          |
| Potential energy = −25880.3488097 ;                                |              |                       | Virial ratio = −2.00000000000  |                          |                          |                          |
| $\rho_0 = 234483.41 ;$                                             |              |                       | $C_{EN} = 1.0000000$           |                          |                          |                          |
|                                                                    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$            | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                 | −2042.698792 | 0.000645              | 0.021967                       | 68.455246                | 9396.387728              |                          |
| 2s                                                                 | −334.966873  | 0.010159              | 0.093066                       | 16.012058                | 1037.770053              |                          |
| 3s                                                                 | −76.528887   | 0.064331              | 0.237557                       | 6.122750                 | 229.267730               |                          |
| 4s                                                                 | −15.611268   | 0.322859              | 0.535228                       | 2.601858                 | 54.286064                |                          |
| 5s                                                                 | −2.060938    | 2.092025              | 1.363463                       | 0.947303                 | 7.789868                 |                          |
| 6s                                                                 | −0.180955    | 23.892760             | 4.577345                       | 0.268195                 | 0.505523                 |                          |
| 2p                                                                 | −319.420993  | 0.007552              | 0.078927                       | 16.009456                | 345.750324               | 11345.735029             |
| 3p                                                                 | −69.417851   | 0.059990              | 0.227763                       | 6.055364                 | 76.500992                | 2389.194360              |
| 4p                                                                 | −12.674672   | 0.339676              | 0.546797                       | 2.502648                 | 17.871653                | 519.728051               |
| 5p                                                                 | −1.188049    | 2.727215              | 1.547512                       | 0.829759                 | 2.271581                 | 56.657653                |
| 3d                                                                 | −56.342527   | 0.047346              | 0.201104                       | 5.951909                 | 43.699192                | 413.214073               |
| 4d                                                                 | −7.391616    | 0.384834              | 0.575988                       | 2.287638                 | 9.253525                 | 76.272030                |
| 4f                                                                 | −0.719265    | 0.706160              | 0.736587                       | 1.767631                 | 4.069896                 | 12.280993                |

(continued on next page)

**Table 1** (continued)Yb,  $Z = 70$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>(<sup>1</sup>S)

Total energy = −13391.4561931 ; Kinetic energy = 13391.4561931

Potential energy = −26782.9123862 ; Virial ratio = −2.00000000000

 $\rho_0 = 244956.58$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −2104.592407 | 0.000627              | 0.021651            | 69.454546                | 9672.402838              |                          |
| 2s | −346.009135  | 0.009852              | 0.091656            | 16.259770                | 1069.976398              |                          |
| 3s | −79.189503   | 0.062211              | 0.233611            | 6.228642                 | 237.243576               |                          |
| 4s | −16.095088   | 0.311877              | 0.526019            | 2.648637                 | 56.276557                |                          |
| 5s | −2.098804    | 2.034469              | 1.344358            | 0.960633                 | 8.004983                 |                          |
| 6s | −0.182463    | 23.511983             | 4.539551            | 0.270460                 | 0.512608                 |                          |
| 2p | −330.197628  | 0.007320              | 0.077709            | 16.257984                | 356.511661               | 11876.497477             |
| 3p | −71.931210   | 0.057944              | 0.223848            | 6.162165                 | 79.185445                | 2512.244015              |
| 4p | −13.083780   | 0.327667              | 0.537009            | 2.549213                 | 18.538354                | 547.835104               |
| 5p | −1.205832    | 2.656526              | 1.526917            | 0.840850                 | 2.329948                 | 58.965839                |
| 3d | −58.585193   | 0.045620              | 0.197427            | 6.060836                 | 45.295942                | 435.789242               |
| 4d | −7.661433    | 0.369900              | 0.564643            | 2.334131                 | 9.628621                 | 80.814695                |
| 4f | −0.732406    | 0.674330              | 0.719312            | 1.811083                 | 4.271847                 | 13.196770                |

Lu,  $Z = 71$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>1</sup>(<sup>2</sup>D)

Total energy = −13851.8080034 ; Kinetic energy = 13851.8080034

Potential energy = −27703.6160067 ; Virial ratio = −2.00000000000

 $\rho_0 = 255751.36$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −2167.731477 | 0.000609              | 0.021343            | 70.453767                | 9952.395299              |                          |
| 2s | −357.572303  | 0.009560              | 0.090290            | 16.507010                | 1102.616769              |                          |
| 3s | −82.268768   | 0.060199              | 0.229805            | 6.334172                 | 245.328348               |                          |
| 4s | −16.937896   | 0.300351              | 0.516244            | 2.700477                 | 58.529574                |                          |
| 5s | −2.317030    | 1.894814              | 1.298354            | 0.996396                 | 8.680786                 |                          |
| 6s | −0.198856    | 20.710428             | 4.258758            | 0.289688                 | 0.612913                 |                          |
| 2p | −341.490124  | 0.007098              | 0.076531            | 16.506094                | 367.421500               | 12422.833964             |
| 3p | −74.860335   | 0.056007              | 0.220077            | 6.268567                 | 81.906226                | 2639.042910              |
| 4p | −13.845079   | 0.314762              | 0.526390            | 2.601716                 | 19.301556                | 579.751423               |
| 5p | −1.375840    | 2.431755              | 1.462590            | 0.879117                 | 2.562164                 | 66.392164                |
| 3d | −61.236416   | 0.043990              | 0.193896            | 6.169239                 | 46.913162                | 459.052948               |
| 4d | −8.264905    | 0.352421              | 0.551381            | 2.389441                 | 10.076525                | 86.187473                |
| 5d | −0.243353    | 7.527112              | 2.485308            | 0.526626                 | 0.602151                 | 3.468697                 |
| 4f | −1.076887    | 0.580465              | 0.674576            | 1.900142                 | 4.651075                 | 14.806814                |

Hf,  $Z = 72$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>2</sup>(<sup>3</sup>F)

Total energy = −14321.2498119 ; Kinetic energy = 14321.2498119

Potential energy = −28642.4996239 ; Virial ratio = −2.00000000000

 $\rho_0 = 266859.04$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −2231.811080 | 0.000592              | 0.021044            | 71.453001                | 10236.386011             |                          |
| 2s | −369.329801  | 0.009281              | 0.088965            | 16.754219                | 1135.743253              |                          |
| 3s | −85.411971   | 0.058284              | 0.226122            | 6.439662                 | 253.544310               |                          |
| 4s | −17.797724   | 0.289287              | 0.506687            | 2.753183                 | 60.866005                |                          |
| 5s | −2.519419    | 1.773712              | 1.256767            | 1.031390                 | 9.370123                 |                          |
| 6s | −0.209002    | 19.060209             | 4.082582            | 0.303160                 | 0.685811                 |                          |
| 2p | −352.976012  | 0.006887              | 0.075387            | 16.754172                | 378.494500               | 12985.619829             |
| 3p | −77.852505   | 0.054166              | 0.216431            | 6.374919                 | 84.671436                | 2769.970861              |
| 4p | −14.621995   | 0.302386              | 0.516005            | 2.655223                 | 20.094748                | 613.372707               |
| 5p | −1.532359    | 2.245875              | 1.406641            | 0.915736                 | 2.794975                 | 74.062520                |
| 3d | −63.948890   | 0.042446              | 0.190489            | 6.277581                 | 48.557699                | 483.114207               |
| 4d | −8.881065    | 0.335760              | 0.538426            | 2.446086                 | 10.544954                | 91.898827                |
| 5d | −0.324432    | 5.775211              | 2.194813            | 0.590772                 | 0.761118                 | 4.652854                 |
| 4f | −1.430271    | 0.512262              | 0.638335            | 1.984529                 | 5.029877                 | 16.482619                |

Table 1 (continued)

Ta,  $Z = 73$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>3</sup> (4F)Total energy = −14799.8125980 ; Kinetic energy = 14799.8125980  
Potential energy = −29599.6251961 ; Virial ratio = −2.00000000000 $\rho_0 = 278284.59$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −2296.842183 | 0.000576              | 0.020753            | 72.452248                | 10524.374752             |                          |
| 2s | −381.292349  | 0.009014              | 0.087677            | 17.001414                | 1169.357756              |                          |
| 3s | −88.628813   | 0.056458              | 0.222555            | 6.545140                 | 261.893548               |                          |
| 4s | −18.684462   | 0.278702              | 0.497373            | 2.806593                 | 63.280798                |                          |
| 5s | −2.720391    | 1.665622              | 1.218294            | 1.066114                 | 10.080597                |                          |
| 6s | −0.217490    | 17.794356             | 3.941820            | 0.314753                 | 0.750109                 |                          |
| 2p | −364.666082  | 0.006685              | 0.074278            | 17.002229                | 389.731114               | 13565.117951             |
| 3p | −80.917469   | 0.052414              | 0.212905            | 6.481249                 | 87.481691                | 2905.117340              |
| 4p | −15.424583   | 0.290574              | 0.505893            | 2.709512                 | 20.915509                | 648.661700               |
| 5p | −1.688385    | 2.084747              | 1.356026            | 0.951718                 | 3.033459                 | 82.126379                |
| 3d | −66.732486   | 0.040982              | 0.187201            | 6.385880                 | 50.229803                | 507.989847               |
| 4d | −9.520499    | 0.320016              | 0.525874            | 2.503661                 | 11.031435                | 97.933844                |
| 5d | −0.387780    | 4.836948              | 2.016485            | 0.641438                 | 0.902130                 | 5.773575                 |
| 4f | −1.804687    | 0.459099              | 0.607627            | 2.065668                 | 5.411079                 | 18.232493                |

W,  $Z = 74$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>4</sup> (5D)Total energy = −15287.5463682 ; Kinetic energy = 15287.5463682  
Potential energy = −30575.0927364 ; Virial ratio = −2.00000000000 $\rho_0 = 290032.50$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −2362.823830 | 0.000560              | 0.020470            | 73.451508                | 10816.361683             |                          |
| 2s | −393.458738  | 0.008758              | 0.086427            | 17.248603                | 1203.461165              |                          |
| 3s | −91.917520   | 0.054717              | 0.219099            | 6.650621                 | 270.377072               |                          |
| 4s | −19.596328   | 0.268595              | 0.488311            | 2.860595                 | 65.770584                |                          |
| 5s | −2.921222    | 1.568551              | 1.182571            | 1.100535                 | 10.810256                |                          |
| 6s | −0.224809    | 16.776313             | 3.824670            | 0.325015                 | 0.807894                 |                          |
| 2p | −376.559177  | 0.006492              | 0.073201            | 17.250273                | 401.131574               | 14161.582925             |
| 3p | −84.053488   | 0.050745              | 0.209492            | 6.587569                 | 90.337307                | 3044.561248              |
| 4p | −16.251207   | 0.279329              | 0.496071            | 2.764440                 | 21.762379                | 685.609466               |
| 5p | −1.845081    | 1.943419              | 1.309852            | 0.987145                 | 3.277509                 | 90.586908                |
| 3d | −69.585560   | 0.039593              | 0.184024            | 6.494149                 | 51.929640                | 533.695655               |
| 4d | −10.181981   | 0.305192              | 0.513756            | 2.561959                 | 11.534755                | 104.289080               |
| 5d | −0.446353    | 4.192439              | 1.882171            | 0.686768                 | 1.038686                 | 6.912088                 |
| 4f | −2.199662    | 0.416065              | 0.580965            | 2.144284                 | 5.795961                 | 20.060751                |

Re,  $Z = 75$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>5</sup> (6S)Total energy = −15784.5331876 ; Kinetic energy = 15784.5331876  
Potential energy = −31569.0663751 ; Virial ratio = −2.00000000000 $\rho_0 = 302107.03$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −2429.751463 | 0.000545              | 0.020195            | 74.450781                | 11112.347082             |                          |
| 2s | −405.824202  | 0.008513              | 0.085212            | 17.495787                | 1238.053862              |                          |
| 3s | −95.273022   | 0.053054              | 0.215747            | 6.756108                 | 278.995310               |                          |
| 4s | −20.528342   | 0.258957              | 0.479509            | 2.915095                 | 68.332421                |                          |
| 5s | −3.119745    | 1.481353              | 1.149453            | 1.134474                 | 11.553840                |                          |
| 6s | −0.230673    | 15.978435             | 3.729895            | 0.333748                 | 0.856674                 |                          |
| 2p | −388.650570  | 0.006307              | 0.072155            | 17.498307                | 412.695991               | 14775.264620             |
| 3p | −87.255518   | 0.049155              | 0.206185            | 6.693887                 | 93.238420                | 3188.375104              |
| 4p | −17.097020   | 0.268642              | 0.486548            | 2.819895                 | 22.634177                | 724.213912               |
| 5p | −2.000529    | 1.819058              | 1.267701            | 1.021869                 | 3.525451                 | 99.397875                |
| 3d | −72.503134   | 0.038272              | 0.180953            | 6.602396                 | 53.657328                | 560.246824               |
| 4d | −10.861012   | 0.291220              | 0.502056            | 2.620929                 | 12.054829                | 110.972210               |
| 5d | −0.514322    | 3.683762              | 1.768667            | 0.730183                 | 1.176732                 | 8.104441                 |
| 4f | −2.611121    | 0.380347              | 0.557438            | 2.220820                 | 6.185174                 | 21.969983                |

(continued on next page)

Table 1 (continued)

Os,  $Z = 76$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>6</sup> (<sup>5</sup>D)

Total energy = −16290.6485954 ; Kinetic energy = 16290.6485954

Potential energy = −32581.2971909 ; Virial ratio = −2.00000000000

 $\rho_0 = 314513.81$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −2497.646279 | 0.000531              | 0.019927            | 75.450065                | 11412.330199             |                          |
| 2s | −418.409954  | 0.008278              | 0.084031            | 17.742983                | 1273.137718              |                          |
| 3s | −98.715591   | 0.051466              | 0.212496            | 6.861634                 | 287.750813               |                          |
| 4s | −21.500388   | 0.249768              | 0.470962            | 2.970054                 | 70.965795                |                          |
| 5s | −3.333636    | 1.400242              | 1.117754            | 1.168928                 | 12.334413                |                          |
| 6s | −0.238767    | 15.083675             | 3.621935            | 0.344264                 | 0.920042                 |                          |
| 2p | −400.961506  | 0.006130              | 0.071138            | 17.746341                | 424.424776               | 15406.425746             |
| 3p | −90.543869   | 0.047637              | 0.202981            | 6.800231                 | 96.185771                | 3336.656132              |
| 4p | −17.981903   | 0.258491              | 0.477322            | 2.875809                 | 23.530277                | 764.490227               |
| 5p | −2.170386    | 1.704887              | 1.227682            | 1.057152                 | 3.786604                 | 108.864599               |
| 3d | −75.505602   | 0.037017              | 0.177983            | 6.710634                 | 55.413026                | 587.659397               |
| 4d | −11.577449   | 0.278258              | 0.490909            | 2.679929                 | 12.586840                | 117.939550               |
| 5d | −0.535781    | 3.377221              | 1.692494            | 0.766125                 | 1.304216                 | 9.277901                 |
| 4f | −3.059248    | 0.349919              | 0.536296            | 2.295887                 | 6.580452                 | 23.967968                |

Ir,  $Z = 77$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>7</sup> (<sup>4</sup>F)

Total energy = −16806.1131497 ; Kinetic energy = 16806.1131497

Potential energy = −33612.2262994 ; Virial ratio = −2.00000000000

 $\rho_0 = 327256.12$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −2566.484813 | 0.000517              | 0.019666            | 76.449361                | 11716.312027             |                          |
| 2s | −431.192234  | 0.008053              | 0.082881            | 17.990182                | 1308.711714              |                          |
| 3s | −102.221814  | 0.049947              | 0.209340            | 6.967179                 | 296.642142               |                          |
| 4s | −22.489451   | 0.241016              | 0.462671            | 3.025382                 | 73.667314                |                          |
| 5s | −3.544825    | 1.327014              | 1.088284            | 1.202839                 | 13.126498                |                          |
| 6s | −0.245373    | 14.373952             | 3.533587            | 0.353317                 | 0.974387                 |                          |
| 2p | −413.468249  | 0.005960              | 0.070149            | 17.994371                | 436.317744               | 16055.304288             |
| 3p | −93.895134   | 0.046188              | 0.199874            | 6.906584                 | 99.178960                | 3489.456366              |
| 4p | −18.883022   | 0.248857              | 0.468393            | 2.932095                 | 24.449717                | 806.440025               |
| 5p | −2.338628    | 1.603562              | 1.190950            | 1.091718                 | 4.051013                 | 118.680423               |
| 3d | −78.569583   | 0.035822              | 0.175109            | 6.818861                 | 57.196741                | 615.947087               |
| 4d | −12.308903   | 0.266081              | 0.480189            | 2.739302                 | 13.133538                | 125.227212               |
| 5d | −0.574001    | 3.085820              | 1.618345            | 0.803071                 | 1.439019                 | 10.548009                |
| 4f | −3.521744    | 0.323804              | 0.517220            | 2.369470                 | 6.981080                 | 26.052518                |

Pt,  $Z = 78$ , [Xe]6s<sup>1</sup>4f<sup>14</sup>5d<sup>9</sup> (<sup>3</sup>D)

Total energy = −17331.0699646 ; Kinetic energy = 17331.0699646

Potential energy = −34662.1399293 ; Virial ratio = −2.00000000000

 $\rho_0 = 340321.87$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −2636.112231 | 0.000504              | 0.019411            | 77.448694                | 12024.298600             |                          |
| 2s | −444.012794  | 0.007837              | 0.081763            | 18.237354                | 1344.771422              |                          |
| 3s | −105.633525  | 0.048495              | 0.206279            | 7.072632                 | 305.659062               |                          |
| 4s | −23.336611   | 0.232708              | 0.454658            | 3.080796                 | 76.422500                |                          |
| 5s | −3.605347    | 1.273714              | 1.065780            | 1.230447                 | 13.784471                |                          |
| 6s | −0.222830    | 15.915972             | 3.707827            | 0.336622                 | 0.849331                 |                          |
| 2p | −426.013215  | 0.005797              | 0.069188            | 18.242396                | 448.375057               | 16722.162079             |
| 3p | −97.151661   | 0.044805              | 0.196861            | 7.012888                 | 102.216451               | 3646.782267              |
| 4p | −19.642774   | 0.239714              | 0.459753            | 2.988709                 | 25.392111                | 850.085424               |
| 5p | −2.371651    | 1.534662              | 1.164446            | 1.118823                 | 4.264528                 | 127.055048               |
| 3d | −81.537857   | 0.034683              | 0.172325            | 6.927084                 | 59.008546                | 645.125114               |
| 4d | −12.899237   | 0.254717              | 0.469929            | 2.798845                 | 13.693595                | 132.828890               |
| 5d | −0.475069    | 3.113442              | 1.614941            | 0.813133                 | 1.487156                 | 11.110287                |
| 4f | −3.842165    | 0.301427              | 0.500027            | 2.441453                 | 7.386028                 | 28.220118                |

Table 1 (continued)

Au,  $Z = 79$ , [Xe]6s<sup>1</sup>4f<sup>14</sup>5d<sup>10</sup> (<sup>2</sup>S)

Total energy = −17865.4000842 ; Kinetic energy = 17865.4000842

Potential energy = −35730.8001685 ; Virial ratio = −2.00000000000

 $\rho_0 = 353747.30$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −2706.832984 | 0.000491              | 0.019164            | 78.448016                | 12336.278197             |                          |
| 2s | −457.182524  | 0.007629              | 0.080675            | 18.484565                | 1381.326464              |                          |
| 3s | −109.260794  | 0.047104              | 0.203302            | 7.178225                 | 314.822752               |                          |
| 4s | −24.353765   | 0.224773              | 0.446867            | 3.136701                 | 79.254753                |                          |
| 5s | −3.808827    | 1.212728              | 1.039978            | 1.263184                 | 14.589912                |                          |
| 6s | −0.220778    | 15.889195             | 3.700595            | 0.337391                 | 0.843045                 |                          |
| 2p | −438.906054  | 0.005641              | 0.068253            | 18.490424                | 460.596664               | 17407.224584             |
| 3p | −100.622567  | 0.043482              | 0.193935            | 7.119272                 | 105.301624               | 3808.836692              |
| 4p | −20.570661   | 0.231043              | 0.451402            | 3.045561                 | 26.356184                | 895.410544               |
| 5p | −2.534336    | 1.450986              | 1.132476            | 1.152265                 | 4.534780                 | 137.526513               |
| 3d | −84.718799   | 0.033598              | 0.169628            | 7.035323                 | 60.848741                | 675.210781               |
| 4d | −13.655497   | 0.243714              | 0.459835            | 2.859616                 | 14.275556                | 140.846536               |
| 5d | −0.521023    | 2.834034              | 1.543263            | 0.850063                 | 1.622799                 | 12.424663                |
| 4f | −4.328347    | 0.281398              | 0.484078            | 2.512740                 | 7.798629                 | 30.485909                |

Hg,  $Z = 80$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>10</sup> (<sup>1</sup>S)

Total energy = −18408.9914949 ; Kinetic energy = 18408.9914949

Potential energy = −36817.9829899 ; Virial ratio = −2.00000000000

 $\rho_0 = 367542.14$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −2778.680221 | 0.000478              | 0.018922            | 79.447325                | 12652.249592             |                          |
| 2s | −470.735051  | 0.007429              | 0.079615            | 18.731826                | 1418.378838              |                          |
| 3s | −113.136600  | 0.045771              | 0.200407            | 7.283992                 | 324.136884               |                          |
| 4s | −25.573382   | 0.217186              | 0.439287            | 3.193123                 | 82.166953                |                          |
| 5s | −4.182017    | 1.142865              | 1.010198            | 1.302578                 | 15.593399                |                          |
| 6s | −0.261046    | 12.799334             | 3.328414            | 0.376067                 | 1.109792                 |                          |
| 2p | −452.180310  | 0.005491              | 0.067342            | 18.738460                | 472.982786               | 18110.745663             |
| 3p | −104.340772  | 0.042215              | 0.191093            | 7.225758                 | 108.435229               | 3975.722192              |
| 4p | −21.698933   | 0.222809              | 0.443321            | 3.102665                 | 27.342277                | 942.460358               |
| 5p | −2.850880    | 1.355169              | 1.095406            | 1.192868                 | 4.873604                 | 150.596398               |
| 3d | −88.145357   | 0.032562              | 0.167014            | 7.143548                 | 62.716895                | 706.211876               |
| 4d | −14.609618   | 0.233642              | 0.450344            | 2.919555                 | 14.862295                | 149.092402               |
| 5d | −0.714203    | 2.409811              | 1.432682            | 0.910323                 | 1.857348                 | 14.706646                |
| 4f | −5.012380    | 0.263365              | 0.469205            | 2.583518                 | 8.219609                 | 32.855320                |

Tl,  $Z = 81$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>10</sup>6p<sup>1</sup> (<sup>2</sup>P)

Total energy = −18961.8248243 ; Kinetic energy = 18961.8248243

Potential energy = −37923.6496485 ; Virial ratio = −2.00000000000

 $\rho_0 = 381684.68$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −2851.547207 | 0.000467              | 0.018687            | 80.446635                | 12972.217042             |                          |
| 2s | −484.560177  | 0.007238              | 0.078582            | 18.979089                | 1455.921382              |                          |
| 3s | −117.150470  | 0.044492              | 0.197591            | 7.389785                 | 333.588038               |                          |
| 4s | −26.883439   | 0.209962              | 0.431943            | 3.249717                 | 85.140447                |                          |
| 5s | −4.618775    | 1.079224              | 0.982189            | 1.341962                 | 16.629451                |                          |
| 6s | −0.361112    | 10.073007             | 2.966905            | 0.421732                 | 1.472377                 |                          |
| 2p | −465.726727  | 0.005347              | 0.066456            | 18.986507                | 485.533771               | 18832.998519             |
| 3p | −108.196801  | 0.041002              | 0.188331            | 7.332307                 | 111.616301               | 4147.473144              |
| 4p | −22.917694   | 0.214973              | 0.435490            | 3.160082                 | 28.351730                | 991.322312               |
| 5p | −3.231369    | 1.265663              | 1.059620            | 1.234762                 | 5.235430                 | 164.793522               |
| 6p | −0.192397    | 17.876087             | 3.926246            | 0.316234                 | 0.312830                 | 7.553060                 |
| 3d | −91.708496   | 0.031573              | 0.164478            | 7.251791                 | 64.613485                | 738.149557               |
| 4d | −15.652961   | 0.224160              | 0.441212            | 2.979688                 | 15.462574                | 157.672994               |
| 5d | −0.968275    | 2.090499              | 1.341196            | 0.968923                 | 2.101579                 | 17.172516                |
| 4f | −5.785240    | 0.247230              | 0.455385            | 2.653660                 | 8.648302                 | 35.327546                |

(continued on next page)

Table 1 (continued)

Pb,  $Z = 82$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>10</sup>6p<sup>2</sup> (<sup>3</sup>P)

Total energy = −19524.0080381 ; Kinetic energy = 19524.0080381

Potential energy = −39048.0160763 ; Virial ratio = −2.00000000000

 $\rho_0 = 396186.26$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −2925.369754 | 0.000455              | 0.018457            | 81.445947                | 13296.180279             |                          |
| 2s | −498.594155  | 0.007053              | 0.077576            | 19.226370                | 1493.956622              |                          |
| 3s | −121.238373  | 0.043266              | 0.194853            | 7.495644                 | 343.180249               |                          |
| 4s | −28.219947   | 0.203072              | 0.424818            | 3.306559                 | 88.180081                |                          |
| 5s | −5.055544    | 1.020357              | 0.955468            | 1.381909                 | 17.715843                |                          |
| 6s | −0.456226    | 8.458071              | 2.726568            | 0.459474                 | 1.811449                 |                          |
| 2p | −479.481368  | 0.005208              | 0.065592            | 19.234568                | 498.249674               | 19574.230355             |
| 3p | −112.126368  | 0.039839              | 0.185645            | 7.438927                 | 114.845162               | 4324.173426              |
| 4p | −24.162410   | 0.207515              | 0.427901            | 3.217773                 | 29.384061                | 1042.012049              |
| 5p | −3.610692    | 1.184411              | 1.025874            | 1.277268                 | 5.616134                 | 180.015331               |
| 6p | −0.255697    | 13.424624             | 3.418863            | 0.363166                 | 0.433458                 | 11.122707                |
| 3d | −95.343943   | 0.030629              | 0.162018            | 7.360055                 | 66.538601                | 771.039167               |
| 4d | −16.720911   | 0.215229              | 0.432427            | 3.039970                 | 16.076042                | 166.590710               |
| 5d | −1.220429    | 1.854022              | 1.267244            | 1.024081                 | 2.346843                 | 19.743583                |
| 4f | −6.582431    | 0.232702              | 0.442489            | 2.723257                 | 9.084932                 | 37.905522                |

Bi,  $Z = 83$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>10</sup>6p<sup>3</sup> (<sup>4</sup>S)

Total energy = −20095.5864271 ; Kinetic energy = 20095.5864271

Potential energy = −40191.1728543 ; Virial ratio = −2.00000000000

 $\rho_0 = 411051.72$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −3000.152768 | 0.000444              | 0.018233            | 82.445260                | 13624.139561             |                          |
| 2s | −512.841770  | 0.006876              | 0.076596            | 19.473672                | 1532.484855              |                          |
| 3s | −125.404955  | 0.042089              | 0.192187            | 7.601573                 | 352.913910               |                          |
| 4s | −29.587497   | 0.196495              | 0.417902            | 3.363634                 | 91.285571                |                          |
| 5s | −5.497783    | 0.966119              | 0.930093            | 1.422190                 | 18.847291                |                          |
| 6s | −0.551690    | 7.330087              | 2.543441            | 0.493369                 | 2.145471                 |                          |
| 2p | −493.449027  | 0.005075              | 0.064750            | 19.482641                | 511.130539               | 20334.690008             |
| 3p | −116.134115  | 0.038725              | 0.183033            | 7.545618                 | 118.121861               | 4505.896338              |
| 4p | −25.437673   | 0.200415              | 0.420547            | 3.275713                 | 30.439079                | 1094.556322              |
| 5p | −3.994714    | 1.110609              | 0.994107            | 1.320171                 | 6.014007                 | 196.229531               |
| 6p | −0.320105    | 10.859221             | 3.084795            | 0.402892                 | 0.550062                 | 14.739553                |
| 3d | −99.056387   | 0.029725              | 0.159629            | 7.468342                 | 68.492295                | 804.895601               |
| 4d | −17.818176   | 0.206808              | 0.423969            | 3.100388                 | 16.702637                | 175.851400               |
| 5d | −1.477157    | 1.668392              | 1.204947            | 1.076800                 | 2.594574                 | 22.429928                |
| 4f | −7.408698    | 0.219561              | 0.430420            | 2.792350                 | 9.529507                 | 40.591151                |

Po,  $Z = 84$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>10</sup>6p<sup>4</sup> (<sup>3</sup>P)

Total energy = −20676.5009150 ; Kinetic energy = 20676.5009150

Potential energy = −41353.0018300 ; Virial ratio = −2.00000000000

 $\rho_0 = 426286.08$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −3075.913471 | 0.000434              | 0.018014            | 83.444577                | 13956.095546             |                          |
| 2s | −527.319921  | 0.006705              | 0.075639            | 19.720997                | 1571.506715              |                          |
| 3s | −129.666880  | 0.040959              | 0.189592            | 7.707575                 | 362.789507               |                          |
| 4s | −31.002593   | 0.190214              | 0.411189            | 3.420930                 | 94.456905                |                          |
| 5s | −5.962098    | 0.916153              | 0.906023            | 1.462686                 | 20.020960                |                          |
| 6s | −0.658255    | 6.457442              | 2.390765            | 0.526022                 | 2.496266                 |                          |
| 2p | −507.646630  | 0.004947              | 0.063930            | 19.730730                | 524.176470               | 21114.629827             |
| 3p | −120.236689  | 0.037655              | 0.180492            | 7.652377                 | 121.446396               | 4692.713192              |
| 4p | −26.759958   | 0.193654              | 0.413420            | 3.333868                 | 31.516357                | 1148.971936              |
| 5p | −4.400146    | 1.045075              | 0.964801            | 1.362668                 | 6.421923                 | 213.204094               |
| 6p | −0.341467    | 9.513473              | 2.884992            | 0.434032                 | 0.659337                 | 18.339235                |
| 3d | −102.862560  | 0.028861              | 0.157309            | 7.576657                 | 70.474656                | 839.734432               |
| 4d | −18.961385   | 0.198859              | 0.415820            | 3.160941                 | 17.342422                | 185.462511               |
| 5d | −1.755303    | 1.515963              | 1.150669            | 1.127987                 | 2.847025                 | 25.252782                |
| 4f | −8.280689    | 0.207618              | 0.419089            | 2.860997                 | 9.982163                 | 43.387044                |

Table 1 (continued)

At,  $Z = 85$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>10</sup>6p<sup>5</sup> (<sup>2</sup>P)

Total energy = −21266.8817131 ; Kinetic energy = 21266.8817131

Potential energy = −42533.7634262 ; Virial ratio = −2.00000000000

 $\rho_0 = 441893.33$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −3152.633219 | 0.000423              | 0.017800            | 84.443897                | 14292.047793             |                          |
| 2s | −542.010114  | 0.006540              | 0.074706            | 19.968342                | 1611.021911              |                          |
| 3s | −134.005628  | 0.039872              | 0.187065            | 7.813646                 | 372.807095               |                          |
| 4s | −32.446784   | 0.184216              | 0.404671            | 3.478424                 | 97.693250                |                          |
| 5s | −6.431043    | 0.870128              | 0.883215            | 1.503279                 | 21.233560                |                          |
| 6s | −0.765378    | 5.779812              | 2.264454            | 0.556524                 | 2.847007                 |                          |
| 2p | −522.055686  | 0.004823              | 0.063130            | 19.978833                | 537.387449               | 21914.295479             |
| 3p | −124.415610  | 0.036629              | 0.178019            | 7.759206                 | 124.818900               | 4884.701196              |
| 4p | −28.110895   | 0.187213              | 0.406512            | 3.392227                 | 32.615865                | 1205.291287              |
| 5p | −4.809864    | 0.985797              | 0.937399            | 1.405051                 | 6.842023                 | 231.031148               |
| 6p | −0.379865    | 8.337897              | 2.702205            | 0.465540                 | 0.776777                 | 22.316452                |
| 3d | −106.743978  | 0.028033              | 0.155054            | 7.684999                 | 72.485687                | 875.569767               |
| 4d | −20.132178   | 0.191351              | 0.407968            | 3.221603                 | 17.995187                | 195.427884               |
| 5d | −2.037918    | 1.389123              | 1.102992            | 1.177645                 | 3.103122                 | 28.201852                |
| 4f | −9.180077    | 0.196725              | 0.408427            | 2.929220                 | 10.442834                | 46.294724                |

Rn,  $Z = 86$ , [Xe]6s<sup>2</sup>4f<sup>14</sup>5d<sup>10</sup>6p<sup>6</sup> (<sup>1</sup>S)

Total energy = −21866.7722409 ; Kinetic energy = 21866.7722409

Potential energy = −43733.5444817 ; Virial ratio = −2.00000000000

 $\rho_0 = 457878.02$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −3230.312838 | 0.000413              | 0.017592            | 85.443220                | 14631.996378             |                          |
| 2s | −556.913115  | 0.006381              | 0.073796            | 20.215708                | 1651.030576              |                          |
| 3s | −138.421866  | 0.038828              | 0.184602            | 7.919786                 | 382.966890               |                          |
| 4s | −33.920746   | 0.178483              | 0.398341            | 3.536100                 | 100.994193               |                          |
| 5s | −6.905818    | 0.827678              | 0.861597            | 1.543905                 | 22.483059                |                          |
| 6s | −0.873993    | 5.232669              | 2.156625            | 0.585507                 | 3.200917                 |                          |
| 2p | −536.676971  | 0.004704              | 0.062350            | 20.226951                | 550.763513               | 22733.935650             |
| 3p | −128.671558  | 0.035644              | 0.175612            | 7.866104                 | 128.239464               | 5081.936219              |
| 4p | −29.491183   | 0.181071              | 0.399811            | 3.450796                 | 33.737918                | 1263.561013              |
| 5p | −5.225212    | 0.929856              | 0.910884            | 1.448390                 | 7.284326                 | 250.103631               |
| 6p | −0.428007    | 7.370016              | 2.543382            | 0.495274                 | 0.888218                 | 26.168890                |
| 3d | −110.701350  | 0.027240              | 0.152862            | 7.793367                 | 74.525417                | 912.416261               |
| 4d | −21.331318   | 0.184254              | 0.400398            | 3.282359                 | 18.660844                | 205.752865               |
| 5d | −2.326319    | 1.281382              | 1.060487            | 1.226078                 | 3.363316                 | 31.282221                |
| 4f | −10.107635   | 0.186753              | 0.398370            | 2.997051                 | 10.911540                | 49.316192                |

Fr,  $Z = 87$ , [Rn]7s<sup>1</sup> (<sup>2</sup>S)

Total energy = −22475.8587125 ; Kinetic energy = 22475.8587125

Potential energy = −44951.7174250 ; Virial ratio = −2.00000000000

 $\rho_0 = 474252.97$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −3309.144859 | 0.000404              | 0.017388            | 86.442556                | 14975.944286             |                          |
| 2s | −572.220473  | 0.006228              | 0.072908            | 20.463119                | 1691.536467              |                          |
| 3s | −143.106572  | 0.037824              | 0.182202            | 8.026027                 | 393.272051               |                          |
| 4s | −35.615036   | 0.172995              | 0.392186            | 3.594025                 | 104.364189               |                          |
| 5s | −7.575756    | 0.788153              | 0.840946            | 1.584810                 | 23.777690                |                          |
| 6s | −1.127303    | 4.550528              | 2.016935            | 0.627099                 | 3.757208                 |                          |
| 7s | −0.117912    | 49.553309             | 6.629737            | 0.181543                 | 0.229130                 |                          |
| 2p | −551.701970  | 0.004590              | 0.061589            | 20.475087                | 564.304939               | 23573.812663             |
| 3p | −133.195189  | 0.034697              | 0.173267            | 7.973061                 | 131.707776               | 5284.474515              |
| 4p | −31.090810   | 0.175215              | 0.393316            | 3.509503                 | 34.881214                | 1323.758844              |
| 5p | −5.834205    | 0.880119              | 0.886486            | 1.490871                 | 7.731322                 | 269.788849               |
| 6p | −0.628530    | 6.064717              | 2.318081            | 0.543595                 | 1.093741                 | 33.363248                |
| 3d | −114.925563  | 0.026480              | 0.150730            | 7.901775                 | 76.594081                | 950.292286               |
| 4d | −22.749141   | 0.177533              | 0.393090            | 3.343248                 | 19.339833                | 216.450001               |
| 5d | −2.808270    | 1.184780              | 1.020925            | 1.274638                 | 3.633604                 | 34.559340                |
| 4f | −11.253710   | 0.177581              | 0.388852            | 3.064581                 | 11.388702                | 52.455830                |

(continued on next page)



Table 1 (continued)

Ra,  $Z = 88$ , [Rn]7s<sup>2</sup> (1S)

Total energy = −23094.3036664 ; Kinetic energy = 23094.3036664

Potential energy = −46188.6073329 ; Virial ratio = −2.00000000000

 $\rho_0 = 491019.47$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −3388.941086 | 0.000395              | 0.017189            | 87.441901                | 15323.890373             |                          |
| 2s | −587.744241  | 0.006081              | 0.072040            | 20.710565                | 1732.538377              |                          |
| 3s | −147.871751  | 0.036857              | 0.179861            | 8.132356                 | 403.721739               |                          |
| 4s | −37.341614   | 0.167743              | 0.386201            | 3.652154                 | 107.801103               |                          |
| 5s | −8.253202    | 0.751412              | 0.821256            | 1.625857                 | 25.113203                |                          |
| 6s | −1.370703    | 4.022840              | 1.900018            | 0.667584                 | 4.347954                 |                          |
| 7s | −0.148771    | 35.021116             | 5.573924            | 0.218512                 | 0.379219                 |                          |
| 2p | −566.942804  | 0.004479              | 0.060846            | 20.723243                | 578.011728               | 24434.173727             |
| 3p | −137.798684  | 0.033786              | 0.170982            | 8.080083                 | 135.224147               | 5492.402355              |
| 4p | −32.722041   | 0.169628              | 0.387014            | 3.568368                 | 36.046296                | 1385.940427              |
| 5p | −6.449906    | 0.834520              | 0.863469            | 1.533273                 | 8.190266                 | 290.372934               |
| 6p | −0.819838    | 5.204268              | 2.152566            | 0.587070                 | 1.299121                 | 40.828423                |
| 3d | −119.228767  | 0.025750              | 0.148656            | 8.010219                 | 78.691673                | 989.211971               |
| 4d | −24.197831   | 0.171165              | 0.386034            | 3.404244                 | 20.031964                | 227.523368               |
| 5d | −3.296896    | 1.100136              | 0.984751            | 1.322683                 | 3.910847                 | 38.004576                |
| 4f | −12.430525   | 0.169121              | 0.379829            | 3.131827                 | 11.874290                | 55.715504                |

Ac,  $Z = 89$ , [Rn]7s<sup>2</sup>6d<sup>1</sup> (2D)

Total energy = −23722.1920622 ; Kinetic energy = 23722.1920622

Potential energy = −47444.3841243 ; Virial ratio = −2.00000000000

 $\rho_0 = 508169.95$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −3469.626572 | 0.000386              | 0.016994            | 88.441266                | 15675.838049             |                          |
| 2s | −603.407615  | 0.005938              | 0.071193            | 20.958030                | 1774.033741              |                          |
| 3s | −152.640817  | 0.035926              | 0.177579            | 8.238712                 | 414.309873               |                          |
| 4s | −39.023932   | 0.162722              | 0.380391            | 3.710354                 | 111.296825               |                          |
| 5s | −8.862401    | 0.717457              | 0.802617            | 1.666587                 | 26.471761                |                          |
| 6s | −1.535632    | 3.675721              | 1.817166            | 0.700218                 | 4.853885                 |                          |
| 7s | −0.161013    | 30.921936             | 5.234096            | 0.233704                 | 0.449856                 |                          |
| 2p | −582.323027  | 0.004373              | 0.060121            | 20.971416                | 591.883944               | 25315.275480             |
| 3p | −142.405714  | 0.032911              | 0.168756            | 8.187136                 | 138.787560               | 5705.744160              |
| 4p | −34.308938   | 0.164296              | 0.380903            | 3.627343                 | 37.232418                | 1450.106994              |
| 5p | −6.998077    | 0.792518              | 0.841686            | 1.575588                 | 8.660614                 | 311.849447               |
| 6p | −0.945999    | 4.675502              | 2.041835            | 0.620858                 | 1.469664                 | 47.289419                |
| 3d | −123.534804  | 0.025051              | 0.146637            | 8.118696                 | 80.818171                | 1029.189581              |
| 4d | −25.602107   | 0.165125              | 0.379214            | 3.465372                 | 20.737600                | 238.985059               |
| 5d | −3.718988    | 1.025218              | 0.951383            | 1.370522                 | 4.196644                 | 41.640086                |
| 6d | −0.251492    | 10.908557             | 3.066223            | 0.413044                 | 0.427487                 | 3.269690                 |
| 4f | −13.562616   | 0.161304              | 0.371265            | 3.198770                 | 12.368061                | 59.095680                |

Th,  $Z = 90$ , [Rn]7s<sup>2</sup>6d<sup>2</sup> (3F)

Total energy = −24359.6224440 ; Kinetic energy = 24359.6224440

Potential energy = −48719.2448879 ; Virial ratio = −2.00000000000

 $\rho_0 = 525715.30$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −3551.263977 | 0.000377              | 0.016804            | 89.440643                | 16031.785022             |                          |
| 2s | −619.274498  | 0.005801              | 0.070366            | 21.205520                | 1816.023614              |                          |
| 3s | −157.477657  | 0.035030              | 0.175353            | 8.345125                 | 425.039778               |                          |
| 4s | −40.726162   | 0.157916              | 0.374745            | 3.768673                 | 114.854791               |                          |
| 5s | −9.468353    | 0.685911              | 0.784883            | 1.707224                 | 27.861885                |                          |
| 6s | −1.690361    | 3.393128              | 1.746498            | 0.730637                 | 5.348487                 |                          |
| 7s | −0.169989    | 28.214259             | 4.996030            | 0.245573                 | 0.507880                 |                          |
| 2p | −597.906324  | 0.004270              | 0.059413            | 21.219605                | 605.921477               | 26217.355968             |
| 3p | −147.080042  | 0.032069              | 0.166586            | 8.294237                 | 142.398609               | 5924.597669              |
| 4p | −35.915381   | 0.159206              | 0.374973            | 3.686429                 | 38.439653                | 1516.292763              |
| 5p | −7.542955    | 0.753897              | 0.821108            | 1.617737                 | 9.141675                 | 334.210073               |
| 6p | −1.065187    | 4.259231              | 1.949821            | 0.651972                 | 1.634991                 | 53.740654                |
| 3d | −127.907309  | 0.024379              | 0.144672            | 8.227204                 | 82.973536                | 1070.238498              |
| 4d | −27.025469   | 0.159395              | 0.372625            | 3.526576                 | 21.456172                | 250.833643               |
| 5d | −4.138488    | 0.959017              | 0.920744            | 1.417729                 | 4.488192                 | 45.437128                |
| 6d | −0.316901    | 8.906283              | 2.781290            | 0.455190                 | 0.525297                 | 4.206359                 |
| 4f | −14.713587   | 0.154061              | 0.363124            | 3.265426                 | 12.869994                | 62.598201                |

Table 1 (continued)

Pa,  $Z = 91$ , [Rn]7s<sup>2</sup>5f<sup>2</sup>6d<sup>1</sup> (<sup>4</sup>K)

Total energy = −25007.1098723 ; Kinetic energy = 25007.1098723

Potential energy = −50014.2197446 ; Virial ratio = −2.00000000000

 $\rho_0 = 543637.92$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −3633.460561 | 0.000369              | 0.016618            | 90.440109                | 16391.754006             |                          |
| 2s | −634.924072  | 0.005668              | 0.069556            | 21.453482                | 1858.580198              |                          |
| 3s | −161.935825  | 0.034167              | 0.173181            | 8.451585                 | 435.909428               |                          |
| 4s | −42.011351   | 0.153350              | 0.369301            | 3.826401                 | 118.424367               |                          |
| 5s | −9.666735    | 0.662226              | 0.771100            | 1.740053                 | 29.000921                |                          |
| 6s | −1.633846    | 3.374929              | 1.740816            | 0.732245                 | 5.346809                 |                          |
| 7s | −0.163733    | 29.709731             | 5.127266            | 0.238184                 | 0.459446                 |                          |
| 2p | −613.277519  | 0.004171              | 0.058720            | 21.468201                | 620.144517               | 27141.826160             |
| 3p | −151.376996  | 0.031258              | 0.164470            | 8.401411                 | 146.058097               | 6149.107868              |
| 4p | −37.104790   | 0.154387              | 0.369271            | 3.744889                 | 39.651732                | 1583.793991              |
| 5p | −7.693145    | 0.726889              | 0.806097            | 1.650003                 | 9.517630                 | 352.730583               |
| 6p | −1.005681    | 4.278802              | 1.952659            | 0.650230                 | 1.617620                 | 53.599617                |
| 3d | −131.905677  | 0.023733              | 0.142757            | 8.335862                 | 85.160942                | 1112.444822              |
| 4d | −28.031813   | 0.154032              | 0.366343            | 3.586988                 | 22.177938                | 262.940627               |
| 5d | −4.203950    | 0.922021              | 0.902452            | 1.448351                 | 4.684274                 | 48.184312                |
| 6d | −0.291102    | 9.171989              | 2.813590            | 0.451938                 | 0.519897                 | 4.206591                 |
| 4f | −15.451392   | 0.147404              | 0.355435            | 3.331434                 | 13.377564                | 66.209657                |
| 5f | −0.611767    | 2.199777              | 1.350167            | 0.970298                 | 1.615949                 | 5.836120                 |

U,  $Z = 92$ , [Rn]7s<sup>2</sup>5f<sup>3</sup>6d<sup>1</sup> (<sup>5</sup>L)

Total energy = −25664.3382676 ; Kinetic energy = 25664.3382676

Potential energy = −51328.6765352 ; Virial ratio = −2.00000000000

 $\rho_0 = 561973.90$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −3716.781476 | 0.000361              | 0.016436            | 91.439552                | 16755.712830             |                          |
| 2s | −650.960245  | 0.005539              | 0.068765            | 21.701296                | 1901.605300              |                          |
| 3s | −166.654983  | 0.033334              | 0.171062            | 8.558086                 | 446.920165               |                          |
| 4s | −43.508322   | 0.148965              | 0.363995            | 3.884444                 | 122.070911               |                          |
| 5s | −10.049791   | 0.637731              | 0.756683            | 1.775682                 | 30.265832                |                          |
| 6s | −1.674926    | 3.253932              | 1.709102            | 0.746224                 | 5.564993                 |                          |
| 7s | −0.165176    | 29.165384             | 5.078784            | 0.240327                 | 0.464689                 |                          |
| 2p | −629.032775  | 0.004075              | 0.058044            | 21.716663                | 634.525527               | 28087.367177             |
| 3p | −155.933735  | 0.030478              | 0.162406            | 8.508604                 | 149.764411               | 6379.220301              |
| 4p | −38.505419   | 0.149770              | 0.363722            | 3.803644                 | 40.889056                | 1653.550011              |
| 5p | −8.022882    | 0.698158              | 0.789986            | 1.685892                 | 9.944439                 | 373.697696               |
| 6p | −1.029528    | 4.123537              | 1.916539            | 0.662754                 | 1.681500                 | 56.468254                |
| 3d | −136.161260  | 0.023113              | 0.140893            | 8.444477                 | 87.375610                | 1155.718369              |
| 4d | −29.248497   | 0.148918              | 0.360248            | 3.647655                 | 22.914470                | 275.468701               |
| 5d | −4.432602    | 0.878904              | 0.881101            | 1.485137                 | 4.923796                 | 51.514346                |
| 6d | −0.300379    | 8.735948              | 2.744927            | 0.463752                 | 0.548210                 | 4.515522                 |
| 4f | −16.397546   | 0.141189              | 0.348092            | 3.397225                 | 13.893398                | 69.947064                |
| 5f | −0.701559    | 1.947676              | 1.274865            | 1.023563                 | 1.787922                 | 6.659530                 |

Np,  $Z = 93$ , [Rn]7s<sup>2</sup>5f<sup>4</sup>6d<sup>1</sup> (<sup>6</sup>L)

Total energy = −26331.4549589 ; Kinetic energy = 26331.4549589

Potential energy = −52662.9099179 ; Virial ratio = −2.00000000000

 $\rho_0 = 580717.66$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −3801.047108 | 0.000353              | 0.016258            | 92.439007                | 17123.671641             |                          |
| 2s | −667.191187  | 0.005415              | 0.067992            | 21.949154                | 1945.129811              |                          |
| 3s | −171.432036  | 0.032532              | 0.168993            | 8.664628                 | 458.071747               |                          |
| 4s | −45.017315   | 0.144766              | 0.358839            | 3.942517                 | 125.773800               |                          |
| 5s | −10.429141   | 0.614865              | 0.742967            | 1.810850                 | 31.539128                |                          |
| 6s | −1.713918    | 3.143912              | 1.679743            | 0.759578                 | 5.775831                 |                          |
| 7s | −0.166567    | 28.657969             | 5.033158            | 0.242397                 | 0.469878                 |                          |
| 2p | −644.982706  | 0.003982              | 0.057382            | 21.965158                | 649.073174               | 29054.723238             |
| 3p | −160.547961  | 0.029726              | 0.160393            | 8.615831                 | 153.518113               | 6615.047535              |
| 4p | −39.917702   | 0.145358              | 0.358338            | 3.862403                 | 42.145196                | 1725.301301              |
| 5p | −8.349355    | 0.671493              | 0.774725            | 1.721251                 | 10.373575                | 395.156485               |
| 6p | −1.051827    | 3.983361              | 1.883324            | 0.674642                 | 1.742786                 | 59.289223                |
| 3d | −140.473594  | 0.022516              | 0.139076            | 8.553111                 | 89.619065                | 1200.106618              |
| 4d | −30.476245   | 0.144063              | 0.354360            | 3.708251                 | 23.662113                | 288.373569               |
| 5d | −4.659317    | 0.839546              | 0.861131            | 1.521204                 | 5.164196                 | 54.926133                |
| 6d | −0.309935    | 8.347171              | 2.682431            | 0.474927                 | 0.575330                 | 4.818343                 |
| 4f | −17.354532   | 0.135399              | 0.341090            | 3.462685                 | 14.416646                | 73.807189                |
| 5f | −0.770080    | 1.772652              | 1.218306            | 1.069421                 | 1.944494                 | 7.443408                 |

(continued on next page)

Table 1 (continued)

Pu,  $Z = 94$ , [Rn]7s<sup>2</sup>5f<sup>6</sup> (<sup>7</sup>F)

Total energy = −27008.7194421 ; Kinetic energy = 27008.7194421

Potential energy = −54017.4388843 ; Virial ratio = −2.00000000000

 $\rho_0 = 599861.44$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −3886.048279 | 0.000346              | 0.016084            | 93.438513                | 17495.642006             |                          |
| 2s | −683.394155  | 0.005295              | 0.067235            | 22.197259                | 1989.187314              |                          |
| 3s | −176.032343  | 0.031758              | 0.166973            | 8.771210                 | 469.363295               |                          |
| 4s | −46.307145   | 0.140755              | 0.353843            | 4.000335                 | 129.511574               |                          |
| 5s | −10.587780   | 0.595646              | 0.731149            | 1.842126                 | 32.687112                |                          |
| 6s | −1.636153    | 3.157899              | 1.682277            | 0.757361                 | 5.702800                 |                          |
| 7s | −0.159929    | 30.416857             | 5.186307            | 0.234250                 | 0.421717                 |                          |
| 2p | −660.907270  | 0.003893              | 0.056735            | 22.213862                | 663.796904               | 30044.708606             |
| 3p | −164.985869  | 0.029001              | 0.158428            | 8.723106                 | 157.319597               | 6856.696820              |
| 4p | −41.110645   | 0.141156              | 0.353131            | 3.920875                 | 43.413343                | 1798.756556              |
| 5p | −8.461448    | 0.649944              | 0.762025            | 1.751794                 | 10.750877                | 414.806456               |
| 6p | −0.975678    | 4.055659              | 1.898280            | 0.668297                 | 1.697467                 | 58.083175                |
| 3d | −144.610821  | 0.021942              | 0.137304            | 8.661822                 | 91.892883                | 1245.660165              |
| 4d | −31.484373   | 0.139476              | 0.348700            | 3.768451                 | 24.417016                | 301.605483               |
| 5d | −4.691281    | 0.812324              | 0.846593            | 1.549199                 | 5.356392                 | 57.805820                |
| 4f | −18.093399   | 0.129994              | 0.334403            | 3.527846                 | 14.947406                | 77.792574                |
| 5f | −0.654375    | 1.743282              | 1.204473            | 1.086277                 | 2.008870                 | 7.814078                 |

Am,  $Z = 95$ , [Rn]7s<sup>2</sup>5f<sup>7</sup> (<sup>8</sup>S)

Total energy = −27695.8872166 ; Kinetic energy = 27695.8872166

Potential energy = −55391.7744332 ; Virial ratio = −2.00000000000

 $\rho_0 = 619433.54$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −3972.193926 | 0.000338              | 0.015914            | 94.437992                | 17871.601240             |                          |
| 2s | −700.004792  | 0.005179              | 0.066495            | 22.445205                | 2033.711384              |                          |
| 3s | −180.915313  | 0.031011              | 0.165000            | 8.877827                 | 480.795953               |                          |
| 4s | −47.830984   | 0.136898              | 0.348970            | 4.058429                 | 133.324766               |                          |
| 5s | −10.952653   | 0.575512              | 0.718655            | 1.876355                 | 33.970946                |                          |
| 6s | −1.668161    | 3.063507              | 1.656714            | 0.769194                 | 5.887188                 |                          |
| 7s | −0.161309    | 29.909022             | 5.141664            | 0.236247                 | 0.427119                 |                          |
| 2p | −677.236792  | 0.003806              | 0.056103            | 22.462424                | 678.678050               | 31056.464670             |
| 3p | −169.705203  | 0.028302              | 0.156510            | 8.830395                 | 161.167880               | 7104.156143              |
| 4p | −42.537061   | 0.137122              | 0.348059            | 3.979608                 | 44.706276                | 1874.538611              |
| 5p | −8.774365    | 0.626654              | 0.748214            | 1.786104                 | 11.182630                | 437.138016               |
| 6p | −0.992523    | 3.936687              | 1.869823            | 0.678531                 | 1.748997                 | 60.570517                |
| 3d | −149.026799  | 0.021390              | 0.135578            | 8.770487                 | 94.193850                | 1292.320046              |
| 4d | −32.725076   | 0.135089              | 0.343200            | 3.828877                 | 25.186351                | 315.271435               |
| 5d | −4.907078    | 0.778457              | 0.828742            | 1.584031                 | 5.597976                 | 61.364340                |
| 4f | −19.062694   | 0.124915              | 0.327987            | 3.592851                 | 15.486543                | 81.910075                |
| 5f | −0.722194    | 1.608114              | 1.158335            | 1.128014                 | 2.158660                 | 8.600343                 |

Cm,  $Z = 96$ , [Rn]7s<sup>2</sup>5f<sup>7</sup>6d<sup>1</sup> (<sup>9</sup>D)

Total energy = −28392.7711729 ; Kinetic energy = 28392.7711729

Potential energy = −56785.5423459 ; Virial ratio = −2.00000000000

 $\rho_0 = 639440.77$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −4059.515709 | 0.000331              | 0.015747            | 95.437442                | 18251.547986             |                          |
| 2s | −717.055916  | 0.005067              | 0.065772            | 22.692987                | 2078.700017              |                          |
| 3s | −186.114113  | 0.030289              | 0.163072            | 8.984490                 | 492.371179               |                          |
| 4s | −49.621354   | 0.133186              | 0.344214            | 4.116823                 | 137.215738               |                          |
| 5s | −11.552378   | 0.554540              | 0.705485            | 1.913836                 | 35.409343                |                          |
| 6s | −1.824316    | 2.864134              | 1.602700            | 0.796613                 | 6.371924                 |                          |
| 7s | −0.171577    | 27.101634             | 4.891334            | 0.249305                 | 0.491511                 |                          |
| 2p | −694.003880  | 0.003722              | 0.055485            | 22.710837                | 693.716008               | 32090.185212             |
| 3p | −174.739115  | 0.027628              | 0.154638            | 8.937707                 | 165.063368               | 7357.514370              |
| 4p | −44.229462   | 0.133246              | 0.343113            | 4.038631                 | 46.024838                | 1952.725167              |
| 5p | −9.316185    | 0.601831              | 0.733336            | 1.824481                 | 11.675443                | 462.527345               |
| 6p | −1.114342    | 3.630658              | 1.797039            | 0.707284                 | 1.914000                 | 67.541037                |
| 3d | −153.754620  | 0.020859              | 0.133894            | 8.879108                 | 96.522004                | 1340.099352              |
| 4d | −34.230821   | 0.130890              | 0.337849            | 3.889564                 | 25.970657                | 329.386599               |
| 5d | −5.333598    | 0.739332              | 0.808012            | 1.625791                 | 5.892426                 | 65.674394                |
| 6d | −0.300695    | 8.012293              | 2.619245            | 0.487281                 | 0.600152                 | 5.173010                 |
| 4f | −20.294897   | 0.120167              | 0.321858            | 3.657503                 | 16.032614                | 86.152196                |
| 5f | −0.968851    | 1.408903              | 1.089331            | 1.193093                 | 2.398483                 | 9.859440                 |

Table 1 (continued)

Bk,  $Z = 97$ , [Rn]7s<sup>2</sup>5f<sup>8</sup>6d<sup>1</sup> (<sup>8</sup>H)Total energy =  $-29099.8316144$  ; Kinetic energy =  $29099.8316144$ Potential energy =  $-58199.6632288$  ; Virial ratio =  $-2.00000000000$  $\rho_0 = 659861.82$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -4147.580550 | 0.000324              | 0.015583            | 96.436938                | 18635.505746             |                          |
| 2s | -734.087216  | 0.004958              | 0.065064            | 22.940998                | 2124.219983              |                          |
| 3s | -191.144846  | 0.029593              | 0.161188            | 9.091197                 | 504.087236               |                          |
| 4s | -51.200532   | 0.129633              | 0.339598            | 4.174993                 | 141.143680               |                          |
| 5s | -11.938003   | 0.536595              | 0.693947            | 1.947769                 | 36.731462                |                          |
| 6s | -1.865158    | 2.778277              | 1.578359            | 0.809067                 | 6.579323                 |                          |
| 7s | -0.173886    | 26.493562             | 4.835216            | 0.252220                 | 0.502473                 |                          |
| 2p | -710.753631  | 0.003642              | 0.054880            | 22.959446                | 708.929585               | 33147.262577             |
| 3p | -179.605363  | 0.026977              | 0.152809            | 9.045071                 | 169.006855               | 7616.917135              |
| 4p | -45.710503   | 0.129546              | 0.338322            | 4.097402                 | 47.356079                | 2032.727071              |
| 5p | -9.649728    | 0.581254              | 0.720657            | 1.858469                 | 12.120185                | 486.255196               |
| 6p | -1.138552    | 3.521149              | 1.769479            | 0.718397                 | 1.974388                 | 70.512040                |
| 3d | -158.315920  | 0.020346              | 0.132251            | 8.987808                 | 98.880535                | 1389.086849              |
| 4d | -35.524917   | 0.126908              | 0.332693            | 3.949902                 | 26.762610                | 343.849950               |
| 5d | -5.570008    | 0.710239              | 0.791929            | 1.660211                 | 6.142207                 | 69.490825                |
| 6d | -0.283443    | 8.232939              | 2.649108            | 0.482002                 | 0.582084                 | 5.016314                 |
| 4f | -21.316714   | 0.115740              | 0.316014            | 3.721674                 | 16.584593                | 90.514172                |
| 5f | -0.988592    | 1.334503              | 1.059682            | 1.228213                 | 2.540309                 | 10.672907                |

Cf,  $Z = 98$ , [Rn]7s<sup>2</sup>5f<sup>10</sup> (<sup>5</sup>I)Total energy =  $-29817.4189160$  ; Kinetic energy =  $29817.4189160$ Potential energy =  $-59634.8378321$  ; Virial ratio =  $-2.00000000000$  $\rho_0 = 680698.46$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -4236.351589 | 0.000318              | 0.015423            | 97.436484                | 19023.475813             |                          |
| 2s | -751.061089  | 0.004853              | 0.064370            | 23.189234                | 2170.271629              |                          |
| 3s | -195.970133  | 0.028919              | 0.159347            | 9.197928                 | 515.942269               |                          |
| 4s | -52.531850   | 0.126231              | 0.335118            | 4.232908                 | 145.105870               |                          |
| 5s | -12.076210   | 0.521379              | 0.683936            | 1.978011                 | 37.923569                |                          |
| 6s | -1.769886    | 2.803193              | 1.584193            | 0.804870                 | 6.461787                 |                          |
| 7s | -0.165547    | 28.433776             | 5.009780            | 0.242392                 | 0.444362                 |                          |
| 2p | -727.448558  | 0.003563              | 0.054288            | 23.208248                | 724.318957               | 34227.977133             |
| 3p | -184.266562  | 0.026349              | 0.151023            | 9.152467                 | 172.997794               | 7882.408068              |
| 4p | -46.943550   | 0.126011              | 0.333682            | 4.155888                 | 48.699117                | 2114.519364              |
| 5p | -9.741983    | 0.564428              | 0.710003            | 1.887965                 | 12.512424                | 508.044017               |
| 6p | -1.047447    | 3.605152              | 1.788329            | 0.709715                 | 1.911473                 | 68.583636                |
| 3d | -162.673303  | 0.019852              | 0.130647            | 9.096576                 | 101.269333               | 1439.296230              |
| 4d | -36.570823   | 0.123132              | 0.327723            | 4.009879                 | 27.561878                | 358.659962               |
| 5d | -5.583823    | 0.689581              | 0.779952            | 1.687376                 | 6.345014                 | 72.757607                |
| 4f | -22.091422   | 0.111580              | 0.310410            | 3.785573                 | 17.143815                | 95.006248                |
| 5f | -0.806768    | 1.349911              | 1.060484            | 1.235808                 | 2.582853                 | 10.996525                |

Es,  $Z = 99$ , [Rn]7s<sup>2</sup>5f<sup>11</sup> (<sup>4</sup>I)Total energy =  $-30544.9721855$  ; Kinetic energy =  $30544.9721855$ Potential energy =  $-61089.9443711$  ; Virial ratio =  $-2.00000000000$  $\rho_0 = 701983.64$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -4326.294960 | 0.000311              | 0.015266            | 98.436002                | 19415.433903             |                          |
| 2s | -768.470955  | 0.004751              | 0.063692            | 23.437315                | 2216.789721              |                          |
| 3s | -201.106192  | 0.028269              | 0.157546            | 9.304698                 | 527.939402               |                          |
| 4s | -54.125634   | 0.122953              | 0.330744            | 4.291071                 | 149.142452               |                          |
| 5s | -12.446889   | 0.505303              | 0.673278            | 2.011291                 | 39.262395                |                          |
| 6s | -1.800654    | 2.728907              | 1.562868            | 0.815901                 | 6.641839                 |                          |
| 7s | -0.166842    | 28.004833             | 4.970757            | 0.244289                 | 0.449782                 |                          |
| 2p | -744.576698  | 0.003487              | 0.053709            | 23.456909                | 739.865658               | 35331.429955             |
| 3p | -189.237321  | 0.025743              | 0.149277            | 9.259881                 | 177.035808               | 8154.003707              |
| 4p | -48.438531   | 0.122611              | 0.329154            | 4.214613                 | 50.066664                | 2198.748429              |
| 5p | -10.061071   | 0.546093              | 0.698341            | 1.921251                 | 12.962468                | 532.795158               |
| 6p | -1.063364    | 3.511976              | 1.764715            | 0.719212                 | 1.961464                 | 71.167098                |
| 3d | -167.337593  | 0.019376              | 0.129081            | 9.205299                 | 103.685306               | 1490.667567              |
| 4d | -37.877709   | 0.119508              | 0.322883            | 4.070067                 | 28.375457                | 373.925860               |
| 5d | -5.807280    | 0.663962              | 0.765298            | 1.720991                 | 6.597828                 | 76.750906                |
| 4f | -23.124979   | 0.107647              | 0.305011            | 3.849343                 | 17.711210                | 99.635895                |
| 5f | -0.843070    | 1.276833              | 1.031424            | 1.271146                 | 2.728793                 | 11.857159                |

(continued on next page)

Table 1 (continued)

| Fm, $Z = 100$ , $[\text{Rn}]7s^25f^{12}(^3\text{H})$                  |                  |                       |                     |                          |                          |                          |
|-----------------------------------------------------------------------|------------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| Total energy = $-31282.7775990$ ; Kinetic energy = $31282.7775990$    |                  |                       |                     |                          |                          |                          |
| Potential energy = $-62565.5551980$ ; Virial ratio = $-2.00000000000$ |                  |                       |                     |                          |                          |                          |
| $\rho_0 = 723708.20$ ; $C_{\text{EN}} = 1.0000000$                    |                  |                       |                     |                          |                          |                          |
|                                                                       | $E_{\text{ORB}}$ | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                    | -4417.185990     | 0.000305              | 0.015112            | 99.435528                | 19811.391820             |                          |
| 2s                                                                    | -786.078755      | 0.004652              | 0.063027            | 23.685428                | 2263.806481              |                          |
| 3s                                                                    | -206.303632      | 0.027640              | 0.155786            | 9.411502                 | 540.077561               |                          |
| 4s                                                                    | -55.735276       | 0.119803              | 0.326484            | 4.349236                 | 153.233921               |                          |
| 5s                                                                    | -12.817797       | 0.490060              | 0.663014            | 2.044342                 | 40.613998                |                          |
| 6s                                                                    | -1.830638        | 2.659050              | 1.542545            | 0.826669                 | 6.818954                 |                          |
| 7s                                                                    | -0.168104        | 27.596465             | 4.933317            | 0.246143                 | 0.455138                 |                          |
| 2p                                                                    | -761.902641      | 0.003414              | 0.053142            | 23.705594                | 755.578795               | 36458.434009             |
| 3p                                                                    | -194.269050      | 0.025157              | 0.147570            | 9.367323                 | 181.121218               | 8431.808062              |
| 4p                                                                    | -49.949034       | 0.119348              | 0.324749            | 4.273323                 | 51.452555                | 2285.142424              |
| 5p                                                                    | -10.380470       | 0.528762              | 0.687135            | 1.954290                 | 13.416754                | 558.150241               |
| 6p                                                                    | -1.078675        | 3.424686              | 1.742297            | 0.728443                 | 2.010378                 | 73.743232                |
| 3d                                                                    | -172.062084      | 0.018917              | 0.127553            | 9.314034                 | 106.130016               | 1543.252096              |
| 4d                                                                    | -39.199520       | 0.116048              | 0.318189            | 4.130192                 | 29.199950                | 389.602723               |
| 5d                                                                    | -6.031402        | 0.639957              | 0.751306            | 1.754314                 | 6.853203                 | 80.852160                |
| 4f                                                                    | -24.172891       | 0.103949              | 0.299835            | 3.912766                 | 18.285098                | 104.393526               |
| 5f                                                                    | -0.878134        | 1.210829              | 1.004412            | 1.305966                 | 2.876859                 | 12.751577                |
| Md, $Z = 101$ , $[\text{Rn}]7s^25f^{13}(^2\text{F})$                  |                  |                       |                     |                          |                          |                          |
| Total energy = $-32030.9329688$ ; Kinetic energy = $32030.9329688$    |                  |                       |                     |                          |                          |                          |
| Potential energy = $-64061.8659376$ ; Virial ratio = $-2.00000000000$ |                  |                       |                     |                          |                          |                          |
| $\rho_0 = 745876.48$ ; $C_{\text{EN}} = 1.0000000$                    |                  |                       |                     |                          |                          |                          |
|                                                                       | $E_{\text{ORB}}$ | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                    | -4509.022098     | 0.000299              | 0.014962            | 100.435064               | 20211.349710             |                          |
| 2s                                                                    | -803.881765      | 0.004556              | 0.062376            | 23.933573                | 2311.322197              |                          |
| 3s                                                                    | -211.559685      | 0.027031              | 0.154064            | 9.518337                 | 552.356485               |                          |
| 4s                                                                    | -57.358286       | 0.116773              | 0.322333            | 4.407393                 | 157.379398               |                          |
| 5s                                                                    | -13.187003       | 0.475605              | 0.653131            | 2.077129                 | 41.976350                |                          |
| 6s                                                                    | -1.859205        | 2.593949              | 1.523355            | 0.837057                 | 6.990466                 |                          |
| 7s                                                                    | -0.169317        | 27.212316             | 4.897828            | 0.247935                 | 0.460374                 |                          |
| 2p                                                                    | -779.423690      | 0.003343              | 0.052586            | 23.954303                | 771.458454               | 37609.244364             |
| 3p                                                                    | -199.358987      | 0.024591              | 0.145902            | 9.474789                 | 185.253946               | 8715.887634              |
| 4p                                                                    | -51.472575       | 0.116215              | 0.320463            | 4.332008                 | 52.856502                | 2373.713784              |
| 5p                                                                    | -10.698300       | 0.512382              | 0.676372            | 1.987043                 | 13.874530                | 584.080050               |
| 6p                                                                    | -1.092903        | 3.343960              | 1.721287            | 0.737274                 | 2.057254                 | 76.270377                |
| 3d                                                                    | -176.844018      | 0.018474              | 0.126060            | 9.422780                 | 108.603440               | 1597.063676              |
| 4d                                                                    | -40.533767       | 0.112740              | 0.313636            | 4.190244                 | 30.035199                | 405.693270               |
| 5d                                                                    | -6.254452        | 0.617471              | 0.737954            | 1.787289                 | 7.110598                 | 85.054821                |
| 4f                                                                    | -25.232636       | 0.100461              | 0.294859            | 3.975915                 | 18.865911                | 109.283612               |
| 5f                                                                    | -0.917868        | 1.149412              | 0.978716            | 1.340670                 | 3.027961                 | 13.683220                |
| No, $Z = 102$ , $[\text{Rn}]7s^25f^{14}(^1\text{S})$                  |                  |                       |                     |                          |                          |                          |
| Total energy = $-32789.5121404$ ; Kinetic energy = $32789.5121404$    |                  |                       |                     |                          |                          |                          |
| Potential energy = $-65579.0242809$ ; Virial ratio = $-2.00000000000$ |                  |                       |                     |                          |                          |                          |
| $\rho_0 = 768492.90$ ; $C_{\text{EN}} = 1.0000000$                    |                  |                       |                     |                          |                          |                          |
|                                                                       | $E_{\text{ORB}}$ | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                    | -4601.802521     | 0.000293              | 0.014814            | 101.434610               | 20615.307614             |                          |
| 2s                                                                    | -821.879184      | 0.004464              | 0.061739            | 24.181750                | 2359.336926              |                          |
| 3s                                                                    | -216.873560      | 0.026442              | 0.152379            | 9.625201                 | 564.776068               |                          |
| 4s                                                                    | -58.994019       | 0.113858              | 0.318288            | 4.465535                 | 161.578455               |                          |
| 5s                                                                    | -13.554080       | 0.461886              | 0.643611            | 2.109653                 | 43.348910                |                          |
| 6s                                                                    | -1.886259        | 2.533319              | 1.505255            | 0.847050                 | 7.155752                 |                          |
| 7s                                                                    | -0.170482        | 26.851181             | 4.864220            | 0.249664                 | 0.465496                 |                          |
| 2p                                                                    | -797.139052      | 0.003274              | 0.052042            | 24.203037                | 787.504652               | 38784.111954             |
| 3p                                                                    | -204.506342      | 0.024043              | 0.144271            | 9.582278                 | 189.433957               | 9006.310970              |
| 4p                                                                    | -53.008505       | 0.113207              | 0.316292            | 4.390661                 | 54.278367                | 2464.482033              |
| 5p                                                                    | -11.014131       | 0.496882              | 0.666026            | 2.019509                 | 14.335615                | 610.580432               |
| 6p                                                                    | -1.105981        | 3.269439              | 1.701642            | 0.745687                 | 2.101873                 | 78.736401                |
| 3d                                                                    | -181.682604      | 0.018046              | 0.124601            | 9.531535                 | 111.105563               | 1652.116233              |
| 4d                                                                    | -41.879795       | 0.109577              | 0.309218            | 4.250221                 | 30.881133                | 422.201526               |
| 5d                                                                    | -6.475998        | 0.596373              | 0.725200            | 1.819920                 | 7.369894                 | 89.357962                |
| 4f                                                                    | -26.303590       | 0.096971              | 0.289846            | 4.040800                 | 19.469371                | 114.422727               |
| 5f                                                                    | -0.963492        | 1.092150              | 0.954385            | 1.373342                 | 3.166603                 | 14.538688                |

Table 1 (continued)

Lr,  $Z = 103$ , [Rn]7s<sup>2</sup>5f<sup>14</sup>6d<sup>1</sup> (<sup>2</sup>D)

Total energy = −33557.9504126 ; Kinetic energy = 33557.9504126

Potential energy = −67115.9008251 ; Virial ratio = −2.00000000000

 $\rho_0 = 791579.98$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −4695.804658 | 0.000287              | 0.014669            | 102.434122               | 21023.251875             |                          |
| 2s | −840.363165  | 0.004374              | 0.061114            | 24.429772                | 2407.816608              |                          |
| 3s | −222.548914  | 0.025872              | 0.150730            | 9.732110                 | 577.339199               |                          |
| 4s | −60.942845   | 0.111044              | 0.314333            | 4.523921                 | 165.852856               |                          |
| 5s | −14.199976   | 0.447425              | 0.633484            | 2.145460                 | 44.890874                |                          |
| 6s | −2.064767    | 2.380578              | 1.460119            | 0.874540                 | 7.690658                 |                          |
| 7s | −0.183297    | 24.048905             | 4.601212            | 0.265080                 | 0.546574                 |                          |
| 2p | −815.337993  | 0.003207              | 0.051510            | 24.451629                | 803.707690               | 39982.646260             |
| 3p | −210.013960  | 0.023513              | 0.142675            | 9.689792                 | 193.661447               | 9303.140431              |
| 4p | −54.856988   | 0.110306              | 0.312215            | 4.449556                 | 55.725137                | 2557.834581              |
| 5p | −11.602624   | 0.480157              | 0.654780            | 2.055982                 | 14.862317                | 640.605052               |
| 6p | −1.247405    | 3.026844              | 1.638932            | 0.775271                 | 2.287801                 | 87.221415                |
| 3d | −186.878762  | 0.017633              | 0.123176            | 9.640248                 | 113.634858               | 1708.384580              |
| 4d | −43.537332   | 0.106533              | 0.304906            | 4.310420                 | 31.741607                | 439.193800               |
| 5d | −6.951918    | 0.571762              | 0.710314            | 1.858931                 | 7.683314                 | 94.501824                |
| 6d | −0.267398    | 7.945686              | 2.586831            | 0.494657                 | 0.596986                 | 5.346384                 |
| 4f | −27.683911   | 0.093804              | 0.285174            | 4.104066                 | 20.069272                | 119.622621               |
| 5f | −1.248362    | 0.984938              | 0.910382            | 1.432049                 | 3.421980                 | 16.086903                |

Rf,  $Z = 104$ , [Rn]7s<sup>2</sup>5f<sup>14</sup>6d<sup>2</sup> (<sup>3</sup>F)

Total energy = −34336.6215955 ; Kinetic energy = 34336.6215955

Potential energy = −68673.2431910 ; Virial ratio = −2.00000000000

 $\rho_0 = 815125.09$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −4790.761257 | 0.000282              | 0.014527            | 103.433642               | 21435.195271             |                          |
| 2s | −859.053569  | 0.004286              | 0.060502            | 24.677796                | 2456.789506              |                          |
| 3s | −228.295547  | 0.025320              | 0.149116            | 9.839055                 | 590.043996               |                          |
| 4s | −62.916090   | 0.108332              | 0.310473            | 4.582352                 | 170.185907               |                          |
| 5s | −14.848505   | 0.433530              | 0.623599            | 2.181572                 | 46.473101                |                          |
| 6s | −2.230989    | 2.247963              | 1.419500            | 0.900996                 | 8.223698                 |                          |
| 7s | −0.192030    | 22.261097             | 4.424044            | 0.276542                 | 0.609256                 |                          |
| 2p | −833.742853  | 0.003142              | 0.050988            | 24.700221                | 820.075691               | 41205.620784             |
| 3p | −215.592378  | 0.023001              | 0.141114            | 9.797333                 | 197.936439               | 9606.463506              |
| 4p | −56.729599   | 0.107513              | 0.308240            | 4.508494                 | 57.191757                | 2653.545001              |
| 5p | −12.193057   | 0.464121              | 0.643811            | 2.092826                 | 15.403849                | 671.829160               |
| 6p | −1.378908    | 2.824457              | 1.584310            | 0.803202                 | 2.470339                 | 95.749920                |
| 3d | −192.144824  | 0.017234              | 0.121783            | 9.748975                 | 116.192866               | 1765.920341              |
| 4d | −45.218587   | 0.103613              | 0.300710            | 4.370658                 | 32.614321                | 456.641388               |
| 5d | −7.428230    | 0.548410              | 0.695873            | 1.898439                 | 8.007151                 | 99.890844                |
| 6d | −0.341986    | 6.463344              | 2.347763            | 0.541999                 | 0.723883                 | 6.769874                 |
| 4f | −29.087197   | 0.090796              | 0.280658            | 4.167235                 | 20.677433                | 124.970953               |
| 5f | −1.532670    | 0.899542              | 0.872804            | 1.488133                 | 3.676591                 | 17.678268                |

Db,  $Z = 105$ , [Rn]7s<sup>2</sup>5f<sup>14</sup>6d<sup>3</sup> (<sup>4</sup>F)

Total energy = −35125.5446447 ; Kinetic energy = 35125.5446447

Potential energy = −70251.0892895 ; Virial ratio = −2.00000000000

 $\rho_0 = 839133.34$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −4886.680834 | 0.000276              | 0.014388            | 104.433168               | 21851.137631             |                          |
| 2s | −877.958854  | 0.004202              | 0.059903            | 24.925826                | 2506.256268              |                          |
| 3s | −234.121659  | 0.024786              | 0.147535            | 9.946039                 | 602.890975               |                          |
| 4s | −64.921953   | 0.105716              | 0.306704            | 4.640831                 | 174.577800               |                          |
| 5s | −15.507954   | 0.420197              | 0.613964            | 2.217933                 | 48.093983                |                          |
| 6s | −2.394741    | 2.129266              | 1.381995            | 0.926933                 | 8.763400                 |                          |
| 7s | −0.199440    | 20.867577             | 4.280643            | 0.286498                 | 0.665280                 |                          |
| 2p | −852.362114  | 0.003079              | 0.050477            | 24.948814                | 836.608805               | 42453.291727             |
| 3p | −221.249812  | 0.022505              | 0.139587            | 9.904904                 | 202.259080               | 9916.359509              |
| 4p | −58.634527   | 0.104824              | 0.304363            | 4.567474                 | 58.678239                | 2751.643365              |
| 5p | −12.793788   | 0.448782              | 0.633138            | 2.129956                 | 15.959192                | 704.226658               |
| 6p | −1.509113    | 2.648028              | 1.534896            | 0.830327                 | 2.654069                 | 104.511554               |
| 3d | −197.489036  | 0.016848              | 0.120421            | 9.857717                 | 118.779631               | 1824.738695              |
| 4d | −46.931707   | 0.100809              | 0.296627            | 4.430929                 | 33.499203                | 474.549117               |
| 5d | −7.913593    | 0.526344              | 0.681925            | 1.938263                 | 8.340151                 | 105.512307               |
| 6d | −0.400170    | 5.590281              | 2.190397            | 0.580256                 | 0.836941                 | 8.099940                 |
| 4f | −30.521648   | 0.087935              | 0.276290            | 4.230317                 | 21.293895                | 130.470061               |
| 5f | −1.825614    | 0.828694              | 0.839792            | 1.542322                 | 3.932026                 | 19.319854                |

(continued on next page)

Table 1 (continued)

Sg,  $Z = 106$ , [Rn]7s<sup>2</sup>5f<sup>14</sup>6d<sup>4</sup> (<sup>5</sup>D)

Total energy = −35924.7569387 ; Kinetic energy = 35924.7569387

Potential energy = −71849.5138774 ; Virial ratio = −2.00000000000

 $\rho_0 = 863609.28$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −4983.563644 | 0.000271              | 0.014252            | 105.432699               | 22271.078985             |                          |
| 2s | −897.079200  | 0.004119              | 0.059315            | 25.173864                | 2556.217173              |                          |
| 3s | −240.027297  | 0.024268              | 0.145988            | 10.053061                | 615.880316               |                          |
| 4s | −66.960473   | 0.103193              | 0.303024            | 4.699355                 | 179.028504               |                          |
| 5s | −16.178397   | 0.407415              | 0.604581            | 2.254505                 | 49.752254                |                          |
| 6s | −2.557374    | 2.022070              | 1.347133            | 0.952429                 | 9.310205                 |                          |
| 7s | −0.205899    | 19.734299             | 4.160232            | 0.295368                 | 0.716175                 |                          |
| 2p | −871.195973  | 0.003018              | 0.049976            | 25.197411                | 853.307103               | 43725.910755             |
| 3p | −226.986321  | 0.022024              | 0.138091            | 10.012506                | 206.629424               | 10232.902593             |
| 4p | −60.571812   | 0.102233              | 0.300580            | 4.626495                 | 60.184569                | 2852.158404              |
| 5p | −13.404960   | 0.434127              | 0.622764            | 2.167313                 | 16.527703                | 737.787463               |
| 6p | −1.639159    | 2.492084              | 1.489703            | 0.856815                 | 2.839555                 | 113.530830               |
| 3d | −202.911483  | 0.016475              | 0.119089            | 9.966474                 | 121.395179               | 1884.854446              |
| 4d | −48.676733   | 0.098117              | 0.292650            | 4.491230                 | 34.396239                | 492.922751               |
| 5d | −8.408375    | 0.505510              | 0.668469            | 1.978326                 | 8.681796                 | 111.363337               |
| 6d | −0.453946    | 4.960030              | 2.067731            | 0.614655                 | 0.945693                 | 9.425754                 |
| 4f | −31.987330   | 0.085212              | 0.272062            | 4.293319                 | 21.918683                | 136.122147               |
| 5f | −2.127830    | 0.768526              | 0.810318            | 1.594985                 | 4.188967                 | 21.014768                |

Bh,  $Z = 107$ , [Rn]7s<sup>2</sup>5f<sup>14</sup>6d<sup>5</sup> (<sup>6</sup>S)

Total energy = −36734.3244057 ; Kinetic energy = 36734.3244057

Potential energy = −73468.6488114 ; Virial ratio = −2.00000000000

 $\rho_0 = 888557.23$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −5081.406844 | 0.000266              | 0.014117            | 106.432237               | 22695.019438             |                          |
| 2s | −916.411703  | 0.004040              | 0.058738            | 25.421909                | 2606.672286              |                          |
| 3s | −246.009513  | 0.023766              | 0.144472            | 10.160121                | 629.012003               |                          |
| 4s | −69.028728   | 0.100758              | 0.299429            | 4.757922                 | 183.537817               |                          |
| 5s | −16.857026   | 0.395167              | 0.595449            | 2.291250                 | 51.446628                |                          |
| 6s | −2.717509    | 1.925089              | 1.314720            | 0.977411                 | 9.861252                 |                          |
| 7s | −0.211181    | 18.833434             | 4.061642            | 0.302999                 | 0.759919                 |                          |
| 2p | −890.241537  | 0.002959              | 0.049485            | 25.446010                | 870.170607               | 45023.726544             |
| 3p | −232.798963  | 0.021559              | 0.136627            | 10.120138                | 211.047465               | 10556.163518             |
| 4p | −62.538541   | 0.099736              | 0.296888            | 4.685554                 | 61.710692                | 2955.116719              |
| 5p | −14.023840   | 0.420134              | 0.612691            | 2.204851                 | 17.108836                | 772.504629               |
| 6p | −1.767792    | 2.353607              | 1.448276            | 0.882621                 | 3.025918                 | 122.768880               |
| 3d | −208.409234  | 0.016114              | 0.117785            | 10.075248                | 124.039521               | 1946.282045              |
| 4d | −50.450767   | 0.095530              | 0.288777            | 4.551561                 | 35.305453                | 511.768732               |
| 5d | −8.910044    | 0.485771              | 0.655455            | 2.018674                 | 9.032516                 | 117.454694               |
| 6d | −0.516506    | 4.445901              | 1.961800            | 0.647583                 | 1.054421                 | 10.785634                |
| 4f | −33.481347   | 0.082617              | 0.267967            | 4.356243                 | 22.551793                | 141.929210               |
| 5f | −2.436949    | 0.716614              | 0.783719            | 1.646339                 | 4.447681                 | 22.764201                |

Hs,  $Z = 108$ , [Rn]7s<sup>2</sup>5f<sup>14</sup>6d<sup>6</sup> (<sup>5</sup>D)

Total energy = −37554.1214298 ; Kinetic energy = 37554.1214298

Potential energy = −75108.2428596 ; Virial ratio = −2.00000000000

 $\rho_0 = 913983.02$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −5180.227370 | 0.000261              | 0.013986            | 107.431779               | 23122.958582             |                          |
| 2s | −935.973286  | 0.003962              | 0.058173            | 25.669966                | 2657.622540              |                          |
| 3s | −252.084838  | 0.023279              | 0.142986            | 10.267225                | 642.287070               |                          |
| 4s | −71.143071   | 0.098406              | 0.295917            | 4.816540                 | 188.106668               |                          |
| 5s | −17.559799   | 0.383427              | 0.586560            | 2.328169                 | 53.177674                |                          |
| 6s | −2.888838    | 1.834156              | 1.283567            | 1.002680                 | 10.435518                |                          |
| 7s | −0.218225    | 17.836261             | 3.950771            | 0.312035                 | 0.815519                 |                          |
| 2p | −909.515753  | 0.002901              | 0.049003            | 25.694617                | 887.199521               | 46346.999356             |
| 3p | −238.704293  | 0.021108              | 0.135193            | 10.227806                | 215.513501               | 10886.231091             |
| 4p | −64.551041   | 0.097327              | 0.293284            | 4.744659                 | 63.256824                | 3060.559259              |
| 5p | −14.666337   | 0.406773              | 0.602910            | 2.242551                 | 17.702411                | 808.388592               |
| 6p | −1.907031    | 2.225506              | 1.408804            | 0.908723                 | 3.220668                 | 132.569736               |
| 3d | −213.998892  | 0.015764              | 0.116510            | 10.184039                | 126.712737               | 2009.037515              |
| 4d | −52.270064   | 0.093043              | 0.285004            | 4.611913                 | 36.226694                | 531.090417               |
| 5d | −9.434450    | 0.467435              | 0.643081            | 2.058721                 | 9.387617                 | 123.720672               |
| 6d | −0.534843    | 4.128393              | 1.889686            | 0.674803                 | 1.154865                 | 12.111194                |
| 4f | −35.020049   | 0.080142              | 0.263999            | 4.419105                 | 23.193345                | 147.894298               |
| 5f | −2.768813    | 0.670886              | 0.759338            | 1.696815                 | 4.709582                 | 24.576593                |

Table 1 (continued)

Mt, Z = 109, [Rn]7s<sup>2</sup>5f<sup>14</sup>6d<sup>7</sup> (<sup>4</sup>F)

Total energy = −38384.3424294 ; Kinetic energy = 38384.3424294

Potential energy = −76768.6848587 ; Virial ratio = −2.00000000000

 $\rho_0 = 939889.81$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −5280.007240 | 0.000256              | 0.013857            | 108.431328               | 23554.896910             |                          |
| 2s | −955.745884  | 0.003887              | 0.057618            | 25.918031                | 2709.067277              |                          |
| 3s | −258.235442  | 0.022807              | 0.141530            | 10.374367                | 655.704692               |                          |
| 4s | −73.285831   | 0.096135              | 0.292484            | 4.875198                 | 192.734065               |                          |
| 5s | −18.269481   | 0.372180              | 0.577913            | 2.365213                 | 54.943241                |                          |
| 6s | −3.057598    | 1.751377              | 1.254480            | 1.027454                 | 11.013393                |                          |
| 7s | −0.224082    | 17.034109             | 3.858960            | 0.319894                 | 0.863957                 |                          |
| 2p | −929.000552  | 0.002845              | 0.048530            | 25.943227                | 904.393715               | 47695.968448             |
| 3p | −244.684474  | 0.020671              | 0.133788            | 10.335503                | 220.027295               | 11223.162593             |
| 4p | −66.591674   | 0.095004              | 0.289764            | 4.803798                 | 64.822733                | 3168.503184              |
| 5p | −15.315353   | 0.394019              | 0.593418            | 2.280373                 | 18.307890                | 845.430984               |
| 6p | −2.044736    | 2.110678              | 1.372377            | 0.934198                 | 3.416201                 | 142.593708               |
| 3d | −219.662607  | 0.015426              | 0.151261            | 10.292847                | 129.414778               | 2073.134087              |
| 4d | −54.117073   | 0.090651              | 0.281327            | 4.672292                 | 37.160084                | 550.895932               |
| 5d | −9.964785    | 0.450100              | 0.631149            | 2.098875                 | 9.750328                 | 130.212201               |
| 6d | −0.569105    | 3.821121              | 1.818623            | 1.260246                 | 1.260246                 | 13.524958                |
| 4f | −36.585816   | 0.077780              | 0.260151            | 4.481899                 | 23.843249                | 154.018669               |
| 5f | −3.106745    | 0.630531              | 0.736985            | 1.746304                 | 4.973785                 | 26.446865                |

Ds, Z = 110, [Rn]7s<sup>1</sup>5f<sup>14</sup>6d<sup>9</sup> (<sup>3</sup>D)

Total energy = −39225.1624771 ; Kinetic energy = 39225.1624771

Potential energy = −78450.3249542 ; Virial ratio = −2.00000000000

 $\rho_0 = 966260.81$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −5380.604208 | 0.000252              | 0.013730            | 109.430893               | 23990.838172             |                          |
| 2s | −975.585459  | 0.003814              | 0.057074            | 26.166090                | 2761.003158              |                          |
| 3s | −264.317778  | 0.022349              | 0.140104            | 10.481498                | 669.258533               |                          |
| 4s | −75.313381   | 0.093943              | 0.289132            | 4.933809                 | 197.412609               |                          |
| 5s | −18.841985   | 0.361442              | 0.569534            | 2.402176                 | 56.731930                |                          |
| 6s | −3.089990    | 1.693775              | 1.233276            | 1.046502                 | 11.460202                |                          |
| 7s | −0.202432    | 18.980960             | 4.062057            | 0.303883                 | 0.745786                 |                          |
| 2p | −948.552228  | 0.002791              | 0.048067            | 26.191838                | 921.753086               | 49070.881390             |
| 3p | −250.596178  | 0.020248              | 0.132413            | 10.443201                | 224.587720               | 11566.962867             |
| 4p | −68.517327   | 0.092764              | 0.286328            | 4.862941                 | 66.407711                | 3278.940243              |
| 5p | −15.827926   | 0.381828              | 0.584197            | 2.318334                 | 18.925625                | 883.667798               |
| 6p | −2.060692    | 2.036122              | 1.347337            | 0.952973                 | 3.562124                 | 150.492015               |
| 3d | −225.257152  | 0.015098              | 0.114039            | 10.401666                | 132.145576               | 2138.584837              |
| 4d | −55.849279   | 0.088349              | 0.277741            | 4.732716                 | 38.105948                | 571.197825               |
| 5d | −10.359073   | 0.433852              | 0.619714            | 2.138971                 | 10.119485                | 136.916370               |
| 6d | −0.480983    | 3.833500              | 1.812841            | 0.710420                 | 1.296341                 | 14.109486                |
| 4f | −38.035935   | 0.075524              | 0.256419            | 4.544616                 | 24.501386                | 160.303174               |
| 5f | −3.308616    | 0.595486              | 0.716745            | 1.794391                 | 5.238116                 | 28.362083                |

Rg, Z = 111, [Rn]7s<sup>1</sup>5f<sup>14</sup>6d<sup>10</sup> (<sup>2</sup>S)

Total energy = −40076.3544159 ; Kinetic energy = 40076.3544159

Potential energy = −80152.7088319 ; Virial ratio = −2.00000000000

 $\rho_0 = 993141.48$  ;  $C_{EN} = 1.0000000$ 

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −5482.299830 | 0.000247              | 0.013606            | 110.430453               | 24430.775032             |                          |
| 2s | −995.776977  | 0.003742              | 0.056539            | 26.414173                | 2813.437055              |                          |
| 3s | −270.615664  | 0.021904              | 0.138705            | 10.588710                | 682.961188               |                          |
| 4s | −77.509748   | 0.091822              | 0.285851            | 4.992534                 | 202.156515               |                          |
| 5s | −19.562481   | 0.351118              | 0.561356            | 2.439408                 | 58.563991                |                          |
| 6s | −3.252602    | 1.623890              | 1.207641            | 1.070210                 | 12.037488                |                          |
| 7s | −0.201006    | 18.926525             | 4.052361            | 0.304754                 | 0.742930                 |                          |
| 2p | −968.455127  | 0.002739              | 0.047612            | 26.440458                | 939.277932               | 50471.991803             |
| 3p | −256.722820  | 0.019837              | 0.131064            | 10.550955                | 229.197056               | 11917.836982             |
| 4p | −70.611031   | 0.090599              | 0.282970            | 4.922144                 | 68.013077                | 3391.969477              |
| 5p | −16.487131   | 0.370207              | 0.575264            | 2.356322                 | 19.553919                | 923.030006               |
| 6p | −2.194268    | 1.939280              | 1.315227            | 0.977507                 | 3.759224                 | 160.932339               |
| 3d | −231.065778  | 0.014781              | 0.112842            | 10.510506                | 134.905333               | 2205.407844              |
| 4d | −57.748535   | 0.086133              | 0.274243            | 4.793165                 | 39.063919                | 591.994394               |
| 5d | −10.898725   | 0.417865              | 0.608334            | 2.180089                 | 10.503960                | 143.972338               |
| 6d | −0.524288    | 3.535302              | 1.743429            | 0.738186                 | 1.399571                 | 15.517663                |
| 4f | −39.652685   | 0.073367              | 0.252797            | 4.607283                 | 25.168001                | 166.752229               |
| 5f | −3.656138    | 0.563046              | 0.697543            | 1.842320                 | 5.507730                 | 30.353465                |

(continued on next page)



Table 1 (continued)

| Uub, $Z = 112$ , [Rn]7s <sup>2</sup> 5f <sup>14</sup> 6d <sup>10</sup> (1S)                 |              |                       |                     |                          |                          |                          |
|---------------------------------------------------------------------------------------------|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| Total energy = −40937.7978561 ; Kinetic energy = 40937.7978561                              |              |                       |                     |                          |                          |                          |
| Potential energy = −81875.5957122 ; Virial ratio = −2.00000000000                           |              |                       |                     |                          |                          |                          |
| $\rho_0 = 1020542.6 ; C_{EN} = 1.0000000$                                                   |              |                       |                     |                          |                          |                          |
|                                                                                             | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                                          | −5585.118173 | 0.000243              | 0.013483            | 111.430008               | 24874.706804             |                          |
| 2s                                                                                          | −1016.344743 | 0.003673              | 0.056015            | 26.662285                | 2866.370177              |                          |
| 3s                                                                                          | −277.153044  | 0.021472              | 0.137334            | 10.696014                | 696.814563               |                          |
| 4s                                                                                          | −79.898632   | 0.089768              | 0.282638            | 5.051397                 | 206.968036               |                          |
| 5s                                                                                          | −20.454185   | 0.341180              | 0.553366            | 2.476938                 | 60.441773                |                          |
| 6s                                                                                          | −3.563905    | 1.539904              | 1.176727            | 1.099901                 | 12.788771                |                          |
| 7s                                                                                          | −0.238168    | 15.233405             | 3.643744            | 0.339775                 | 0.986597                 |                          |
| 2p                                                                                          | −988.733509  | 0.002687              | 0.047166            | 26.689089                | 956.968386               | 51899.554963             |
| 3p                                                                                          | −263.088318  | 0.019438              | 0.129742            | 10.658773                | 233.855687               | 12275.881317             |
| 4p                                                                                          | −72.896386   | 0.088508              | 0.279685            | 4.981417                 | 69.639175                | 3507.640330              |
| 5p                                                                                          | −17.315951   | 0.359109              | 0.566599            | 2.394359                 | 20.193094                | 963.549891               |
| 6p                                                                                          | −2.461834    | 1.824229              | 1.276685            | 1.008304                 | 4.016082                 | 174.415865               |
| 3d                                                                                          | −237.112440  | 0.014474              | 0.111670            | 10.619368                | 137.694068               | 2273.617819              |
| 4d                                                                                          | −59.838343   | 0.083999              | 0.270834            | 4.853588                 | 40.033215                | 613.277022               |
| 5d                                                                                          | −11.606420   | 0.403229              | 0.597668            | 2.220242                 | 10.886668                | 151.118521               |
| 6d                                                                                          | −0.696094    | 3.085551              | 1.637112            | 0.783541                 | 1.581319                 | 18.008168                |
| 4f                                                                                          | −41.459662   | 0.071302              | 0.249277            | 4.669914                 | 25.843232                | 173.369184               |
| 5f                                                                                          | −4.171938    | 0.532941              | 0.679272            | 1.890209                 | 5.783234                 | 32.426494                |
| Uut, $Z = 113$ , [Rn]7s <sup>2</sup> 5f <sup>14</sup> 6d <sup>10</sup> 7p <sup>1</sup> (2P) |              |                       |                     |                          |                          |                          |
| Total energy = −41809.5353119 ; Kinetic energy = 41809.5353119                              |              |                       |                     |                          |                          |                          |
| Potential energy = −83619.0706239 ; Virial ratio = −2.00000000000                           |              |                       |                     |                          |                          |                          |
| $\rho_0 = 1048434.8 ; C_{EN} = 1.0000000$                                                   |              |                       |                     |                          |                          |                          |
|                                                                                             | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                                          | −5688.953835 | 0.000238              | 0.013363            | 112.429564               | 25322.636152             |                          |
| 2s                                                                                          | −1037.181523 | 0.003606              | 0.055500            | 26.910404                | 2919.797432              |                          |
| 3s                                                                                          | −283.823034  | 0.021053              | 0.135988            | 10.803350                | 710.810399               |                          |
| 4s                                                                                          | −82.372799   | 0.087782              | 0.279495            | 5.110281                 | 211.837113               |                          |
| 5s                                                                                          | −21.408723   | 0.331658              | 0.545599            | 2.514496                 | 62.349907                |                          |
| 6s                                                                                          | −3.922116    | 1.463351              | 1.147706            | 1.129362                 | 13.556457                |                          |
| 7s                                                                                          | −0.326582    | 12.119262             | 3.264974            | 0.379056                 | 1.300495                 |                          |
| 2p                                                                                          | −1009.280640 | 0.002638              | 0.046728            | 26.937728                | 974.824443               | 53353.826860             |
| 3p                                                                                          | −269.586265  | 0.019051              | 0.128446            | 10.766633                | 238.562746               | 12641.118820             |
| 4p                                                                                          | −75.267059   | 0.086486              | 0.276474            | 5.040747                 | 71.285785                | 3625.973726              |
| 5p                                                                                          | −18.207827   | 0.348475              | 0.558166            | 2.432545                 | 20.844893                | 1005.336407              |
| 6p                                                                                          | −2.776992    | 1.716125              | 1.239377            | 1.039977                 | 4.289008                 | 188.933251               |
| 7p                                                                                          | −0.184086    | 20.399451             | 4.213437            | 0.291293                 | 0.286278                 | 9.935346                 |
| 3d                                                                                          | −243.290836  | 0.014176              | 0.110521            | 10.728249                | 140.511776               | 2343.229341              |
| 4d                                                                                          | −62.012936   | 0.081943              | 0.267507            | 4.914040                 | 41.014702                | 635.068013               |
| 5d                                                                                          | −12.376408   | 0.389349              | 0.587364            | 2.260449                 | 11.276513                | 158.496814               |
| 6d                                                                                          | −0.913580    | 2.735574              | 1.547505            | 0.827515                 | 1.769030                 | 20.656218                |
| 4f                                                                                          | −43.350937   | 0.069326              | 0.245857            | 4.732508                 | 26.527032                | 180.155597               |
| 5f                                                                                          | −4.749946    | 0.505536              | 0.662122            | 1.937679                 | 6.062900                 | 34.571927                |
| Uuq, $Z = 114$ , [Rn]7s <sup>2</sup> 5f <sup>14</sup> 6d <sup>10</sup> 7p <sup>2</sup> (3P) |              |                       |                     |                          |                          |                          |
| Total energy = −42691.6571511 ; Kinetic energy = 42691.6571511                              |              |                       |                     |                          |                          |                          |
| Potential energy = −85383.3143021 ; Virial ratio = −2.00000000000                           |              |                       |                     |                          |                          |                          |
| $\rho_0 = 1076831.4 ; C_{EN} = 1.0000000$                                                   |              |                       |                     |                          |                          |                          |
|                                                                                             | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
| 1s                                                                                          | −5793.753853 | 0.000234              | 0.013245            | 113.429121               | 25774.562993             |                          |
| 2s                                                                                          | −1058.234593 | 0.003541              | 0.054995            | 27.158534                | 2973.720308              |                          |
| 3s                                                                                          | −290.572759  | 0.020646              | 0.134668            | 10.910731                | 724.950825               |                          |
| 4s                                                                                          | −84.879361   | 0.085860              | 0.276420            | 5.169216                 | 216.766417               |                          |
| 5s                                                                                          | −22.373353   | 0.322516              | 0.538036            | 2.552150                 | 64.292417                |                          |
| 6s                                                                                          | −4.275164    | 1.392327              | 1.120009            | 1.159103                 | 14.356004                |                          |
| 7s                                                                                          | −0.410516    | 10.267291             | 3.013633            | 0.411081                 | 1.588872                 |                          |
| 2p                                                                                          | −1030.043684 | 0.002589              | 0.046299            | 27.186376                | 992.846127               | 54835.055351             |
| 3p                                                                                          | −276.163652  | 0.018676              | 0.127175            | 10.874536                | 243.318380               | 13013.630369             |
| 4p                                                                                          | −77.669916   | 0.084531              | 0.273333            | 5.100133                 | 72.952909                | 3746.999867              |
| 5p                                                                                          | −19.109537   | 0.338288              | 0.549965            | 2.470844                 | 21.508759                | 1048.378642              |
| 6p                                                                                          | −3.086296    | 1.617719              | 1.204211            | 1.071919                 | 4.573985                 | 204.321692               |
| 7p                                                                                          | −0.241721    | 15.655764             | 3.707843            | 0.331066                 | 0.390218                 | 14.268192                |
| 3d                                                                                          | −249.547973  | 0.013887              | 0.109396            | 10.837151                | 143.358502               | 2414.257730              |
| 4d                                                                                          | −64.219125   | 0.079960              | 0.264258            | 4.974517                 | 42.008342                | 657.372602               |
| 5d                                                                                          | −13.155431   | 0.376179              | 0.577412            | 2.300675                 | 11.673158                | 166.105161               |
| 6d                                                                                          | −1.125263    | 2.469165              | 1.473999            | 0.868710                 | 1.955668                 | 23.368672                |
| 4f                                                                                          | −45.273367   | 0.067431              | 0.242531            | 4.795069                 | 27.219442                | 187.113986               |
| 5f                                                                                          | −5.336976    | 0.480471              | 0.645973            | 1.984777                 | 6.346806                 | 36.791189                |

Table 1 (continued)

Uup,  $Z = 115$ ,  $[\text{Rn}]7s^25f^{14}6d^{10}7p^3$  ( $^4S$ )Total energy =  $-43584.1991337$ ; Kinetic energy =  $43584.1991337$ Potential energy =  $-87168.3982675$ ; Virial ratio =  $-2.00000000000$  $\rho_0 = 1105737.3$ ;  $C_{\text{EN}} = 1.0000000$ 

|    | $E_{\text{ORB}}$ | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|------------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -5899.523013     | 0.000230              | 0.013130            | 114.428679               | 26230.487465             |                          |
| 2s | -1079.508693     | 0.003477              | 0.054499            | 27.406678                | 3028.138925              |                          |
| 3s | -297.406919      | 0.020250              | 0.133373            | 11.018157                | 739.235969               |                          |
| 4s | -87.422990       | 0.083999              | 0.273409            | 5.228201                 | 221.756079               |                          |
| 5s | -23.352750       | 0.313735              | 0.530668            | 2.589895                 | 66.269206                |                          |
| 6s | -4.628409        | 1.326615              | 1.093669            | 1.188969                 | 15.183551                |                          |
| 7s | -0.494316        | 8.971703              | 2.822711            | 0.439479                 | 1.868253                 |                          |
| 2p | -1051.027380     | 0.002542              | 0.045877            | 27.435033                | 1011.033447              | 56343.489122             |
| 3p | -282.825175      | 0.018311              | 0.125929            | 10.982482                | 248.122589               | 13393.488488             |
| 4p | -80.109620       | 0.082640              | 0.270260            | 5.159573                 | 74.640539                | 3870.748809              |
| 5p | -20.025749       | 0.328525              | 0.541987            | 2.509246                 | 22.184589                | 1092.689271              |
| 6p | -3.395389        | 1.527826              | 1.171035            | 1.104028                 | 4.870049                 | 220.551606               |
| 7p | -0.300122        | 12.859700             | 3.370700            | 0.364500                 | 0.489283                 | 18.556780                |
| 3d | -255.888563      | 0.013606              | 0.108293            | 10.946075                | 146.234261               | 2486.717714              |
| 4d | -66.461595       | 0.078048              | 0.261087            | 5.035020                 | 43.014157                | 680.197232               |
| 5d | -13.948224       | 0.363674              | 0.567794            | 2.340917                 | 12.076581                | 173.946601               |
| 6d | -1.337111        | 2.255579              | 1.411380            | 0.907908                 | 2.142486                 | 26.157665                |
| 4f | -47.231643       | 0.065614              | 0.239296            | 4.857601                 | 27.920471                | 194.246549               |
| 5f | -5.937787        | 0.457475              | 0.630738            | 2.031513                 | 6.634869                 | 39.084681                |

Uuh,  $Z = 116$ ,  $[\text{Rn}]7s^25f^{14}6d^{10}7p^4$  ( $^3P$ )Total energy =  $-44487.1002441$ ; Kinetic energy =  $44487.1002441$ Potential energy =  $-88974.2004882$ ; Virial ratio =  $-2.00000000000$  $\rho_0 = 1135157.7$ ;  $C_{\text{EN}} = 1.0000000$ 

|    | $E_{\text{ORB}}$ | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|------------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -6006.276636     | 0.000226              | 0.013016            | 115.428240               | 26690.409964             |                          |
| 2s | -1101.018964     | 0.003415              | 0.054012            | 27.654835                | 3083.053623              |                          |
| 3s | -304.340559      | 0.019866              | 0.132103            | 11.125628                | 753.666009               |                          |
| 4s | -90.018651       | 0.082197              | 0.270462            | 5.287239                 | 226.806307               |                          |
| 5s | -24.361772       | 0.305296              | 0.523489            | 2.627730                 | 68.280399                |                          |
| 6s | -4.996692        | 1.265784              | 1.068634            | 1.218890                 | 16.037123                |                          |
| 7s | -0.587215        | 7.968610              | 2.664024            | 0.466536                 | 2.157695                 |                          |
| 2p | -1072.246885     | 0.002496              | 0.045462            | 27.683700                | 1029.386457              | 57879.380220             |
| 3p | -289.585866      | 0.017956              | 0.124706            | 11.090468                | 252.975352               | 13780.764385             |
| 4p | -82.601113       | 0.080811              | 0.267253            | 5.219064                 | 76.348624                | 3997.248162              |
| 5p | -20.971284       | 0.319167              | 0.534227            | 2.547728                 | 22.872024                | 1138.266614              |
| 6p | -3.719157        | 1.447914              | 1.140486            | 1.135593                 | 5.170661                 | 237.314445               |
| 7p | -0.318112        | 11.380102             | 3.168580            | 0.390487                 | 0.581734                 | 22.770742                |
| 3d | -262.327677      | 0.013334              | 0.107212            | 11.055020                | 149.139086               | 2560.624534              |
| 4d | -68.755341       | 0.076202              | 0.257989            | 5.095551                 | 44.032200                | 703.549010               |
| 5d | -14.769704       | 0.351784              | 0.558490            | 2.381187                 | 12.486889                | 182.026353               |
| 6d | -1.563991        | 2.076911              | 1.356258            | 0.945845                 | 2.331479                 | 29.048417                |
| 4f | -49.240774       | 0.063870              | 0.236147            | 4.920106                 | 28.630147                | 201.555669               |
| 5f | -6.567278        | 0.436300              | 0.616327            | 2.077926                 | 6.927183                 | 41.454027                |

Uus,  $Z = 117$ ,  $[\text{Rn}]7s^25f^{14}6d^{10}7p^5$  ( $^2P$ )Total energy =  $-45400.4748133$ ; Kinetic energy =  $45400.4748133$ Potential energy =  $-90800.9496267$ ; Virial ratio =  $-2.00000000000$  $\rho_0 = 1165096.6$ ;  $C_{\text{EN}} = 1.0000000$ 

|    | $E_{\text{ORB}}$ | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|------------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | -6113.998867     | 0.000222              | 0.012904            | 116.427802               | 27154.330201             |                          |
| 2s | -1122.749661     | 0.003355              | 0.053533            | 27.903005                | 3138.464189              |                          |
| 3s | -311.357940      | 0.019492              | 0.130856            | 11.233143                | 768.240921               |                          |
| 4s | -92.650623       | 0.080451              | 0.267575            | 5.346325                 | 231.916991               |                          |
| 5s | -25.384797       | 0.297184              | 0.516493            | 2.665641                 | 70.325411                |                          |
| 6s | -5.365249        | 1.209458              | 1.044856            | 1.248790                 | 16.914298                |                          |
| 7s | -0.680180        | 7.186249              | 2.532720            | 0.491594                 | 2.443464                 |                          |
| 2p | -1093.686450     | 0.002452              | 0.045055            | 27.932376                | 1047.905131              | 59442.974719             |
| 3p | -296.430008      | 0.017612              | 0.123506            | 11.198495                | 257.876721               | 14175.534031             |
| 4p | -85.128705       | 0.079041              | 0.264310            | 5.278607                 | 78.077217                | 4126.531740              |
| 5p | -21.930581       | 0.310194              | 0.526676            | 2.586289                 | 23.571089                | 1185.129655              |
| 6p | -4.043072        | 1.375241              | 1.111864            | 1.166940                 | 5.478257                 | 254.737367               |
| 7p | -0.351830        | 10.072991             | 2.982518            | 0.416661                 | 0.680175                 | 27.356377                |
| 3d | -268.849589      | 0.013070              | 0.106152            | 11.163986                | 152.072972               | 2635.992417              |
| 4d | -71.084655       | 0.074421              | 0.254962            | 5.156109                 | 45.062448                | 727.433525               |
| 5d | -15.604312       | 0.340474              | 0.549489            | 2.421467                 | 12.903902                | 190.344757               |
| 6d | -1.791485        | 1.925972              | 1.307453            | 0.982506                 | 2.521726                 | 32.027321                |
| 4f | -51.285055       | 0.062196              | 0.233082            | 4.982586                 | 29.348460                | 209.043387               |
| 5f | -7.209941        | 0.416751              | 0.602675            | 2.124013                 | 7.223602                 | 43.899131                |

(continued on next page)

Table 1 (continued)

Uuo, Z = 118, [Rn]7s<sup>2</sup>5f<sup>14</sup>6d<sup>10</sup>7p<sup>6</sup> (<sup>1</sup>S)  
Total energy = −46324.3558151 ;      Kinetic energy = 46324.3558151  
Potential energy = −92648.7116302 ;      Virial ratio = −2.00000000000  
 $\rho_0 = 1195558.5$  ;       $C_{EN} = 1.0000000$

|    | $E_{ORB}$    | $\langle r^2 \rangle$ | $\langle r \rangle$ | $\langle r^{-1} \rangle$ | $\langle r^{-2} \rangle$ | $\langle r^{-3} \rangle$ |
|----|--------------|-----------------------|---------------------|--------------------------|--------------------------|--------------------------|
| 1s | −6222.690665 | 0.000219              | 0.012794            | 117.427366               | 27622.248213             |                          |
| 2s | −1144.701722 | 0.003296              | 0.053063            | 28.151187                | 3194.370670              |                          |
| 3s | −318.459965  | 0.019128              | 0.129632            | 11.340701                | 782.960770               |                          |
| 4s | −95.319797   | 0.078759              | 0.264747            | 5.405460                 | 237.088162               |                          |
| 5s | −26.422739   | 0.289383              | 0.509673            | 2.703622                 | 72.404003                |                          |
| 6s | −5.735380    | 1.157222              | 1.022258            | 1.278631                 | 17.813655                |                          |
| 7s | −0.774032    | 6.551828              | 2.420578            | 0.515226                 | 2.728567                 |                          |
| 2p | −1115.347016 | 0.002408              | 0.044655            | 28.181060                | 1066.589480              | 61034.521436             |
| 3p | −303.358508  | 0.017277              | 0.122329            | 11.306562                | 262.826716               | 14577.871338             |
| 4p | −87.693285   | 0.077326              | 0.261429            | 5.338200                 | 79.826370                | 4258.633276              |
| 5p | −22.904566   | 0.301581              | 0.519324            | 2.624945                 | 24.282180                | 1233.318062              |
| 6p | −4.368537    | 1.305680              | 1.083887            | 1.199102                 | 5.802531                 | 273.324293               |
| 7p | −0.394359    | 8.987302              | 2.820324            | 0.441102                 | 0.771420                 | 31.666617                |
| 3d | −275.455220  | 0.012814              | 0.105112            | 11.272972                | 155.035927               | 2712.836006              |
| 4d | −73.450439   | 0.072700              | 0.252004            | 5.216692                 | 46.104908                | 751.856963               |
| 5d | −16.453015   | 0.329710              | 0.540777            | 2.461751                 | 13.327566                | 198.904259               |
| 6d | −2.020924    | 1.796023              | 1.263633            | 1.018136                 | 2.713649                 | 35.099567                |
| 4f | −53.365396   | 0.060588              | 0.230097            | 5.045040                 | 30.075413                | 216.711863               |
| 5f | −7.866750    | 0.398652              | 0.589719            | 2.169791                 | 7.524104                 | 46.420760                |